

**Essays in Theoretical Microeconomics
and Empirical Macroeconomics
with Implications for Social Policy
All Around the World**

Inauguraldissertation

zur Erlangung des Grades eines Doktors
der Wirtschaftswissenschaften

durch

die Rechts- und Staatswissenschaftliche Fakultät der
Rheinischen Friedrich-Wilhelms-Universität Bonn

vorgelegt von

Ů. R. Bèta Zâne-Ål

aus Summacumlaudeville

2025

Dekan:	Prof. Dr. Martin Böse
Erstreferent:	Lou E. Vui-Tøn
Zweitreferent:	Prof. Dr. Văe-Rée Smart
Tag der mündlichen Prüfung:	1. Oktober 2025

Acknowledgements

I blame all of you. Writing this book has been an exercise in sustained suffering. The casual reader may, perhaps, exempt herself from excessive guilt, but for those of you who have played the larger role in prolonging my agonies with your encouragement and support, well ... you know who you are, and you owe me.

—Brendan Pietsch, assistant professor of religious studies
at Nazarbayev University in Astana, Kazakhstan

Source: [https://www.timeshighereducation.com/
blog/best-academic-acknowledgements-ever](https://www.timeshighereducation.com/blog/best-academic-acknowledgements-ever)

Commenting is on!
To switch it off, activate
`\PassOptionsToPackage`
`{final}{changes}` and
`\PassOptionsToPackage`
`{disable}{todonotes}`
in the master file.

For doctoral students of the Bonn Graduate School of Economics: Regarding the formal requirements for submitting your dissertation to the Faculty of Law and Economics at the University of Bonn, please see

- <https://www.rsf.uni-bonn.de/de/promotion>.

In particular, please consult

- [https://www.rsf.uni-bonn.de/de/promotion/dateien/
promotionsordnung-wirtschaftswissenschaften-2005.pdf](https://www.rsf.uni-bonn.de/de/promotion/dateien/promotionsordnung-wirtschaftswissenschaften-2005.pdf) (§ 10 lists the documents that need to be handed in when submitting the dissertation) and
- [https://www.rsf.uni-bonn.de/de/promotion/dateien/
eidesstattliche-erklaerung.pdf](https://www.rsf.uni-bonn.de/de/promotion/dateien/eidesstattliche-erklaerung.pdf).

The leaflet

- [https://www.rsf.uni-bonn.de/de/promotion/dateien/
vorgaben-fuer-dissertationschriften-wirtschaftswissenschaften.pdf](https://www.rsf.uni-bonn.de/de/promotion/dateien/vorgaben-fuer-dissertationschriften-wirtschaftswissenschaften.pdf)

describes the details to be included in the front matter of your dissertation and the details regarding the inclusion of the CV at the end of the document.

Ů. R. Bèta Żâne-Ål
Bonn, September 2025

Contents

Acknowledgements	iii
List of Figures	xi
List of Tables	xiii
Introduction	1
References	2
1 My Job Market Paper	5
1.1 Introduction	5
1.2 Portability	6
1.2.1 Clean Code/Semantic Coding: Separating Content and Formatting	6
1.2.2 Using BibLaTeX (Rather Than BibTeX)	8
1.2.3 Using the <i>tabularray</i> Package (Rather Than <i>tabularx/tabulary/tabu</i> , <i>booktabs</i> , <i>longtable</i> , <i>xltabular</i> , <i>multirow</i> , <i>makecell</i> , <i>colortbl</i> , <i>parnotes</i> , <i>threeparttable</i> , ...)	9
1.3 Methods	10
1.3.1 Structuring Introduction and Discussion: Funnel and Reverse Funnel	10
1.3.2 Some Questions That May Help You Guide Writing Your Paper	10
1.3.3 Structuring Paragraphs in Academic Manuscripts	14
1.3.4 Additional Online Resources on Effective Writing in Economics	16
1.3.5 Design of the Main Experiment	17
1.3.6 Predictions	22
1.4 Results	28
1.4.1 Test of Hypothesis 1.1	28
1.4.2 Test of Hypothesis 1.2	29
1.4.3 Heterogeneity	30
1.4.4 Structural Estimation	32
1.5 Discussion	32

1.5.1	Some Limitations	32
1.5.2	Utility from Money	33
1.6	Conclusion	35
Appendix 1.A	Put More Complicated Derivations and Proofs Here	36
1.A.1	Appendix Subsection	36
1.A.2	Salience	37
Appendix 1.B	Additional Figures	38
Appendix 1.C	Additional <i>siunitx</i> and <i>tabularray</i> Example Tables	40
References		47
2	My Second Paper Has a Long Title That Spans Two Lines	49
2.1	Introduction	49
2.2	Portability	51
2.2.1	Clean Code/Semantic Coding: Separating Content and Formatting	51
2.2.2	Using BibLaTeX (Rather Than BibTeX)	52
2.2.3	Using the <i>tabularray</i> Package (Rather Than <i>tabularx</i> / <i>tabulary</i> / <i>tabu</i> , <i>booktabs</i> , <i>longtable</i> , <i>xltabular</i> , <i>multirow</i> , <i>makecell</i> , <i>colortbl</i> , <i>parnotes</i> , <i>threeparttable</i> , ...)	53
2.3	Methods	54
2.3.1	Structuring Introduction and Discussion: Funnel and Reverse Funnel	54
2.3.2	Some Questions That May Help You Guide Writing Your Paper	57
2.3.3	Structuring Paragraphs in Academic Manuscripts	58
2.3.4	Additional Online Resources on Effective Writing in Economics	61
2.3.5	Design of the Main Experiment	61
2.3.6	Predictions	66
2.4	Results	72
2.4.1	Test of Hypothesis 2.1	73
2.4.2	Test of Hypothesis 2.2	74
2.4.3	Heterogeneity	75
2.4.4	Structural Estimation	76
2.5	Discussion	77
2.5.1	Some Limitations	77
2.5.2	Utility from Money	77
2.6	Conclusion	80
Appendix 2.A	Put More Complicated Derivations and Proofs Here	80
2.A.1	Appendix Subsection	80
2.A.2	Salience	81
Appendix 2.B	Additional Figures	83

Appendix 2.C Additional <i>siunitx</i> and <i>tabularray</i> Example Tables	85
References	92
3 Math Tests	95
3.1 Math Test Serif	95
3.1.1 Overview Serif	95
3.1.2 Formulas Serif	96
3.1.3 Math Alphabets Serif	97
3.1.4 Character Sidebearings Serif	99
3.1.5 Superscript Positioning Serif	100
3.1.6 Subscript Positioning Serif	101
3.1.7 Accent Positioning Serif	102
3.1.8 Differentials Serif	103
3.1.9 Slash Kerning Serif	104
3.1.10 (Big) Operators Serif	105
3.1.11 Radicals Serif	105
3.1.12 Over- and Underbraces Serif	105
3.1.13 Normal and Wide Accents Serif	106
3.1.14 Long Arrows Serif	106
3.1.15 Left and Right Delimiters Serif	106
3.1.16 Big-g-g Delimiters Serif	106
3.1.17 Binary Operators Serif	107
3.1.18 Relations Serif	107
3.1.19 Punctuation Serif	107
3.1.20 Arrows Serif	108
3.1.21 Miscellaneous Symbols Serif	108
3.1.22 Variable-Sized Operators Serif	108
3.1.23 Log-Like Operators Serif	108
3.1.24 Delimiters Serif	109
3.1.25 Large Delimiters Serif	109
3.1.26 Math Mode Accents Serif	109
3.1.27 Miscellaneous Constructions Serif	109
3.1.28 AMS Delimiters Serif	109
3.1.29 AMS Arrows Serif	110
3.1.30 AMS Negated Arrows Serif	110
3.1.31 AMS Greek Serif	110
3.1.32 AMS Hebrew Serif	110
3.1.33 AMS Miscellaneous Serif	111
3.1.34 AMS Binary Operators Serif	111
3.1.35 AMS Relations Serif	112
3.1.36 AMS Negated Relations Serif	113
3.1.37 Math “Torture” Test Serif	113

3.2	Math Test Serif Bold	116
3.2.1	Overview Serif Bold	116
3.2.2	Formulas Serif Bold	116
3.2.3	Math Alphabets Serif Bold	118
3.2.4	Character Sidebearings Serif Bold	120
3.2.5	Superscript Positioning Serif Bold	121
3.2.6	Subscript Positioning Serif Bold	122
3.2.7	Accent Positioning Serif Bold	123
3.2.8	Differentials Serif Bold	124
3.2.9	Slash Kerning Serif Bold	125
3.2.10	(Big) Operators Serif Bold	126
3.2.11	Radicals Serif Bold	126
3.2.12	Over- and Underbraces Serif Bold	126
3.2.13	Normal and Wide Accents Serif Bold	127
3.2.14	Long Arrows Serif Bold	127
3.2.15	Left and Right Delimiters Serif Bold	127
3.2.16	Big-g-g Delimiters Serif Bold	127
3.2.17	Binary Operators Serif Bold	128
3.2.18	Relations Serif Bold	128
3.2.19	Punctuation Serif Bold	128
3.2.20	Arrows Serif Bold	129
3.2.21	Miscellaneous Symbols Serif Bold	129
3.2.22	Variable-Sized Operators Serif Bold	129
3.2.23	Log-Like Operators Serif Bold	129
3.2.24	Delimiters Serif Bold	130
3.2.25	Large Delimiters Serif Bold	130
3.2.26	Math Mode Accents Serif Bold	130
3.2.27	Miscellaneous Constructions Serif Bold	130
3.2.28	AMS Delimiters Serif Bold	130
3.2.29	AMS Arrows Serif Bold	131
3.2.30	AMS Negated Arrows Serif Bold	131
3.2.31	AMS Greek Serif Bold	131
3.2.32	AMS Hebrew Serif Bold	131
3.2.33	AMS Miscellaneous Serif Bold	132
3.2.34	AMS Binary Operators Serif Bold	132
3.2.35	AMS Relations Serif Bold	133
3.2.36	AMS Negated Relations Serif Bold	134
3.2.37	Math “Torture” Test Serif Bold	134
3.3	Math Test Sans Serif	137
3.3.1	Overview Sans Serif	137
3.3.2	Formulas Sans Serif	137

3.3.3	Math Alphabets Sans Serif	139
3.3.4	Character Sidebearings Sans Serif	141
3.3.5	Superscript Positioning Sans Serif	142
3.3.6	Subscript Positioning Sans Serif	143
3.3.7	Accent Positioning Sans Serif	144
3.3.8	Differentials Sans Serif	145
3.3.9	Slash Kerning Sans Serif	146
3.3.10	(Big) Operators Sans Serif	147
3.3.11	Radicals Sans Serif	147
3.3.12	Over- and Underbraces Sans Serif	147
3.3.13	Normal and Wide Accents Sans Serif	148
3.3.14	Long Arrows Sans Serif	148
3.3.15	Left and Right Delimiters Sans Serif	148
3.3.16	Big-g-g Delimiters Sans Serif	148
3.3.17	Binary Operators Sans Serif	149
3.3.18	Relations Sans Serif	149
3.3.19	Punctuation Sans Serif	149
3.3.20	Arrows Sans Serif	150
3.3.21	Miscellaneous Symbols Sans Serif	150
3.3.22	Variable-Sized Operators Sans Serif	150
3.3.23	Log-Like Operators Sans Serif	150
3.3.24	Delimiters Sans Serif	151
3.3.25	Large Delimiters Sans Serif	151
3.3.26	Math Mode Accents Sans Serif	151
3.3.27	Miscellaneous Constructions Sans Serif	151
3.3.28	AMS Delimiters Sans Serif	151
3.3.29	AMS Arrows Sans Serif	152
3.3.30	AMS Negated Arrows Sans Serif	152
3.3.31	AMS Greek Sans Serif	152
3.3.32	AMS Hebrew Sans Serif	152
3.3.33	AMS Miscellaneous Sans Serif	153
3.3.34	AMS Binary Operators Sans Serif	153
3.3.35	AMS Relations Sans Serif	154
3.3.36	AMS Negated Relations Sans Serif	155
3.3.37	Math “Torture” Test Sans Serif	155
3.4	Math Test Sans Serif Bold	158
3.4.1	Overview Sans Serif Bold	158
3.4.2	Formulas Sans Serif Bold	158
3.4.3	Math Alphabets Sans Serif Bold	160
3.4.4	Character Sidebearings Sans Serif Bold	162
3.4.5	Superscript Positioning Sans Serif Bold	163

3.4.6	Subscript Positioning	Sans Serif Bold	164
3.4.7	Accent Positioning	Sans Serif Bold	165
3.4.8	Differentials	Sans Serif Bold	166
3.4.9	Slash Kerning	Sans Serif Bold	167
3.4.10	(Big) Operators	Sans Serif Bold	168
3.4.11	Radicals	Sans Serif Bold	168
3.4.12	Over- and Underbraces	Sans Serif Bold	168
3.4.13	Normal and Wide Accents	Sans Serif Bold	169
3.4.14	Long Arrows	Sans Serif Bold	169
3.4.15	Left and Right Delimiters	Sans Serif Bold	169
3.4.16	Big-g-g Delimiters	Sans Serif Bold	169
3.4.17	Binary Operators	Sans Serif Bold	170
3.4.18	Relations	Sans Serif Bold	170
3.4.19	Punctuation	Sans Serif Bold	170
3.4.20	Arrows	Sans Serif Bold	171
3.4.21	Miscellaneous Symbols	Sans Serif Bold	171
3.4.22	Variable-Sized Operators	Sans Serif Bold	171
3.4.23	Log-Like Operators	Sans Serif Bold	171
3.4.24	Delimiters	Sans Serif Bold	172
3.4.25	Large Delimiters	Sans Serif Bold	172
3.4.26	Math Mode Accents	Sans Serif Bold	172
3.4.27	Miscellaneous Constructions	Sans Serif Bold	172
3.4.28	AMS Delimiters	Sans Serif Bold	172
3.4.29	AMS Arrows	Sans Serif Bold	173
3.4.30	AMS Negated Arrows	Sans Serif Bold	173
3.4.31	AMS Greek	Sans Serif Bold	173
3.4.32	AMS Hebrew	Sans Serif Bold	173
3.4.33	AMS Miscellaneous	Sans Serif Bold	174
3.4.34	AMS Binary Operators	Sans Serif Bold	174
3.4.35	AMS Relations	Sans Serif Bold	175
3.4.36	AMS Negated Relations	Sans Serif Bold	176
3.4.37	Math “Torture” Test	Sans Serif Bold	176

List of Figures

1.1	Illustration of the funneling approach, using the “Introduction” section from sigfridsson<empty citation> .	11
1.2	Illustration of the “reverse funneling” approach, using the “Concluding Remarks” subsection from sigfridsson<empty citation> .	12
1.3	Budget Sets $C_{1:1}^{BAL, I}$ and $C_{1:n}^{UNBAL, I}$	18
1.4	Budget Sets $C_{1:1}^{BAL, II}$ and $C_{n:1}^{UNBAL, II}$	18
1.5	Screenshots of a $BAL_{1:1}^I$ Decision (Top) and an $UNBAL_{1:8}^I$ Decision (Bottom)	19
1.B.1	Earnings Sequences Included in Choice List C_{CL}^{BAL}	38
1.B.2	Earnings Sequences Included in Choice List $C_{CL}^{UNBAL, I}$	39
1.B.3	Earnings Sequences Included in Choice List $C_{CL}^{UNBAL, II}$	40
2.1	Illustration of the funneling approach, using the “Introduction” section from sigfridsson<empty citation> .	55
2.2	Illustration of the “reverse funneling” approach, using the “Concluding Remarks” subsection from sigfridsson<empty citation> .	56
2.3	Budget Sets $C_{1:1}^{BAL, I}$ and $C_{1:n}^{UNBAL, I}$	62
2.4	Budget Sets $C_{1:1}^{BAL, II}$ and $C_{n:1}^{UNBAL, II}$	62
2.5	Screenshots of a $BAL_{1:1}^I$ Decision (Top) and an $UNBAL_{1:8}^I$ Decision (Bottom)	64
2.B.1	Earnings Sequences Included in Choice List C_{CL}^{BAL}	83
2.B.2	Earnings Sequences Included in Choice List $C_{CL}^{UNBAL, I}$	84
2.B.3	Earnings Sequences Included in Choice List $C_{CL}^{UNBAL, II}$	85

List of Tables

1	Characters Contained in the Serif Font: XCharter-TLF	3
2	Characters Contained in the Sans-Serif Font: FiraSans-TLF	4
1.1	A <i>tabularray</i> -Based Example Table	29
1.2	Points Awarded in Our Typeface Competition—Basic Formatting That Produces Irregular Column Widths and a Suboptimal Table Width Test Greek: ϵ, θ, ϕ	33
1.3	Points Awarded in Our Typeface Competition—More Sophisticated Formatting with Homogeneous Column Widths and a Table Width That Equals the Text Width	34
1.C.1	An Example of a Regression Table (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017). Remember to Mention the Dependent Variable!	41
1.C.2	Figure Grouping via <i>siunitx</i> in a Table	41
1.C.3	Overview of the Choice Lists Presented to Subjects (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017)	42
1.C.4	A Really Long Table That Spans Multiple Pages	42
2.1	A <i>tabularray</i> -Based Example Table	73
2.2	Points Awarded in Our Typeface Competition—Basic Formatting That Produces Irregular Column Widths and a Suboptimal Table Width Test Greek: ϵ, θ, ϕ	78
2.3	Points Awarded in Our Typeface Competition—More Sophisticated Formatting with Homogeneous Column Widths and a Table Width That Equals the Text Width	78
2.C.1	An Example of a Regression Table (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017). Remember to Mention the Dependent Variable!	86
2.C.2	Figure Grouping via <i>siunitx</i> in a Table	86
2.C.3	Overview of the Choice Lists Presented to Subjects (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017)	87
2.C.4	A Really Long Table That Spans Multiple Pages	87

Introduction

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. ¹

On November 14, 1885, Senator & Mrs. Leland Stanford called together at their San Francisco mansion the 24 prominent men who had been chosen as the first trustees of The Leland Stanford Junior University. They handed to the board the Founding Grant of the University, which they had executed three days before. This document—with various amendments, legislative acts, and court decrees—remains as the University’s charter. In bold, sweeping language it stipulates that the objectives of the University are “to qualify students for personal success and direct usefulness in life; and to promote the publick welfare by exercising an influence in behalf of humanity and civilization, teaching the blessings of liberty regulated by law, and inculcating love and reverence for the great principles of government as derived from the inalienable rights of man to life, liberty, and the pursuit of happiness.”

¿But aren’t Kafka’s Schloß and Æsop’s Œuvres often naïve vis-à-vis the dæmonic phoenix’s official rôle in fluffy soufflés?

(iTHE DAZED BROWN FOX QUICKLY GAVE 12345–67890 JUMPS!)

Ångelå Beatrice Claire Diana Érica Françoise Ginette Hélène Iris Jackie Kären
Łaura María Nátaĥie Øctave Pauline Quêneau Roxanne Sabine Tǎja Uršula Vivian
Wendy Xanthippe Yvønne Zäzilie

Let us cite some publications: Andersen et al. (2008), Andreoni and Sprenger (2012), Kószegi and Szeidl (2013), and Balakrishnan, Haushofer, and Jakiela (2016). With the options set for BibLaTeX in the preamble, citations in the body text

1. A test footnote.

are automatically sorted chronologically—irrespective of the order of the “citekeys” in your input. Of course, entries are sorted alphabetically by author surname in the list of references.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

[Chapter 1](#) consists of my job market paper. I enjoyed writing that paper a lot. This also holds for the paper that makes up [Chapter 2](#) of this dissertation. [Chapter 3](#) includes a large variety of tests to judge the quality of the typesetting of mathematical formulas.

References

- Andersen, Steffen, Glenn W. Harrison, Morten I. Lau, and E. Elisabet Rutström.** 2008. “Eliciting Risk and Time Preferences.” *Econometrica* 76 (3): 583–618. <https://doi.org/10.1111/j.1468-0262.2008.00848.x>. [1]
- Andreoni, James, and Charles Sprenger.** 2012. “Estimating Time Preferences from Convex Budgets.” *American Economic Review* 102 (7): 3333–56. <https://doi.org/10.1257/aer.102.7.3333>. [1]
- Balakrishnan, Uttara, Johannes Haushofer, and Pamela Jakiela.** 2016. “How Soon Is Now? Evidence of Present Bias from Convex Time Budget Experiments.” IZA Discussion Paper, IZA Discussion Paper Series. <http://ftp.iza.org/dp9653.pdf>. [1]
- Kőszegi, Botond, and Adam Szeidl.** 2013. “A Model of Focusing in Economic Choice.” *Quarterly Journal of Economics* 128 (1): 53–104. <https://doi.org/10.1093/qje/qjs049>. [1]

Table 1. Characters Contained in the Serif Font: XCharter-TLF

	‘0	‘1	‘2	‘3	‘4	‘5	‘6	‘7	
‘00x	` 0	‘ 1	^ 2	~ 3	¨ 4	” 5	° 6	ˇ 7	”0x
‘01x	˘ 8	¯ 9	· 10	¸ 11	˙ 12	, 13	< 14	> 15	
‘02x	“ 16	” 17	„ 18	« 19	» 20	— 21	— 22	23	”1x
‘03x	o 24	l 25	j 26	ffl 27	ffi 28	ff 29	fl 30	fi 31	
‘04x	¸ 32	! 33	" 34	# 35	\$ 36	% 37	& 38	' 39	”2x
‘05x	(40	) 41	* 42	+ 43	, 44	- 45	. 46	/ 47	
‘06x	0 48	1 49	2 50	3 51	4 52	5 53	6 54	7 55	”3x
‘07x	8 56	9 57	: 58	; 59	< 60	= 61	> 62	? 63	
‘10x	@ 64	A 65	B 66	C 67	D 68	E 69	F 70	G 71	”4x
‘11x	H 72	I 73	J 74	K 75	L 76	M 77	N 78	O 79	
‘12x	P 80	Q 81	R 82	S 83	T 84	U 85	V 86	W 87	”5x
‘13x	X 88	Y 89	Z 90	[91	\ 92	] 93	^ 94	_ 95	
‘14x	‘ 96	a 97	b 98	c 99	d 100	e 101	f 102	g 103	”6x
‘15x	h 104	i 105	j 106	k 107	l 108	m 109	n 110	o 111	
‘16x	p 112	q 113	r 114	s 115	t 116	u 117	v 118	w 119	”7x
‘17x	x 120	y 121	z 122	{ 123	124	} 125	~ 126	- 127	
‘20x	Ā 128	Ą 129	Ć 130	Č 131	Ď 132	Ě 133	Ě 134	Ĝ 135	”8x
‘21x	Í 136	Ĺ 137	Ł 138	Ń 139	Ň 140	Ŋ 141	Ŏ 142	Ř 143	
‘22x	Ř 144	Š 145	Š 146	Ş 147	Ť 148	Ţ 149	Ů 150	Ű 151	”9x
‘23x	Ỳ 152	Ž 153	Ž 154	Ž 155	IJ 156	İ 157	đ 158	§ 159	
‘24x	ă 160	ą 161	ć 162	č 163	ď 164	ě 165	ę 166	ğ 167	”Ax
‘25x	í 168	ł 169	ł 170	ń 171	ň 172	ŋ 173	ő 174	ř 175	
‘26x	ř 176	ś 177	š 178	ş 179	ť 180	ţ 181	ű 182	ű 183	”Bx
‘27x	ÿ 184	ź 185	ž 186	ž 187	ij 188	i 189	ı 190	£ 191	
‘30x	À 192	Á 193	Â 194	Ã 195	Ä 196	Å 197	Æ 198	Ç 199	”Cx
‘31x	È 200	É 201	Ê 202	Ë 203	Ì 204	Í 205	Î 206	Ï 207	
‘32x	Ð 208	Ñ 209	Ò 210	Ó 211	Ô 212	Õ 213	Ö 214	Œ 215	”Dx
‘33x	Ø 216	Ù 217	Ú 218	Û 219	Ü 220	Ý 221	Þ 222	ŠŠ 223	
‘34x	à 224	á 225	â 226	ã 227	ä 228	å 229	æ 230	ç 231	”Ex
‘35x	è 232	é 233	ê 234	ë 235	ì 236	í 237	î 238	ï 239	
‘36x	ð 240	ñ 241	ò 242	ó 243	ô 244	õ 245	ö 246	œ 247	”Fx
‘37x	ø 248	ù 249	ú 250	û 251	ü 252	ý 253	þ 254	ß 255	
	”8	”9	”A	”B	”C	”D	”E	”F	

Table 2. Characters Contained in the Sans-Serif Font: FiraSans-TLF

	‘0	‘1	‘2	‘3	‘4	‘5	‘6	‘7	
‘00x	` ₀	´ ₁	^ ₂	~ ₃	¨ ₄	” ₅	° ₆	ˇ ₇	“0x
‘01x	˘ ₈	¯ ₉	· ₁₀	¸ ₁₁	¸ ₁₂	, ₁₃	◀ ₁₄	▶ ₁₅	
‘02x	“ ₁₆	” ₁₇	„ ₁₈	« ₁₉	» ₂₀	– ₂₁	— ₂₂	₂₃	
‘03x	ff ₂₄	l ₂₅	j ₂₆	ff ₂₇	fi ₂₈	fl ₂₉	ffl ₃₀	fj ₃₁	“1x
‘04x	₃₂	! ₃₃	" ₃₄	# ₃₅	\$ ₃₆	% ₃₇	& ₃₈	' ₃₉	
‘05x	(₄₀	) ₄₁	* ₄₂	+ ₄₃	, ₄₄	- ₄₅	. ₄₆	/ ₄₇	
‘06x	0 ₄₈	1 ₄₉	2 ₅₀	3 ₅₁	4 ₅₂	5 ₅₃	6 ₅₄	7 ₅₅	“3x
‘07x	8 ₅₆	9 ₅₇	: ₅₈	; ₅₉	< ₆₀	= ₆₁	> ₆₂	? ₆₃	
‘10x	@ ₆₄	A ₆₅	B ₆₆	C ₆₇	D ₆₈	E ₆₉	F ₇₀	G ₇₁	
‘11x	H ₇₂	I ₇₃	J ₇₄	K ₇₅	L ₇₆	M ₇₇	N ₇₈	O ₇₉	“4x
‘12x	P ₈₀	Q ₈₁	R ₈₂	S ₈₃	T ₈₄	U ₈₅	V ₈₆	W ₈₇	
‘13x	X ₈₈	Y ₈₉	Z ₉₀	[₉₁	\ ₉₂	] ₉₃	^ ₉₄	_ ₉₅	
‘14x	‘ ₉₆	a ₉₇	b ₉₈	c ₉₉	d ₁₀₀	e ₁₀₁	f ₁₀₂	g ₁₀₃	“6x
‘15x	h ₁₀₄	i ₁₀₅	j ₁₀₆	k ₁₀₇	l ₁₀₈	m ₁₀₉	n ₁₁₀	o ₁₁₁	
‘16x	p ₁₁₂	q ₁₁₃	r ₁₁₄	s ₁₁₅	t ₁₁₆	u ₁₁₇	v ₁₁₈	w ₁₁₉	
‘17x	x ₁₂₀	y ₁₂₁	z ₁₂₂	{ ₁₂₃	₁₂₄	} ₁₂₅	~ ₁₂₆	- ₁₂₇	“7x
‘20x	Ā ₁₂₈	Ą ₁₂₉	Ć ₁₃₀	Č ₁₃₁	Ď ₁₃₂	Ě ₁₃₃	Ę ₁₃₄	Ğ ₁₃₅	
‘21x	Ĭ ₁₃₆	Ł ₁₃₇	ł ₁₃₈	Ń ₁₃₉	Ñ ₁₄₀	Ŋ ₁₄₁	Ŏ ₁₄₂	Ŕ ₁₄₃	
‘22x	Ř ₁₄₄	Ś ₁₄₅	Š ₁₄₆	Ş ₁₄₇	Ť ₁₄₈	Ț ₁₄₉	Ů ₁₅₀	Ű ₁₅₁	“9x
‘23x	Ÿ ₁₅₂	Ž ₁₅₃	Ž ₁₅₄	Ž ₁₅₅	ı ₁₅₆	İ ₁₅₇	đ ₁₅₈	š ₁₅₉	
‘24x	ă ₁₆₀	ą ₁₆₁	ć ₁₆₂	č ₁₆₃	ď ₁₆₄	ě ₁₆₅	ę ₁₆₆	ğ ₁₆₇	
‘25x	í ₁₆₈	ĺ ₁₆₉	ł ₁₇₀	ń ₁₇₁	ň ₁₇₂	ŋ ₁₇₃	ő ₁₇₄	ř ₁₇₅	“Ax
‘26x	ř ₁₇₆	ś ₁₇₇	š ₁₇₈	ş ₁₇₉	ť ₁₈₀	ț ₁₈₁	ů ₁₈₂	ű ₁₈₃	
‘27x	ÿ ₁₈₄	ž ₁₈₅	ž ₁₈₆	ž ₁₈₇	ij ₁₈₈	i ₁₈₉	ı ₁₉₀	£ ₁₉₁	
‘30x	À ₁₉₂	Á ₁₉₃	Â ₁₉₄	Ã ₁₉₅	Ä ₁₉₆	Å ₁₉₇	Æ ₁₉₈	Ç ₁₉₉	“Cx
‘31x	È ₂₀₀	É ₂₀₁	Ê ₂₀₂	Ë ₂₀₃	Ì ₂₀₄	Í ₂₀₅	Î ₂₀₆	Ï ₂₀₇	
‘32x	Ð ₂₀₈	Ñ ₂₀₉	Ò ₂₁₀	Ó ₂₁₁	Ô ₂₁₂	Õ ₂₁₃	Ö ₂₁₄	Œ ₂₁₅	
‘33x	Ø ₂₁₆	Ù ₂₁₇	Ú ₂₁₈	Û ₂₁₉	Ü ₂₂₀	Ý ₂₂₁	Þ ₂₂₂	ŠŠ ₂₂₃	“Dx
‘34x	à ₂₂₄	á ₂₂₅	â ₂₂₆	ã ₂₂₇	ä ₂₂₈	å ₂₂₉	æ ₂₃₀	ç ₂₃₁	
‘35x	è ₂₃₂	é ₂₃₃	ê ₂₃₄	ë ₂₃₅	ì ₂₃₆	í ₂₃₇	î ₂₃₈	ï ₂₃₉	
‘36x	ð ₂₄₀	ñ ₂₄₁	ò ₂₄₂	ó ₂₄₃	ô ₂₄₄	õ ₂₄₅	ö ₂₄₆	œ ₂₄₇	“Fx
‘37x	ø ₂₄₈	ù ₂₄₉	ú ₂₅₀	û ₂₅₁	ü ₂₅₂	ý ₂₅₃	þ ₂₅₄	ß ₂₅₅	
	“8	“9	“A	“B	“C	“D	“E	“F	

Chapter 1

My Job Market Paper^{*}

1.1 Introduction

“Most people can save a few dollars a day or even \$10 a day,” she said. “That’s doable. But if you say, ‘Can you save \$300 a month or a couple of thousand dollars a year?’ people will say, ‘Whoa.’ Avoiding that ‘whoa,’ which is the hesitancy that can derail planning, is what consultants like Ms. Davidson are trying to do.”

—New York Times, March 27, 2016

This template uses the [Charter](#) typeface for the body text. Charter is a serif typeface and was designed in 1987 by [Matthew Carter](#). By contrast, all headings, tables, and captions are set in a [sans-serif typeface](#). The sans-serif typeface used in this document is [Fira Sans](#), designed by [Erik Spiekermann](#) and collaborators.

[Anonym 1]

The math settings are adjusted in the preamble to the effect that mathematical formulas are automatically typeset in the same font as the surrounding text. That is, math in a serif environment will be set in a serif font, while math in a sans-serif environment will use the sans-serif font. This is an aesthetic choice that may not please everyone given that a sans-serif font may be used in mathematical formulas to express a particular meaning. These cases are, however, very rare.

Let us cite [a couple of publications](#) ([Lisi 1995](#); [Andersen et al. 2008](#); [Andreoni and Sprenger 2012](#); [Balakrishnan, Haushofer, and Jakiela 2016](#); [Bursztyn et al. 2025](#)). With the options set for BibLaTeX in the preamble, citations in the body text are [sorted chronologically](#)—irrespective of the order of the “citekeys” in your input. In the list of references, entries are sorted alphabetically by author surname. [Let’s cite Andersen et al. \(2008\)](#) once more.

[Holger 1]

Ersetzt: some

[Lou E. 1]

Gelöscht: automatically

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text,

[U. R. 1]

Eingefügt

^{*} This footnote can be used for acknowledgments. This is where you can express your gratitude to referees, editors, and colleagues for their valuable feedback and suggestions that helped improve your manuscript. Financial support by third parties can also be mentioned here.

you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

[Holger 2]

Eingefügt

We already included several references above.

[U. R. 2]

Check whether there are more recent publications!

Some **additional** references: See Sims (2003) and Gabaix (2014) for models of “rational inattention” or “goal-driven attention.” See Bordalo, Gennaioli, and Shleifer (2012, 2013), Kőszegi and Szeidl (2013), Taubinsky (2014), and Bushong, Rabin, and Schwartzstein (2016) for models of “stimulus-driven attention.”

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

[Lou E. 2]

Italics?

[Holger 3]

Gelöscht: in detail

Too wordy.

[Lou E. 3]

Ersetzt: will conclude

Let's use the present tense throughout.

In [Section 1.3](#), we describe the **design** of our study. We present the data analysis and our results in [Section 1.4](#). In [Section 1.5](#), we discuss the plausibility of potential alternative explanations. [Section 1.6 concludes](#).

1.2 Portability

1.2.1 Clean Code/Semantic Coding: Separating Content and Formatting

All written texts are structured into words and sentences. If they are long enough, texts are also structured into paragraphs. Academic texts, in addition, feature sections, subsections, and often subsubsections, as well as figures, tables, lists of ref-

erences, and potentially appendices. These structural elements are associated with headings/titles and captions.

The structure of a document is conveyed to the reader through *formatting*. Effective formatting of academic documents, thus, requires that all elements of a particular type—say, section headings—be identifiable as such. Identifiability is achieved through styling that is *consistent within type* and *distinct across types*. For example, all section headings must be formatted identically, and at the same time, they must look different from subsection headings.

As a consequence, one should refrain from using manual, discretionary formatting whenever possible. In line with this principle, the source code of this template refers to the document’s structure as much as possible. This is also called “semantic coding”: using code that describes the *meaning* of the content and not *how* those elements should look.

- Examples for *semantic* commands are `\author`, `\title`, `\section`, `\emph`, `\url`, `\cite`, `\begin{figure} ... \end{figure}`, `\caption`, `\begin{quote} ... \end{quote}`.
- Examples for LaTeX commands that are *not semantic*—and should be avoided—are `\textbf`, `\textit`, `\newpage`, `\pagebreak`, `\linebreak`, `\`, `\noindent`, `\smallskip`, `\centering`, `\noindent`, `\hspace`, `\vspace`, `\cellcolor`, `\multirow`.

The reason why one should stick to using semantic commands is that they keep the code *portable*: Semantic code can be *reused* across document without any changes (or at least without major adjustments). This is not the case with nonsemantic code, that is, code that includes manual formatting instructions. Instructions like `\bigskip`, `\hspace`, `\` (even worse, `\` `\`), etc. may result in decent formatting in a particular document but are bound to lead to undesirable formatting in another document. This applies, in particular, when you face the task of combining several research papers into one, “cumulative” doctoral thesis. Moreover, manual formatting is error-prone, as it easily leads to inconsistent styling.

This entails two things: First, the formatting should be determined, as much as possible, in the *preamble* of the LaTeX document. This comes at the cost, of course, of a rather long preamble. Second, instead of defining custom commands, one should *redefine* standard LaTeX commands or use (widely adopted) packages that are available on CTAN (<https://ctan.org>).

This template follows the above principles, resulting in clean, portable code: The source code of the body of this template (apart from the boxes with examples) consists almost completely of semantic, basic LaTeX, with commands from popular packages on top.

1.2.2 Using BibLaTeX (Rather Than BibTeX)

This template uses the modern BibLaTeX framework (the *biblatex-chicago* package, <https://ctan.org/pkg/biblatex-chicago>, to be precise) instead of the vintage BibTeX framework to manage the references included in the document. The reason for this choice is that BibLaTeX is much more flexible than BibTeX and also handles multi-language references much better.

Moreover, in line with the principle describe above, BibLaTeX permits much better separation of a document's content (in this case, the *.bib* file) and its formatting (of the citations and the bibliography) than BibTeX. In particular, suppressing output of information that is present in a *.bib* file is cumbersome with BibTeX: It amounts to editing the “bibliography style” *.bst* file, and *.bst* files have a syntax that is completely different from LaTeX and difficult to learn. It is usually easier to just remove the information from the *.bib* file. This, however, impacts the portability of the *.bib* file: In other circumstances one might want that very information to be present. BibLaTeX simplifies skipping information that is present in the *.bib* file.

An example: Researchers frequently present their research in the form of posters. Space on posters is usually quite limited. Hence, one might not want authors' first names to be abbreviated on a poster. In the associated paper, however, the names should be printed in full to minimize ambiguity. With BibTeX, this requires time-consuming adjustments of the *.bst* file. With BibLaTeX, it is much easier: In the *.tex* file for the poster, one can include the command

```
\ExecuteBibliographyOptions{giveninits = true}
```

Similarly, in a paper, the list of references should include URLs or DOIs (digital object identifiers) whenever they exist, to make it easy for readers to locate a cited work. On a poster, by contrast, one might want to suppress DOIs and URLs. With BibTeX, this would require time-consuming adjustments of the *.bst* file. With BibLaTeX, one can simply do, for instance,

```
\AtEveryBibitem{%
  \ifentrytype{article}{\clearfield{doi}\clearfield{url}}{}%
}
```

Using BibLaTeX requires using the program *biber* instead of *bibtex* for compiling the bibliography. When using the Overleaf online service, using *biber* instead of *bibtex* is as simple as it gets: Overleaf automatically chooses *biber* to compile the bibliography when it encounters BibLaTeX's backend = *biber* option.¹ And when using the TeXstudio editor (<https://texstudio.org>), an easy way to make this happen is to include the “magic comment”

1. This applies to the commercial version (<https://www.overleaf.com>) as well as to free versions <https://overleaf-students.astro.uni-bonn.de/> and https://uni-bonn.sciebo.de/apps/overleaf_nextcloud/launcher/launch.

```
% !BIB program = biber
```

at the beginning of the *.tex* file.

1.2.3 Using the *tabularray* Package (Rather Than *tabularx/tabulary/tabu*, *booktabs*, *longtable*, *xltabular*, *multirow*, *makecell*, *colortbl*, *parnotes*, *threeparttable*, ...)

The principle of separating the content from its formatting as much as possible can also be applied to tables. The *tabularray* package (<https://ctan.org/pkg/tabularray>) allows you to do just that. *tabularray* has been around since May 2021, so it is relatively new—but it has already matured into a package that is extremely useful. *tabularray* replaces (or loads) all the other table-related LaTeX packages that you may have used so far—like *tabularx/tabulary/tabu*, *longtable*, *xltabular*, *multirow*, *makecell*, *booktabs*, *colortbl*, *parnotes*, *threeparttable*, ...

Separating the contents of a table from its formatting means that no formatting instructions whatsoever need to be placed in the table cells—neither for drawing rules between table rows nor for highlighting entries nor for increasing spacing nor for changing alignment, and so on. Instead, the placement of rules at the top and bottom of a table or between particular rows, the specification of column widths, the specification of font sizes and cell alignment, desired highlighting via boldface or a background color, etc., can all be coded as arguments to the table environment. This is beneficial in at least three aspects:

- The source code of the table's contents is very clean. This particularly applies to animated Beamer presentations: no need to put `\pause/\only/\cellcolor/` ... in table cells.
- It makes table contents *portable*: A table's contents can be reused easily across multiple documents that are formatted in different ways.
- It also drastically simplifies the inclusion of table contents that are produced by an external program such as Python, R, or Stata.

Further additional helpful features of *tabularray* are:

- *tabularray* makes it easy to bottom-align or top-align particular table rows or individual cells entries, while keeping a different alignment for the other rows/cells.
- Table entries that consist of multiple lines of text become easy to produce: Just enclose them in curly braces, like in this example: `... & {Line 1 \\ Line 2} & ... \\`. No need to use `\multirow` or `\makecell` any more.
- You can set document-wide formatting defaults to achieve consistent formatting of all tables. For instance, you can set defaults such that all tables feature the

same top and bottom rules. Or, you could make all column heads be typeset in boldface document-wide.

When using `\UseTblrLibrary{siunitx}`, `tabularray` loads the `siunitx` package. This is another useful package which permits separating content from formatting. In particular, the `S` column type is able to round numbers to a specified precision. This means that one can include table content produced by an external program with as many decimal digits as that program produces and have LaTeX take care of the rounding.

1.3 Methods

1.3.1 Structuring Introduction and Discussion: Funnel and Reverse Funnel

A general structure that can be found in many research papers is the so-called X-shaped or “hourglass” structure.² This structure is brought about when the so-called “funnel” technique is used to write the introduction,³ and when a “reverse funnel” or “upside-down funnel” is used to write the discussion/concluding section:⁴

- (1) The paper starts with a *broad* perspective at the beginning of the introduction, which becomes *narrower* with every sentence, leading to the paper’s narrow research question.
- (2) The *narrowest* part is when the exact research question is posed, the research methodology is described in detail, and the results are derived/presented.
- (3) When the results are discussed and related to the literature in the concluding section, the perspective becomes *broad* again.

An example of the “funnel” technique can be found in Figure 1.1, and of the “reverse funnel” technique in Figure 1.2.

1.3.2 Some Questions That May Help You Guide Writing Your Paper

You may find the following list of questions helpful as a guide to making sure that your manuscript covers all important aspects. The questions are primarily intended for doctoral and advanced researchers, but they may also be helpful to students writing term papers and theses.

Your introduction might answer the following questions (not necessarily in this order, but this order will give rise to a “funnel” structure):

2. See https://skillsearthsciences.sites.uu.nl/writing/structure/#attachment_2387, Fig. 1, and <https://pmc.ncbi.nlm.nih.gov/articles/PMC5405644/pdf/JHRS-10-3.pdf>, Figure 1.

3. See <https://read.aupress.ca/read/read-think-write/section/9e945e9b-62cf-4dbb-b8cd-823549ede292#fig1401>, Figure 14.1, and <https://kib.ki.se/en/write-cite/academic-writing/structure-academic-texts>

4. See <https://read.aupress.ca/read/read-think-write/section/9e945e9b-62cf-4dbb-b8cd-823549ede292#fig1402>, Figure 14.2, and <https://kib.ki.se/en/write-cite/academic-writing/writing-conclusion>.

Molecular simulation methods have become an important technique in many areas of chemistry through the recent advent of effective and widespread software for molecular mechanics, molecular dynamics, and Monte Carlo simulations, and procedures for the estimation of free energy differences. In all such methods, a proper description of the electrostatic interactions within the simulated system is of key importance, ... However, in most implementations, ..., the simplest possible solution is used, ... Thus, most classical simulation methods need a point-charge parameterisation of the molecules of interest.

Unfortunately, atomic charges are not observables, i.e. they cannot unambiguously be determined by experiments or quantum chemical calculations. Therefore, a large number of methods have been suggested for the estimation of point charges [1]. Several groups have tried to derive the charges directly from experimental quantities, ... Yet, relevant data is usually missing or too scarce to allow a determination of all charges in interesting molecules. Instead, most techniques derive atomic charges from quantum chemical calculations.

The simplest way to determine quantum chemical charges is the Mulliken population analysis. ... Although, Mulliken charges are known to strongly depend on the basis set [6] and to reproduce electrostatic moments poorly [1], they are still widely used due to their simplicity. Several other ... methods have been devised to cure the problems of the Mulliken charges [1], ...

...

In this paper, we make a critical analysis of the four most popular potential-based point-charge methods, ... It is shown that these methods may give widely different results and possible explanations for this are discussed. Two alternative methods for deriving atomic charges are suggested, which avoid the arbitrariness in the selection of potential points, and their performance is judged using ...

Paragraph 1

Broad introduction with general description of the research field ("many areas of chemistry"), research topic ("molecular simulation methods"), and recent development ("through the recent advent of ..."). Followed by a more detailed description of crucial aspects in this context ("In all such methods, a proper description ... is of key importance") and of shortcomings of existing methods.

Paragraph 2

Narrowing down the topic: Description of the state of the art and of issues and limitations observed in the literature.

Paragraph 3

Further description of the state of the art: strengths and weaknesses of existing approaches.

Paragraphs 4–6

Further description of unresolved issues.

Paragraph 7

Narrow description of the specific contribution of this paper: Identification of undesirable arbitrary variation in simulation results of existing methods, which inspired development of an "alternative methods" that avoids this issue.

Figure 1.1. Illustration of the funneling approach, using the "Introduction" section from sigfridsson<empty citation>.

Note: The full text of the article by sigfridsson<empty citation> is available at https://lucris.lub.lu.se/ws/portalfiles/portal/135489685/25_charges.pdf.

Potential-based charges strongly depend on the way the potential points are selected. We have presented and tested a new method that avoids such a dependence, CHELMO. It fits the charges directly to the electrostatic moments and it performs as well, or better, as the best electrostatic potential method judging from the calculated electrostatic potential and moments. ... The major disadvantage of the method is that it cannot be used for large molecules, since there is a limited number of moments. No more than 25 independent charges in a molecule can be determined if all moments up to hexadecapole moments are used. In practice it should not be used for molecules with more than about 20 atoms since otherwise the higher moments may be poorly reproduced ... This could be remedied if moments higher than hexadecapole moments are calculated but such moments are not included in the output of normal quantum chemical packages.

In order to get a method that can be used for larger system, we constructed the CHELP-BOW method, which estimates the charges from a Boltzmann-weighted fit to the electrostatic potential. It has the advantage over the other ... methods that ... The present implementation of the CHELP-BOW method is very simplified, but in a future publication we will refine and thoroughly test the method. However, the results are conclusive enough to show that the method ... has the desired behaviour. ... Clearly, this is the method we recommend for general use.

An important advantage of the methods developed in this paper is that they are general, i.e. they can be used with any quantum chemical program and they can use experimental data (...) as well. The only thing that has to be changed is the input section of the programs. Thus any quantum chemical basis sets can be used, and any level of theory that gives a wave function may be employed. Furthermore, the methods can easily be adapted to ...

Finally, a comment on the traditional electrostatic potential methods. ... Thus, we cannot see any justification to use CHELP charges except for backward comparisons.

Paragraph 1

Narrow recap of the first contribution of the paper: brief description of the issue and how it was solved by the new method. Followed by description of the limitations of the new method. Outlook how these limitations could be overcome.

Paragraph 2

Narrow recap of the central contribution of the paper and how it relates to the literature: how existing methods were further improved upon by the second method proposed in this paper. Followed by outlook on future advances and recommendation for researchers in the field.

Paragraph 3

Broader implications due to the generality of the new method.

Paragraph 4

Other broader implications.

Figure 1.2. Illustration of the “reverse funneling” approach, using the “Concluding Remarks” subsection from sigfridsson<empty citation>.

- (1) What do we already know?
- (2) What do we not know yet (the “knowledge gap”)?
- (3) Why is it interesting/relevant to close the gap?
- (4) *What is the research question?* [R]
- (5) Why is it interesting to answer the research question?
- (6) How do we contribute to closing the knowledge gap: How do we answer the research question? (Which method do we use?)
- (7) Why is the chosen method (more) suitable (than alternative methods) for answering the research question?

And your conclusion/discussion might answer the following questions (again, not necessarily in this order, but this order will give rise to a “reverse funnel” structure):

- (8) What was the knowledge gap before the current study? [Similar to the introduction.]
- (9) *How did we contribute to closing that gap, and what did we find?* [A] (Which method/dataset did we use, and what are the results?—Recap and take-home message.)
- (10) How do our results relate to (confirm/contradict/complement/extend) the results from previous and other contemporaneous studies?
- (11) What part of the gap is still open? (Phrased a bit more negatively: What are the limitations of our approach?)
- (12) Next steps/avenues for future research: How could we go about closing the remaining knowledge gap (by removing the limitations of our current approach)?⁵

In her book *The Little Book of Research Writing*,⁶ Varanya Chaubey proposes a simpler list of only three items—the “RAP method”:

- The “R” stands for “research question.”
- The “A” stands for “answer” (to the research question, of course).
- The “P” stands for “positioning statement.” (How does our manuscript relate to the literature: Does it corroborate or extend or contradict others’ findings?)

Relating the “RAP” approach to the the list of questions above, question 4 is the “R,” the answer to question 9 is the “A,” and the remaining questions spell out the “P.”

5. John Cochrane might do away with question 12. See his “Writing Tips for Ph.D. Students,” https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf.

6. For a quick summary of the book, see https://mauve-porcupine-8992.squarespace.com/s/Chaubey_Research_Writing-bxddd.pdf. See also <https://www.econscribe.org/about> and <https://arxiv.org/pdf/2012.07787>.

1.3.3 Structuring Paragraphs in Academic Manuscripts

Obviously, there is no single “right” way of constructing paragraphs. However, a few basic rules can help you write paragraphs that are *easy to comprehend* by your readers. A common approach is to compose sentences such that they reflect the following functional sequence:⁷

- (1) *topic sentence (TS)*—which, if necessary, connects to the previous paragraph;
- (2) *supporting sentences (SS)*—which elaborate on the topic sentence; and
- (3) *concluding sentence or connecting sentence (CS)*—which refers back to the topic sentence and/or provides a link to the subsequent paragraph.

There are various ways in which the supporting sentences can be filled with content. The University of Sheffield suggests⁸ that the supporting sentences could consist of

- (1) “explanation or definitions (optional),”
- (2) “evidence and examples,” and
- (3) “comment” (an interpretation and evaluation of the evidence by you).

The University of Hull and Newcastle University (UK) consider the mnemonic “PEEL” helpful, although they define its meaning in slightly different ways: “Point, Evidence, Explanation, Link”⁹ versus “Point, Evidence, Evaluation, Link”¹⁰. One may view “Explanation”—“why the point is important and how it helps with your overall argument”—as narrower than “Evaluation”—which is a general critical reflection on the informativeness of the evidence and can include an explanation of its importance. “PEEL” relates to TS/SS/CS as follows:

- (1) “P,” the Point, corresponds to the topic sentence.
- (2) “E,” Evidence, is the first part of the supporting sentences.
- (3) “E,” Evaluation/Explanation, forms the second part of the supporting sentences.
- (4) “L,” Link, refers to the connecting sentence that links to the next paragraph.

The Wilfrid Laurier University (Canada) suggests a different mnemonic: “MEAL.”¹¹

7. See <https://libguides.newcastle.edu.au/writing-paragraphs/structure>, https://www.une.edu.au/library/students/academic-writing/write-paragraphs/paragraphs/Academic-paragraphs_v2.pdf, <https://learningessentials.auckland.ac.nz/writing-effectively/paragraph/>

8. See <https://sheffield.ac.uk/study-skills/writing/academic/paragraph>.

9. See <https://libguides.hull.ac.uk/writing/paras>.

10. See <https://www.ncl.ac.uk/academic-skills-kit/writing/academic-writing/paragraphing/>

11. See <https://students.wlu.ca/academics/support-and-advising/student-success/assets/resources/writing/how-to-structure-an-academic-paragraph.html>.

- (1) “M: Main Point Sentence” corresponds to the topic sentence.
- (2) “E: Evidence” is the first part of the supporting sentences.
- (3) “A: Analysis/Synthesis” forms the second part of the supporting sentences.
- (4) “L: Linking-Back Sentence” refers to the concluding sentence that summarizes the paragraph and links back to the topic sentence or even the topic of the entire paper.

The first three components are largely identical according to these approaches. A difference exists in the meaning of “L”: While “PEEL” stresses the link to the *next* step in the chain of thought, “MEAL” stresses underscoring the message of the *current* paragraph. That is, the latter advocates *closure*: summarizing the paragraph and linking *back* to the topic sentence.

Yet slightly different is the approach put forth by Academic Writing UK:¹²

- (1) Topic sentence—key topic in this paragraph.
- (2) Development—the main idea/topic discussed in more detail.
- (3) Example—support/evidence/data/statistics that show your development is valid/credible.
- (4) Summary—overall main point summarized/evaluated.

It is perfectly fine to mix the different approaches. For some paragraphs, a closer explanation of the topic sentence (“development”) may be necessary, while it is unnecessary for others. And in some paragraphs, you may want underscore the central message by reiterating it in the final sentence (concluding sentence), while for other paragraphs, you will use the final sentence to set the stage for the next step in your chain of reasoning (connecting sentence).

One thing that *all* these suggestions have in common is, however, that they start with a *topic sentence* and end with a *concluding/connecting sentence*. Hence, no matter how you fill the supporting sentences, it is probably best to start a paragraph with a topic sentence and end with a concluding/connecting sentence.

Let us illustrate the “MEAL”/“PEEL” structure, with an optional explanation/definition included, through some example sentences from sigfridsson<empty citation>:

... Thus, most classical simulation methods need a point-charge parameterisation of the molecules of interest.

L: Concluding sentence

Continued on next page

12. See <https://academic-englishuk.com/paragraphing/>.

(Continued)

Unfortunately, atomic charges are not observables, i.e. they cannot unambiguously be determined by experiments or quantum chemical calculations. Therefore, a large number of methods have been suggested for the estimation of point charges [1]. Several groups have tried to derive the charges directly from experimental quantities, e.g. dipole moments, electrostatic potentials, or free energy differences [2–4]. Yet, relevant data is usually missing or too scarce to allow a determination of all charges in interesting molecules. Instead, most techniques derive atomic charges from quantum chemical calculations.

The simplest way to determine quantum chemical charges is the Mulliken population analysis. ...

M/P: Topic sentence
(reference to previous paragraph)

Optional explanation/definition

Supporting sentences:
E: Evidence
A/E: Analysis/Evaluation

L: Connecting sentence
(link to next paragraph)

M/P: Topic sentence
(reference to previous paragraph)

The rule to finish a paragraph with a concluding sentence entails, in particular, that a theoretical or empirical finding should never be the last thing that you mention in a paragraph. An “evidence” sentence should always be followed by an interpretation in which you tell the reader what we learn from that finding. Here is an example from sigfridsson<empty citation>:

... With these radii, the charges almost coincide with those obtained by the CHELPG method (within 0.02 e). Thus, we can conclude that the main difference between the Merz–Kollman and the CHELPG method is the 1.4 times larger effective radii of the former method, whereas the sampling schemes are almost equivalent.

Evidence:
Supporting sentence with empirical finding

Concluding sentence with interpretation of that finding

To reiterate, there is no single “right” way of constructing paragraphs. However, keeping the TS/SS/CS structure in mind as a guideline is bound to help you produce a text that enables readers to follow your chain of reasoning.

1.3.4 Additional Online Resources on Effective Writing in Economics

Additional useful recommendations and tips can be found in Plamen Nikolov’s “Writing Tips for Crafting Effective Economics Research Papers – 2023-2024 Edition” (<https://docs.iza.org/dp16276.pdf>) and in John H. Cochrane’s “Writing Tips for Ph.D. Students” (https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf).

In this section, we first present the design of the experiment (1.3.5) and derive behavioral predictions (1.3.6).

1.3.5 Design of the Main Experiment

1.3.5.1 General Features

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.3.5.2 More Specific Features

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Let’s test the euro symbol: €, €1,234.56, €1,234.56. Let’s also test text superscripts: t^{th} and text subscripts: CO_2 and H_2O . $\sigma_\epsilon, c^\alpha$. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Let’s test the footnote settings.¹³

13. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

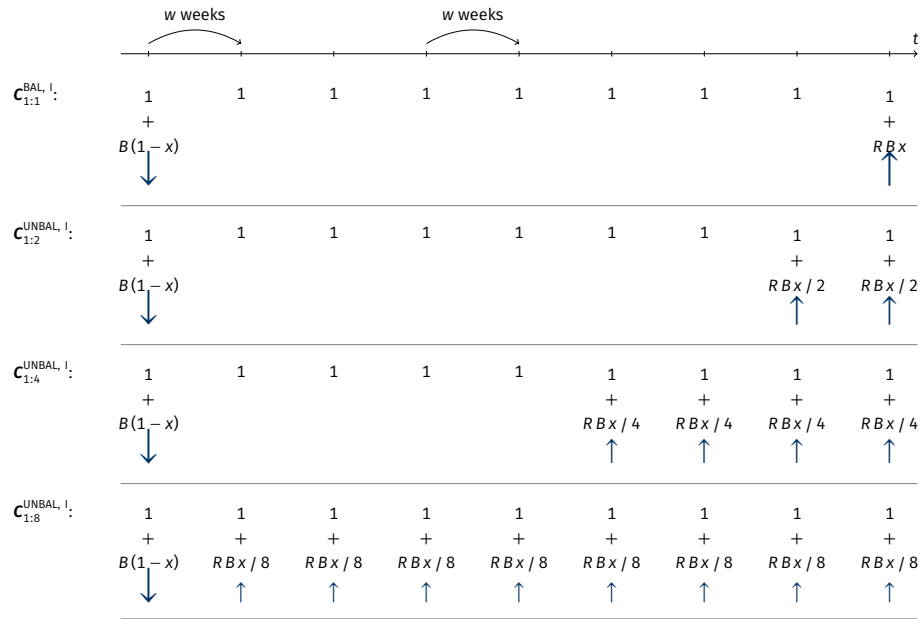


Figure 1.3. Budget Sets $C_{1:1}^{BAL, I}$ and $C_{1:n}^{UNBAL, I}$

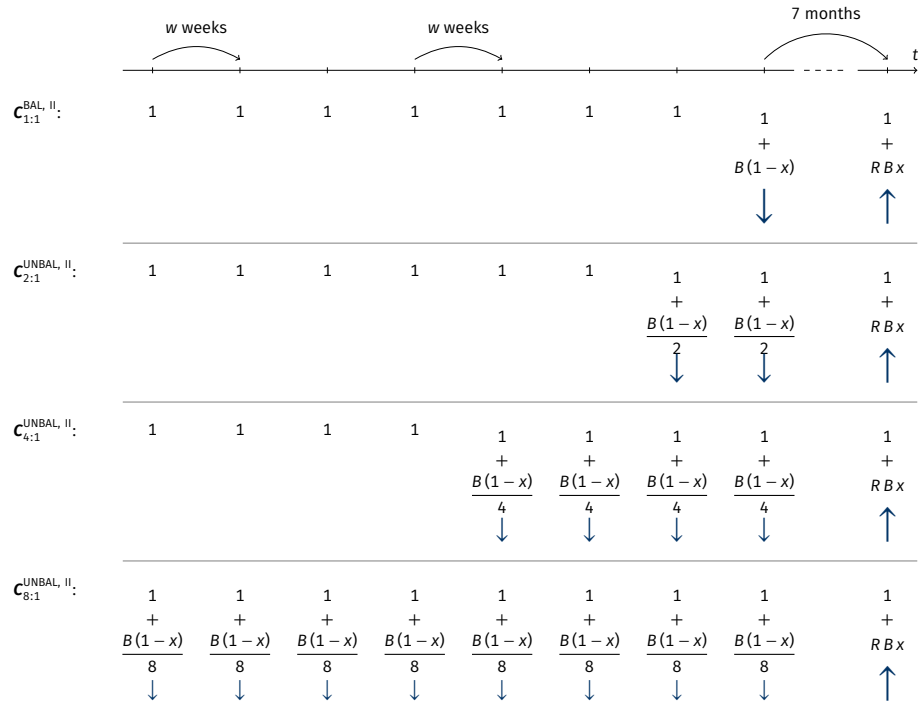


Figure 1.4. Budget Sets $C_{1:1}^{BAL, II}$ and $C_{n:1}^{UNBAL, II}$

Notes: For the values of B , R , and w that we used, see [Section 1.3.5.4](#). The savings rate x is individuals' choice variable: they choose some $x \in X = \{0, 1/100, 2/100, \dots, 1\}$ in each trial. The arrows indicate whether and in which direction payments at the respective payment dates change if x is increased. σ_c, c^a . This figure was taken from Dertwinkel-Kalt et al. (2017).

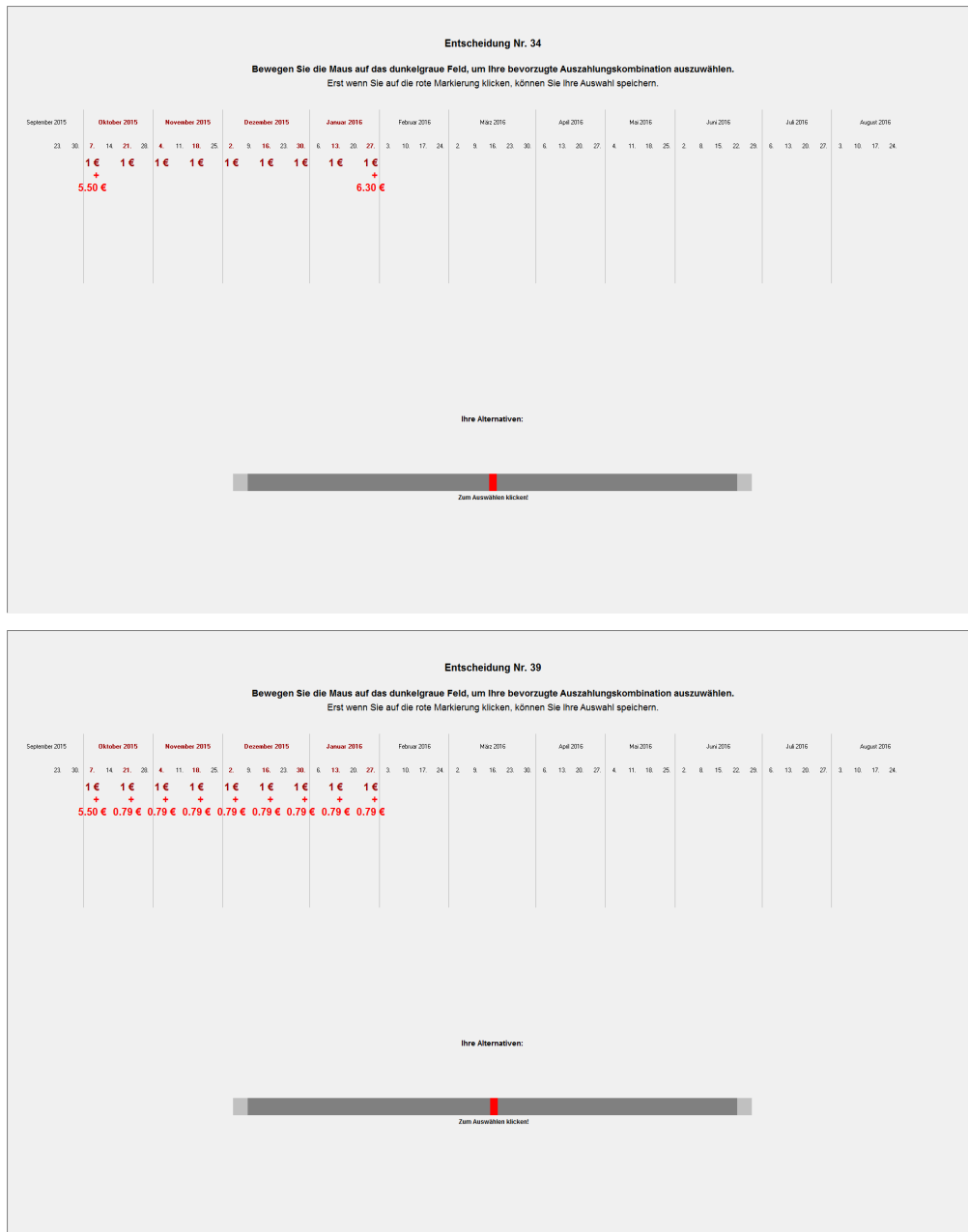


Figure 1.5. Screenshots of a $BAL_{1:1}^I$ Decision (Top) and an $UNBAL_{1:8}^I$ Decision (Bottom)

Note: This figure was taken from Dertwinkel-Kalt et al. (2017).

Figure 1.5 shows an exemplary decision screen with $B = €11$ and $r \approx 15\%$ for both $BAL_{1:1}^I$ (upper panel) and $UNBAL_{1:8}^I$ (lower panel). Through a slider, subjects

choose their preferred $x \in X$.¹⁴ The slider position in Figure 1.5 indicates $x = 0.5$, i.e., the earliest payment is reduced by €5.50. Since $r \approx 15\%$ in this example, this slider position amounts to €6.30 that are paid at later payment dates. While these €6.30 are paid in a single bank transfer on the latest payment date in $BAL_{1,1}^I$, the amount is dispersed in equal parts over the last 8 payment dates in $UNBAL_{1,8}^I$ —i.e., 8 consecutive payments of €0.79.¹⁵

1.3.5.3 Some More Details

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Here’s a bulleted list:

- Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.
- Hello, here is some text without a meaning. $d\Omega = \sin\vartheta d\vartheta d\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.

14. The slider had no initial position—it appeared only after subjects first positioned the mouse cursor over the slider bar. This was done to avoid default effects.

15. We always rounded the second decimal place up so that the sum of the payments included in a dispersed payoff was always at least as great as the respective concentrated payoff.

- Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. If you read this text, you will get no information. $d\Omega = \sin\vartheta d\vartheta d\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

1.3.5.4 Procedure

Describe the sequence of events in your study. You could do this with the help of an enumerated list:

- (1) Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.
- (2) Hello, here is some text without a meaning. $d\Omega = \sin\vartheta d\vartheta d\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.
- (3) Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. If you read this text, you will get no information. $d\Omega = \sin\vartheta d\vartheta d\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages.

By discounted utility we understand any intertemporal utility function that (1) is time-separable and that (2) values a payment farther in the future at most as much as an equal-sized payment closer in the future. Importantly, the predictions derived below hold for all three frequently used types of discounting—exponential, hyperbolic, and quasi-hyperbolic.

In the following, we assume that individuals base their decisions on utility derived from receiving monetary payments c_t at various dates t . This is an assumption that is frequently made in experiments on intertemporal decision making. One way to justify this assumption is that individuals anticipate to consume the payments they receive within a short period around date t . Given that the maximum payment was below €20 and that any two payment dates were separated by at least two weeks, this assumption seems reasonable (see the arguments in favor of this view in Halevy 2014). Kőszegi and Szeidl (2013) themselves make the same assumption of “money in the utility function”: “in some applications we also assume that monetary transactions induce direct utility consequences, so that for instance an agent making a payment experiences an immediate utility loss. The idea that people experience monetary transactions as immediate utility is both intuitively compelling and supported in the literature: ... some evidence on individuals’ attitudes toward money, such as narrow bracketing (...) and laboratory evidence on hyperbolic discounting (...), is difficult to explain without it.” Last but not least, the papers by McClure et al. (2004) and McClure et al. (2007) demonstrate that brain activation, as measured by functional magnetic resonance imaging, is similar for primary and

get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

monetary rewards. Additionally, we make the standard assumption that utility from money is increasing in its argument but not convex: $u'(c_t) \geq 0$ and $u''(c_t) \leq 0$.

1.3.6.1 Discounted Utility

Individuals make their allocation decisions by comparing the aggregated consumption utility of each earnings sequence $\mathbf{c} \in \mathbf{C}$. Discounted utility assumes that the utility of each period enters overall utility additively. That is, utility derived from the payment to be received at future date t can be expressed as $u_t(c_t) := D(t) u(c_t)$. Here, $D(t)$ denotes the individual's discount function for conversion of future utility into present utility. The discount function satisfies $0 \leq D(t)$ and $D'(t) \leq 0$, such that a payment further in the future is valued at most as much as an equal-sized payment closer in the future.¹⁷

The utility of earnings sequence $\mathbf{c}$ with payments c_t in periods $t = 1, \dots, T$ is as follows:

\$\$ \dots \$\$:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) = \sum_{t=1}^T D(t) u(c_t).$$

\[\dots \] with manual \tag{\dots}:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) = \sum_{t=1}^T D(t) u(c_t). \quad (\text{II})$$

\begin{equation} \dots \end{equation}:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) = \sum_{t=1}^T D(t) u(c_t). \quad (1.1)$$

\begin{equation*} \dots \end{equation*}:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) = \sum_{t=1}^T D(t) u(c_t).$$

\begin{eqnarray} \dots \end{eqnarray}:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) \quad (1.2)$$

$$= \sum_{t=1}^T D(t) u(c_t). \quad (1.3)$$

17. Normalization such that $D(t) \leq 1$ is not necessary in our case. Provided that t is a metric time measure, where $t = 0$ stands for the present, examples are $D(t) := \delta^t$ with some $\delta > 0$ for exponential discounting and $D(t) := (1 + \alpha t)^{-\gamma/\alpha}$ with some $\alpha, \gamma > 0$ for generalized hyperbolic discounting.

`\begin{eqnarray*} ... \end{eqnarray*}`:

$$\begin{aligned} U(\mathbf{c}) &= \sum_{t=1}^T u_t(c_t) \\ &= \sum_{t=1}^T D(t) u(c_t). \end{aligned}$$

`\begin{align} ... \end{align}`, equation number in the final line only:

$$\begin{aligned} U(\mathbf{c}) &= \sum_{t=1}^T u_t(c_t) \\ &= \sum_{t=1}^T D(t) u(c_t). \end{aligned} \tag{1.4}$$

`\begin{align} ... \end{align}`, equation number in each line:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) \tag{1.5}$$

$$= \sum_{t=1}^T D(t) u(c_t). \tag{1.6}$$

`\begin{align*} ... \end{align*}`:

$$\begin{aligned} U(\mathbf{c}) &= \sum_{t=1}^T u_t(c_t) \\ &= \sum_{t=1}^T D(t) u(c_t). \end{aligned}$$

`\begin{alignat}{2} ... \end{alignat}`:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) \tag{1.7}$$

$$= \sum_{t=1}^T D(t) u(c_t). \tag{1.8}$$

`\begin{alignat*}{2} ... \end{alignat*}`:

$$\begin{aligned} U(\mathbf{c}) &= \sum_{t=1}^T u_t(c_t) \\ &= \sum_{t=1}^T D(t) u(c_t). \end{aligned}$$

Individuals choose how much to allocate to the different periods by maximizing their utility over all possible earnings sequences available within a given budget set $\mathbf{C}$, see equations (II), (1.1), (1.2), (1.3), (1.4), (1.5), and (1.6). See also [Equation 1.8](#). We use the superscript ^{DU} to indicate decisions based on discounted utility.

A Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Another Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.3.6.2 Focus-Weighted Utility

In this section, we extend the model of discounted utility through “focus weights,” as proposed by Kőszegi and Szeidl (2013). Period- t weights g_t scale period- t consumption utility u_t . Individuals are assumed to maximize focus-weighted utility, which is defined as follows:

$$\tilde{U}(\mathbf{c}, \mathbf{C}) := \sum_{t=1}^T g_t(\mathbf{C}) u_t(c_t). \quad (1.9)$$

In contrast to discounted utility $U(\mathbf{c})$, focus-weighted utility $\tilde{U}(\mathbf{c}, \mathbf{C})$ has two arguments: the earnings sequence $\mathbf{c}$ and the choice set $\mathbf{C}$. The latter dependence is due to the weights g_t . These are given by a strictly increasing weighting function g that

takes as its argument the difference between the maximum and the minimum attainable utility in period t over all possible earnings sequences in set C :

$$g_t(C) := g[\Delta_t(C)] \quad \text{with} \quad \Delta_t(C) := \max_{c \in C} u_t(c_t) - \min_{c \in C} u_t(c_t). \quad (1.10)$$

If the underlying consumption utility function is characterized by discounted utility, then $u_t(c_t) := D(t) u(c_t)$. That is, focused thinkers put more weight on period t than on period t' if the discounted-utility distance between the best and worst alternative is larger for period t than for period t' .

A Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Yet Another Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.3.6.3 Hypotheses

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. This gives rise to our first hypothesis:

Hypothesis 1.1. *This environment can be used to clearly state your hypothesis and set them apart from the body text.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Based on this, we can state our second hypothesis:

Hypothesis 1.2. *This environment can be used to clearly state your hypothesis and set them apart from the body text.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.4 Results

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. With this, we can test our hypotheses.

1.4.1 Test of Hypothesis 1.1

Our first result supports [Hypothesis 1.1](#). Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information

Table 1.1. A tabularray-Based Example Table

Dependent variable	$\hat{d}$
Estimate	0.123*** (0.011)
Observations	750
Subjects	250

Notes: Standard errors in parentheses, clustered on the subject level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. The analysis we conducted to obtain [Result 1.1](#) is described in detail in ???. Let's reference a section, a subsection, and a figure from the appendices: [Section 1.C](#), [Section 1.A.2](#), [Figure 1.B.1](#).

Result 1.1. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.4.2 Test of [Hypothesis 1.2](#)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference

between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. We thereby test [Hypothesis 1.2](#).

Result 1.2. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.*

Our second result provides evidence in support of [Hypothesis 1.2](#). Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

1.4.3 Heterogeneity

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{i=n} x_i = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text,

you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\int_0^\infty e^{-\alpha x^2} dx = \frac{1}{2} \sqrt{\int_{-\infty}^\infty e^{-\alpha x^2} dx \int_{-\infty}^\infty e^{-\alpha y^2} dy} = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\sum_{k=0}^{\infty} a_0 q^k = \lim_{n \rightarrow \infty} \sum_{k=0}^n a_0 q^k = \lim_{n \rightarrow \infty} a_0 \frac{1 - q^{n+1}}{1 - q} = \frac{a_0}{1 - q}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all*

letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.4.4 Structural Estimation

Inspect the variance–covariance matrix Σ :

$$\Sigma := \text{Cov}(X) = \begin{bmatrix} \text{Var}(X_1) & \cdots & \text{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \cdots & \text{Var}(X_n) \end{bmatrix}.$$

1.5 Discussion

1.5.1 Some Limitations

Let’s reference some tables: [Table 1.2](#) and ???. And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information

Table 1.2. Points Awarded in Our Typeface Competition—Basic Formatting That Produces Irregular Column Widths and a Suboptimal Table Width Test Greek: ε , θ , ϕ

	Utopia	Computer Modern	Charter	Times Roman	Palatino
Yoël	1	1	2	0	1
Çelik	2	0	2	1	0
Anità	1	2	1	2	0
Uğur	1	2	0	1	0
Håkan	1	0	2	0	1
Allison	2	0	1	2	1
Pía	1	0	2	1	0
David	1	0	2	1	1
Sum	10	5	12	8	4

about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Right?

1.5.2 Utility from Money

In deriving our predictions (Section 1.3.6), we assume that subjects base their decisions on utility derived from receiving monetary payments c_t at various dates t . We also make the standard assumption that utility from money is increasing in its argu-

ment but not convex, i.e., $u'(c_t) > 0$ and $u''(c_t) \leq 0$. Both assumptions are frequently made in studies on intertemporal decision making.

One way to justify the assumption of utility being based on money—rather than consumption—is that individuals anticipate to consume the payments that they receive at date t within a short period around t . Given that the maximum payment was below €20 and that any two payment dates were separated by at least two weeks, this seems reasonable (see the arguments in favor of this view in Halevy 2014).

A second justification is consistency within the discipline: Halevy (2014) points out that “in the domain of risk and uncertainty ... preferences are often defined over payments.” In line with this, Kőszegi and Szeidl (2013, p. 62) make the same assumption of “money in the utility function”:

in some applications we also assume that monetary transactions induce *direct* utility consequences, so that for instance an agent making a payment experiences an immediate utility loss. The idea that people experience monetary transactions as immediate utility is both intuitively compelling and supported in the literature: ... some evidence on individuals’ attitudes toward money, such as narrow bracketing (...) and laboratory evidence on hyperbolic discounting (...), is difficult to explain without it.

Last but not least, the papers by McClure et al. (2004) and McClure et al. (2007) demonstrate that brain activation, as measured by functional magnetic resonance imaging, is similar for primary and monetary rewards.

Table 1.3. Points Awarded in Our Typeface Competition—More Sophisticated Formatting with Homogeneous Column Widths and a Table Width That Equals the Text Width

	Utopia ^a	Computer Modern ^b	Charter ^c	Times Roman ^d	Palatino ^e
Yoël	1	1	2	0	1
Çelik	2	0	2	1	0
Anità	1	2	1	2	0
Uğur	1	2	0	1	0
Håkan	1	0	2	0	1
Allison	2	0	1	2	1
Pía	1	0	2	1	0
David	1	0	2	1	1
Sum	10	5	12	8	4

^a `\usepackage{fourier}`

^b The $\TeX$ standard serif font.

^c `\usepackage[charter]{mathdesign}`

^d `\usepackage{newtxtext, newtxmath}`

^e `\usepackage[sc]{mathpazo}`

Let us now discuss the second assumption: that utility from money is nonconvex. We find that subjects allocate more money to the concentrated payoffs in the unbalanced than in the associated balanced budget sets—which we call concentration bias. One might argue that this relative preference for concentrated payoffs can be explained by the per-period utility function over money being convex.

Obtaining evidence on the shape of utility over money is nontrivial because it requires that at least two monetary amounts be compared with each other without the one clearly dominating the other. Thus, estimates of the curvature of the utility function over money can be obtained in two ways: the monetary amounts must be paid in different states of the world, i.e., comprise a lottery, or they have to be paid at different points in time.¹⁸ Both methods entail particular theoretical assumptions.

Andersen et al. (2008) advocate the former approach and argue that when estimating time preference parameters, one should control for the curvature of the utility function through a measure of the curvature that is based on observed choices under risk. Their study and numerous other studies on risk attitudes consistently reveal that the vast majority of subjects is risk-averse even over small stakes. Hence, for the vast majority of subjects, utility over money is concave according to this methodology (ruling out probability weighting). Others, most notably Andreoni and Sprenger (2012), have argued that the degree of curvature measured via risky choices probably overstates the degree of curvature effective in intertemporal choices, but they also find that utility is concave (albeit close to linear). Given this unambiguous evidence from previous studies, it is implausible that our subjects exhibit convex utility over money.

1.6 Conclusion

Cite some more papers (Yaari 1965; Warner and Pleeter 2001; Davidoff, Brown, and Diamond 2005; Benartzi, Previtro, and Thaler 2011). Let's cite a book: Luce (1959). Let's cite a contribution to a collected volume: Harrison and Rutström (2008) and a collection (an edited volume) itself: Kagel and Roth (2016). Now let's cite presentations at conferences: Vosgerau et al. (2008) and Beute and Kort (2012). Attema et al. (2016) propose a method for “measuring discounting without measuring utility”¹⁹.

18. As a matter of fact, the latter was the motivation behind Samuelson (1937): “Under the following four assumptions, it is believed possible to arrive theoretically at a precise measure of the marginal utility of *money income* ...” (p. 155; emphasis in the original).

19. The basic idea of their method is intriguingly simple: Imagine an individual who is indifferent between, say, Option A: \$10 today and Option B: \$10 in one year plus \$10 in two years. With a constant annual discount factor δ , this indifference translates to $u(\$10) = \delta u(\$10) + \delta^2 u(\$10)$, so that $u(\$10)$ cancels out, and δ can be readily calculated as the solution to $1 = \delta + \delta^2$.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Appendix 1.A Put More Complicated Derivations and Proofs Here

1.A.1 Appendix Subsection

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

(1) Erster Listenpunkt, Stufe 1

- a. Erster Listenpunkt, Stufe 2
 - i. Erster Listenpunkt, Stufe 3
 - ii. Zweiter Listenpunkt, Stufe 3
 - iii. Dritter Listenpunkt, Stufe 3
 - iv. Vierter Listenpunkt, Stufe 3
- b. Zweiter Listenpunkt, Stufe 2
- c. Dritter Listenpunkt, Stufe 2
- d. Vierter Listenpunkt, Stufe 2

(2) Zweiter Listenpunkt, Stufe 1

(3) Dritter Listenpunkt, Stufe 1

(4) Vierter Listenpunkt, Stufe 1

The typeset math below follows the ISO recommendations that only variables be set in italic. Note the use of upright shapes for “d,” “e,” and “ π .” (These are entered as `\mathup{d}`, `\mathup{e}`, and `\mathup{\pi}`, respectively.)

Theorem 1.1 (Simplest form of the Central Limit Theorem). Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables with mean 0 and variance 1 on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then

$$\mathbb{P}\left(\frac{X_1 + \dots + X_n}{\sqrt{n}} \leq y\right) \rightarrow \mathfrak{N}(y) := \int_{-\infty}^y \frac{e^{-v^2/2}}{\sqrt{2\pi}} dv \quad \text{as } n \rightarrow \infty,$$

or, equivalently, letting $S_n := \sum_{k=1}^n X_k$,

$$\mathbb{E}f(S_n/\sqrt{n}) \rightarrow \int_{-\infty}^{\infty} f(v) \frac{e^{-v^2/2}}{\sqrt{2\pi}} dv \quad \text{as } n \rightarrow \infty, \text{ for every } f \in \mathcal{BC}(\mathbb{R}).$$

1.A.2 Saliency

Saliency theory (Bordalo, Gennaioli, and Shleifer 2012, 2013) represents a behavioral model according to which the most distinctive features of the available alternatives receive a particularly large share of attention and are therefore over-weighted. More precisely, a particular attribute out of all attributes of an alternative becomes the more salient, the more it differs from that attribute's average level over all available alternatives.

Formally, alternatives are assumed to be uniquely characterized by the values they take in $T \geq 1$ attributes (or, “dimensions”). Utility is assumed to be additively separable in attributes, and saliency attaches a decision weight to each attribute of each good which indicates how salient the respective attribute is for that good. Suppose an agent chooses one alternative from some finite choice set $\mathcal{C}$. Let t index the T different attributes, and let k index the K available alternatives. Let $u_t(\cdot)$ denote the function which assigns utility to values in dimension t . Denote by a_t^k the level of attribute t of good k and define $u_t^k := u_t(a_t^k)$ as the utility that dimension t of good k yields. Let $\bar{u}_t$ be the average utility level, across all K goods, of dimension t . The saliency of each dimension of good k is determined by a symmetric and continuous saliency function $\sigma(\cdot, \cdot)$ that satisfies the following two properties:

(1) *Ordering.* Let $\mu := \text{sgn}(u_t^k - \bar{u}_t)$. Then for any $\varepsilon, \varepsilon' \geq 0$ with $\varepsilon + \varepsilon' > 0$, it holds that

$$\sigma(u_t^k + \mu \varepsilon, \bar{u}_t - \mu \varepsilon') > \sigma(u_t^k, \bar{u}_t). \quad (1.A.1)$$

(2) *Diminishing sensitivity.* For any $u_t^k, \bar{u}_t \geq 0$ and all $\varepsilon > 0$, it holds that

$$\sigma(u_t^k + \varepsilon, \bar{u}_t + \varepsilon) < \sigma(u_t^k, \bar{u}_t). \quad (1.A.2)$$

Following the smooth saliency characterization proposed in Bordalo, Gennaioli, and Shleifer (2012, p. 1255), each dimension t of good k receives weight $\Delta^{-\sigma(u_t^k, \bar{u}_t)}$, where $\Delta \in (0, 1]$ is a constant that captures an agent's susceptibility to saliency. $\Delta = 1$ gives rise to a rational decision maker, and the smaller Δ , the stronger is the saliency bias. We call an agent with $\Delta < 1$ a salient thinker.

A reference with a large number of authors is Henrich et al. (2005).

Appendix 1.B Additional Figures

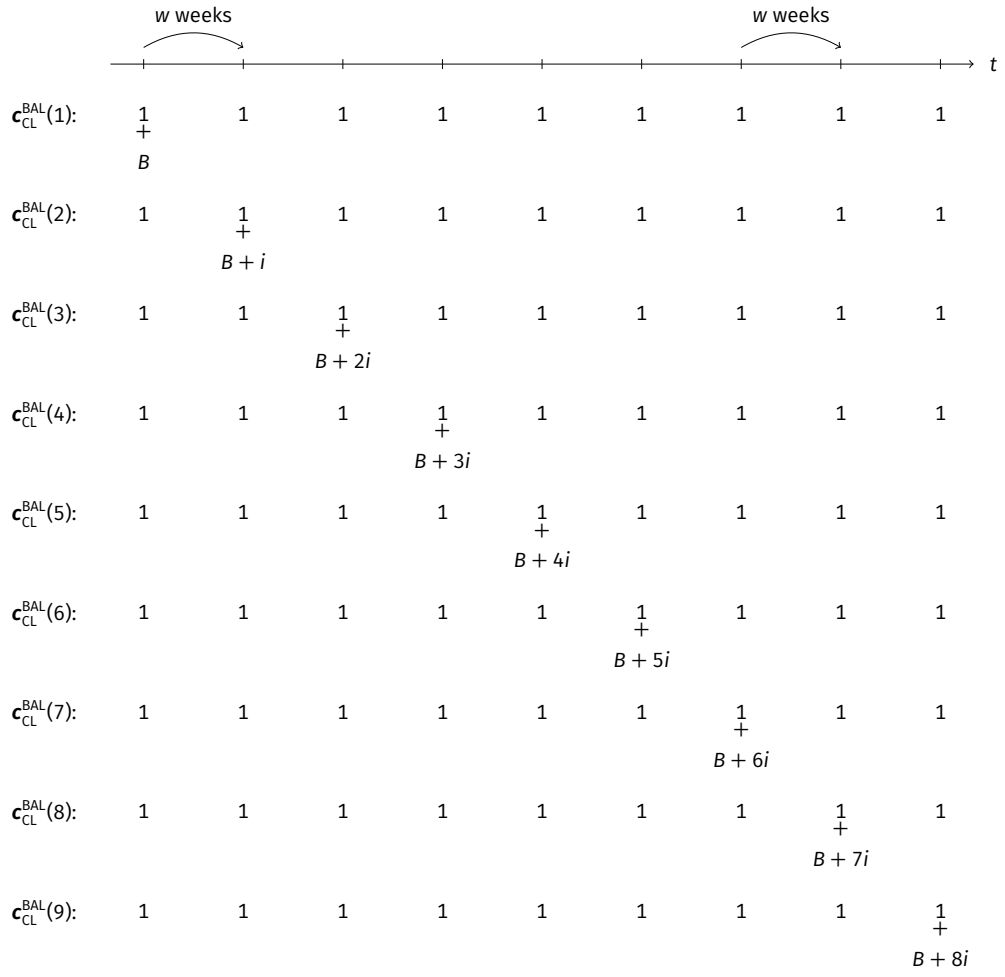


Figure 1.B.1. Earnings Sequences Included in Choice List c_{CL}^{BAL}

Notes: For the values of B , i , and w that we used see [Section 1.3](#). Figure taken from Dertwinkel-Kalt et al. (2017).

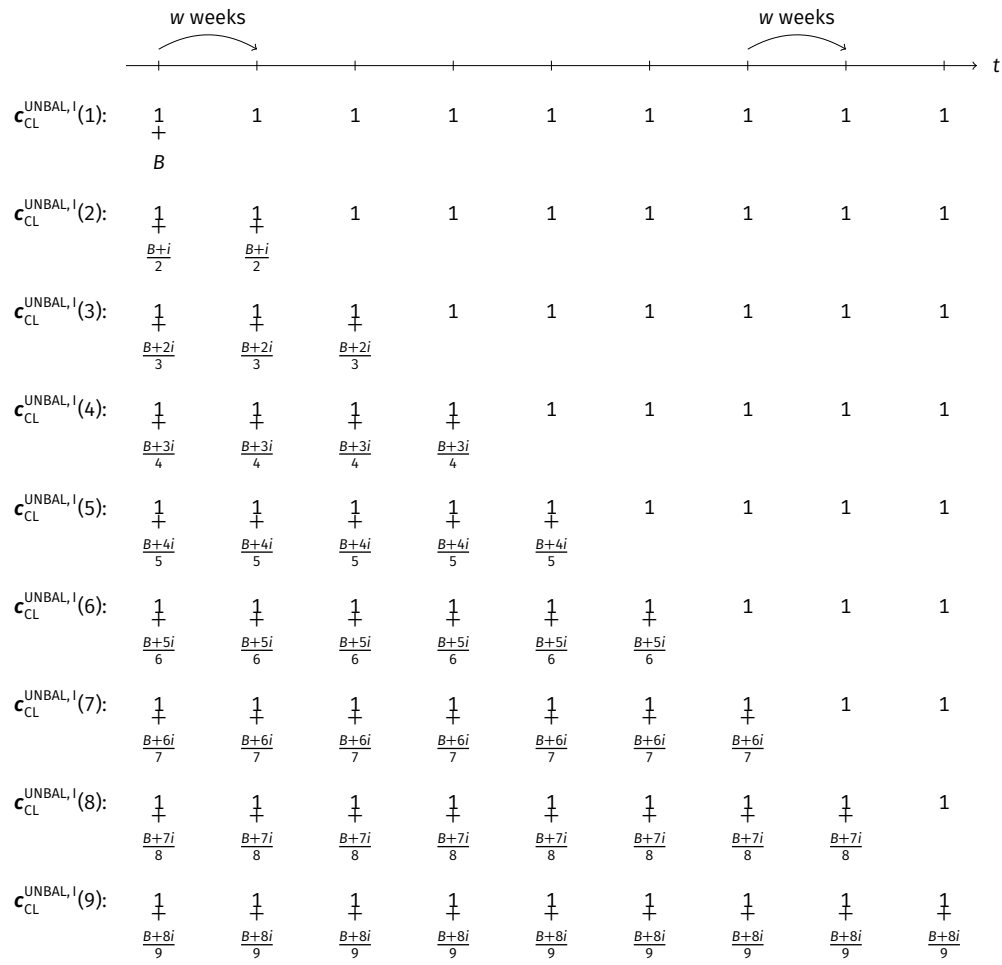


Figure 1.B.2. Earnings Sequences Included in Choice List $c_{CL}^{UNBAL, l}$

Notes: For the values of B , i , and w that we used see [Section 1.3](#). Figure taken from Dertwinkel-Kalt et al. (2017).

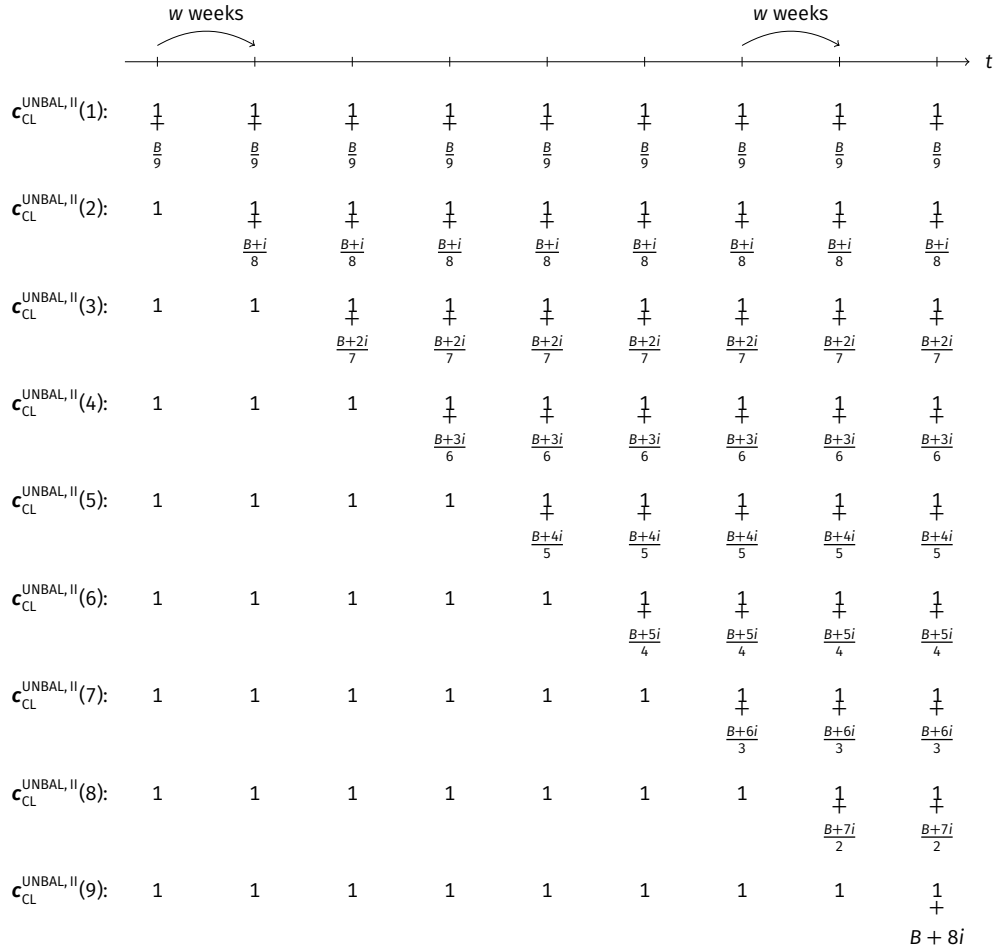


Figure 1.B.3. Earnings Sequences Included in Choice List $c_{CL}^{UNBAL, II}$

Notes: For the values of B , i , and w that we used see [Section 1.3](#). Figure taken from Dertwinkel-Kalt et al. (2017).

Appendix 1.C Additional *siunitx* and *tabularray* Example Tables

Table 1.C.1. An Example of a Regression Table (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017). Remember to Mention the Dependent Variable!

	(1)	(2)	(3)	(4)	(5)
Treatment	-0.390 (+0.352)	-0.228 (-0.205)	-0.729* [+0.377]	-0.449* [-0.245]	-0.453** {+0.204}
Female	0.948*** (0.354)	0.061 (0.233)	0.188 (0.372)	0.305 (0.226)	0.385* (0.222)
Female $\times$ Treatment	0.169 (0.514)	0.251 (0.325)	0.892* (0.533)	0.454 (0.341)	0.439 (0.307)
Final high school grade	-0.101 (0.198)	0.013 (0.144)	0.076 (0.224)	0.117 (0.146)	0.039 (0.133)
Trait self-control	-0.016 (0.016)	0.002 (0.010)	-0.016 (0.015)	0.000 (0.010)	-0.007 (0.009)
Constant	2.357*** (0.239)	1.512*** (0.144)	-0.322 (0.265)	2.158*** (0.161)	1.437*** (0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment $\times$ (1 + Female)	-0.221	0.023	0.163	0.004	-0.014
p_F [Treatment $\times$ (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Notes: Dependent variable: $m_{\dots}$. Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

Table 1.C.2. Figure Grouping via *siunitx* in a Table

(1)	(2)	(3)
-0.100* (2.871)	-0.100 01* (2.871 23)	-123 456.444*** [+50 000.123]

Table 1.C.3. Overview of the Choice Lists Presented to Subjects (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017)

	Alternative A				Alternative B			
	$c_{A,1}$	$p_{A,1}$	$c_{A,2}$	$p_{A,2}$	$c_{B,1}$	$p_{B,1}$	$c_{B,2}$	$p_{B,2}$
<i>Choice List I: risky/risky</i> ($x = €22.00$, $r = €7.50$, $k = €11.50$; 25 rows)								
Top row	€ 3.00	50%	€22.00	50%	€ 3.00	50%	€ 7.00	50%
Center row	€ 3.00	50%	€22.00	50%	€ 9.00	50%	€13.00	50%
Row with $m = 0$	€ 3.00	50%	€22.00	50%	€10.50	50%	€14.50	50%
Bottom row	€ 3.00	50%	€22.00	50%	€15.00	50%	€19.00	50%
<i>Choice List II: safe/risky</i> ($x = €16.00$, $r = €5.00$, $k = €5.00$; 19 rows)								
Top row	€11.00	100%			€11.00	50%	€21.00	50%
Center row	€11.00	100%			€ 6.50	50%	€16.50	50%
Row with $m = 0$	€11.00	100%			€ 6.00	50%	€16.00	50%
Bottom row	€11.00	100%			€ 2.00	50%	€12.00	50%
<i>Choice List III: "long shot"</i> ($x = €14.00$, $r = -€36.00$, $k = €7.00$; 21 rows)								
Top row	€ 7.00	90%	€50.00	10%	€ 7.00	90%	€10.00	10%
Row with $m = 0$	€ 7.00	90%	€50.00	10%	€11.00	90%	€14.00	10%
Center row	€ 7.00	90%	€50.00	10%	€12.00	90%	€15.00	10%
Bottom row	€ 7.00	90%	€50.00	10%	€17.00	90%	€20.00	10%
<i>Choice List IV: delayed payoffs</i> ($x = €18.00$, $r = €6.00$, $k = €8.50$, paid in one week; 20 rows)								
Top row	€ 9.50	50%	€12.00	50%	€ 9.50	50%	€24.00	50%
Above-center row	€ 9.50	50%	€12.00	50%	€ 5.00	50%	€19.50	50%
Below-center row	€ 9.50	50%	€12.00	50%	€ 4.50	50%	€19.00	50%
Row with $m = 0$	€ 9.50	50%	€12.00	50%	€ 3.50	50%	€18.00	50%
Bottom row	€ 9.50	50%	€12.00	50%	€ 0.00	50%	€14.50	50%

Table 1.C.4. A Really Long Table That Spans Multiple Pages

	(1)	(2)	(3)	(4)
Row 1	0.0070	0.1356	0.1560	0.8979
Row 2	0.4223	0.7311	0.4213	0.6900
Row 3	0.0767	0.5110	0.7399	0.9491
Row 4	0.5954	0.1685	0.3778	0.9960
Row 5	0.6465	0.0524	0.8895	0.1544
Row 6	0.3838	0.7069	0.1773	0.5785
Row 7	0.1537	0.5442	0.6361	0.0327
Row 8	0.0879	0.1812	0.3082	0.2942

Continued on next page.

Table 1.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 9	0.2720	0.2565	0.6214	0.8944
Row 10	0.4873	0.3064	0.9913	0.0591
Row 11	0.8387	0.1713	0.6747	0.7455
Row 12	0.0645	0.4891	0.2892	0.1013
Row 13	0.0989	0.3798	0.5795	0.3725
Row 14	0.3256	0.7080	0.0262	0.8709
Row 15	0.7867	0.8768	0.0690	0.6081
Row 16	0.2713	0.4399	0.5838	0.6107
Row 17	0.5236	0.1527	0.4402	0.8002
Row 18	0.4851	0.4619	0.4040	0.2711
Row 19	0.1742	0.8151	0.2757	0.4184
Row 20	0.0495	0.3288	0.2759	0.1452
Row 21	0.1678	0.2403	0.1993	0.3676
Row 22	0.4977	0.9472	0.2810	0.2493
Row 23	0.6777	0.6516	0.3573	0.1413
Row 24	0.3668	0.3075	0.8724	0.3945
Row 25	0.5877	0.5670	0.0417	0.5213
Row 26	0.3599	0.5485	0.2407	0.6362
Row 27	0.1029	0.9796	0.5696	0.8696
Row 28	0.3070	0.8169	0.4015	0.4386
Row 29	0.4453	0.0670	0.3726	0.3257
Row 30	0.2648	0.9977	0.8864	0.0755
Row 31	0.4085	0.2017	0.5406	0.1333
Row 32	0.4861	0.4466	0.3472	0.2486
Row 33	0.5996	0.8639	0.1837	0.7636
Row 34	0.4446	0.3755	0.6901	0.4208
Row 35	0.9616	0.3585	0.0074	0.2867
Row 36	0.5168	0.5752	0.5778	0.0060
Row 37	0.7978	0.0283	0.7998	0.9952
Row 38	0.0561	0.3133	0.1207	0.6922
Row 39	0.5237	0.1488	0.9217	0.2268
Row 40	0.0944	0.7939	0.6252	0.9836
Row 41	0.3179	0.6226	0.4493	0.4277
Row 42	0.7175	0.7267	0.8016	0.6880
Row 43	0.0192	0.4807	0.7610	0.9808
Row 44	0.9923	0.8888	0.4494	0.0645

Continued on next page.

Table 1.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 45	0.3938	0.8529	0.0496	0.0429
Row 46	0.1135	0.6166	0.5899	0.7500
Row 47	0.0654	0.1640	0.1952	0.0431
Row 48	0.8895	0.0549	0.1105	0.1284
Row 49	0.6817	0.8942	0.6597	0.3661
Row 50	0.6690	0.8817	0.2343	0.1903
Row 51	0.4091	0.0874	0.4726	0.1381
Row 52	0.9061	0.9039	0.7439	0.2061
Row 53	0.5282	0.2135	0.5223	0.7846
Row 54	0.6505	0.7404	0.8748	0.2078
Row 55	0.5824	0.8443	0.3242	0.8253
Row 56	0.0151	0.9929	0.4812	0.5010
Row 57	0.7296	0.8420	0.1535	0.4273
Row 58	0.8102	0.8068	0.1832	0.8830
Row 59	0.1650	0.5545	0.1820	0.0791
Row 60	0.5882	0.5750	0.9195	0.8993
Row 61	0.0638	0.5132	0.5994	0.0877
Row 62	0.9916	0.8032	0.0564	0.3218
Row 63	0.5555	0.4078	0.7056	0.9225
Row 64	0.8680	0.5577	0.2992	0.0941
Row 65	0.2939	0.7801	0.7039	0.7295
Row 66	0.0829	0.6756	0.5386	0.0644
Row 67	0.3868	0.4199	0.0308	0.5947
Row 68	0.0943	0.2663	0.0379	0.0887
Row 69	0.0050	0.1396	0.8348	0.2830
Row 70	0.9585	0.8018	0.4472	0.9477
Row 71	0.8153	0.2659	0.7030	0.4096
Row 72	0.7532	0.4214	0.3914	0.2360
Row 73	0.6419	0.2074	0.7386	0.0653
Row 74	0.4215	0.7004	0.3193	0.9282
Row 75	0.1307	0.8242	0.1305	0.8925
Row 76	0.5812	0.6879	0.4844	0.0464
Row 77	0.1080	0.5293	0.2700	0.4844
Row 78	0.3073	0.7945	0.8300	0.3479
Row 79	0.4777	0.5842	0.2233	0.3206
Row 80	0.7218	0.7687	0.0432	0.7268

Continued on next page.

Table 1.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 81	0.1427	0.8696	0.7573	0.1263
Row 82	0.0244	0.6493	0.6750	0.9651
Row 83	0.1925	0.4131	0.3064	0.0508
Row 84	0.8678	0.3827	0.7732	0.3896
Row 85	0.6830	0.0868	0.0773	0.1712
Row 86	0.2699	0.5507	0.1200	0.4458
Row 87	0.3873	0.8615	0.0624	0.4357
Row 88	0.0610	0.0065	0.1505	0.0287
Row 89	0.3380	0.6846	0.1305	0.8998
Row 90	0.4337	0.2892	0.9326	0.7977
Row 91	0.7618	0.7254	0.6185	0.5718
Row 92	0.2404	0.2312	0.6645	0.7351
Row 93	0.8908	0.4011	0.6728	0.4192
Row 94	0.7596	0.5054	0.3343	0.1696
Row 95	0.9736	0.2894	0.8395	0.7554
Row 96	0.2555	0.3570	0.6331	0.3460
Row 97	0.5865	0.8620	0.9528	0.8383
Row 98	0.1753	0.9843	0.5822	0.7130
Row 99	0.2085	0.7513	0.4976	0.6609
Row 100	0.8550	0.6317	0.2716	0.3482
Row 101	0.0003	0.2699	0.1657	0.9740
Row 102	0.8108	0.7631	0.4779	0.7736
Row 103	0.1700	0.7518	0.6194	0.2642
Row 104	0.9089	0.7737	0.1760	0.1838
Row 105	0.2693	0.6957	0.8645	0.7214
Row 106	0.7675	0.7649	0.1831	0.5527
Row 107	0.6605	0.6763	0.6069	0.6509
Row 108	0.9355	0.8627	0.1932	0.1369
Row 109	0.2459	0.2674	0.5147	0.3251
Row 110	0.1111	0.9926	0.6565	0.3905
Row 111	0.3883	0.7516	0.0597	0.2444
Row 112	0.3873	0.8884	0.8992	0.4628
Row 113	0.7374	0.3370	0.2922	0.8778
Row 114	0.9644	0.3383	0.7343	0.4642
Row 115	0.8793	0.1624	0.6602	0.6129
Row 116	0.7910	0.7928	0.9132	0.4582

Continued on next page.

Table 1.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 117	0.4158	0.6584	0.0655	0.3760
Row 118	0.6719	0.8505	0.2902	0.3726
Row 119	0.6456	0.6116	0.7580	0.3331
Row 120	0.9372	0.5338	0.9066	0.8391
Row 121	0.1427	0.6179	0.7094	0.5079
Row 122	0.1748	0.9789	0.1452	0.5829
Row 123	0.7514	0.2678	0.7714	0.1895
Row 124	0.4058	0.7714	0.4468	0.5559
Row 125	0.0799	0.6205	0.4477	0.3788
Row 126	0.3297	0.7600	0.5485	0.8005
Row 127	0.8873	0.3812	0.9346	0.4062
Row 128	0.5164	0.9326	0.8897	0.6300
Row 129	0.1876	0.8342	0.5704	0.9817
Row 130	0.3990	0.2170	0.8709	0.4717
Row 131	0.4454	0.3671	0.2185	0.9753
Row 132	0.8951	0.9321	0.3854	0.4805
Row 133	0.3442	0.8316	0.8667	0.6898
Row 134	0.0586	0.2090	0.3720	0.1668
Row 135	0.1312	0.5375	0.6314	0.2907
Row 136	0.5138	0.7588	0.2177	0.7461
Row 137	0.4966	0.1501	0.3993	0.0631
Row 138	0.7154	0.8785	0.8362	0.5782
Row 139	0.6265	0.2019	0.9703	0.2705
Row 140	0.5248	0.5235	0.5018	0.9854
Row 141	0.2711	0.5263	0.8829	0.8525
Row 142	0.1335	0.8354	0.0190	0.3996
Row 143	0.7644	0.3912	0.8849	0.7440
Row 144	0.4358	0.2065	0.4528	0.8955
Row 145	0.9038	0.0718	0.7912	0.5230
Row 146	0.1919	0.7559	0.2908	0.2352
Row 147	0.6801	0.3179	0.8315	0.7988
Row 148	0.7810	0.3397	0.5245	0.8478
Row 149	0.1458	0.1098	0.2659	0.2319
Row 150	0.7207	0.1931	0.2071	0.0241

Note: At the very end, you can add some notes to the table.

References

- Andersen, Steffen, Glenn W. Harrison, Morten I. Lau, and E. Elisabet Rutström. 2008. "Eliciting Risk and Time Preferences." *Econometrica* 76 (3): 583–618. <https://doi.org/10.1111/j.1468-0262.2008.00848.x>. [5, 35]
- Andreoni, James, and Charles Sprenger. 2012. "Estimating Time Preferences from Convex Budgets." *American Economic Review* 102 (7): 3333–56. <https://doi.org/10.1257/aer.102.7.3333>. [5, 35]
- Attema, Arthur E., Han Bleichrodt, Yu Gao, Zhenxing Huang, and Peter P. Wakker. 2016. "Measuring Discounting without Measuring Utility." *American Economic Review* 106 (6): 1476–94. <https://doi.org/10.1257/aer.20150208>. [35]
- Balakrishnan, Uttara, Johannes Haushofer, and Pamela Jakiela. 2016. "How Soon Is Now? Evidence of Present Bias from Convex Time Budget Experiments." IZA Discussion Paper, IZA Discussion Paper Series. <http://ftp.iza.org/dp9653.pdf>. [5]
- Benartzi, Shlomo, Alessandro Previtero, and Richard H. Thaler. 2011. "Annuity Puzzles." *Journal of Economic Perspectives* 25 (4): 143–64. <https://doi.org/10.1257/jep.25.4.143>. [35]
- Beute, Femke, and Yvonne A. W. de Kort. 2012. "Always Look on the Bright Side of Life: Ego-Replenishing Effects of Daylight versus Artificial Light." In *Proceedings of Experiencing Light 2012: International Conference on the Effects of Light on Wellbeing*, edited by Y. A. W. de Kort, M. P. J. Aarts, F. Beute, A. Haans, W. A. IJsselstein, D. Lakens, K. C. H. J. Smolders, and L. van Rijswijk, 1–4. Eindhoven, The Netherlands: Eindhoven University of Technology. <http://2012.experiencinglight.nl/doc/41.pdf>. [35]
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer. 2012. "Salience Theory of Choice Under Risk." *Quarterly Journal of Economics* 127 (3): 1243–85. <https://doi.org/10.1093/qje/qjs018>. [6, 37]
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer. 2013. "Salience and Consumer Choice." *Journal of Political Economy* 121 (5): 803–43. <https://doi.org/10.1086/673885>. [6, 37]
- Bursztyn, Leonardo, Benjamin Handel, Rafael Jiménez-Durán, and Christopher Roth. 2025. "When Product Markets Become Collective Traps: The Case of Social Media." *American Economic Review* 115 (12): 4105–36. <https://doi.org/10.1257/aer.20231468>. [5]
- Bushong, Benjamin, Matthew Rabin, and Joshua Schwartzstein. 2016. "A Model of Relative Thinking." Working paper. Cambridge, MA, USA: Harvard University. <http://people.hbs.edu/jschwartzstein/RelativeThinking.pdf>. [6]
- Davidoff, Thomas, Jeffrey R. Brown, and Peter A. Diamond. 2005. "Annuities and Individual Welfare." *American Economic Review* 95 (5): 1573–90. <https://doi.org/10.1257/000282805775014281>. [35]
- Dertwinkel-Kalt, Markus, Holger Gerhardt, Gerhard Riener, Frederik Schwerter, and Louis Strang. 2017. "Concentration Bias in Intertemporal Choice." Working paper. Bonn, Germany, et al.: University of Bonn et al. https://www.dropbox.com/s/dv20mcu0qkygmjz/Concentration_Bias_in_Intertemporal_Choice.pdf. [18, 19, 38–40]
- Gabaix, Xavier. 2014. "A Sparsity-Based Model of Bounded Rationality." *Quarterly Journal of Economics* 129 (4): 1661–710. <https://doi.org/10.1093/qje/qju024>. [6]
- Gerhardt, Holger, Hannah Schildberg-Hörisch, and Jana Willrodt. 2017. "Does self-control depletion affect risk attitudes?" *European Economic Review* 100: 463–87. <https://doi.org/10.1016/j.euroecorev.2017.09.004>. [41, 42]

- Halevy, Yoram.** 2014. "Some Comments on the Use of Monetary and Primary Rewards in the Measurement of Time Preferences." Working paper. University of British Columbia. http://faculty.arts.ubc.ca/yhalevy/monetary_primary.pdf. [23, 34]
- Harrison, Glenn W., and E. Elisabet Rutström.** 2008. "Risk Aversion in the Laboratory." Chap. 1 in *Risk Aversion in Experiments*, edited by Glenn W. Harrison and James C. Cox, 12: 41–196. Research in Experimental Economics. Bingley, UK: Emerald Group. [https://doi.org/10.1016/S0193-2306\(08\)00003-3](https://doi.org/10.1016/S0193-2306(08)00003-3). [35]
- Henrich, Joseph, Robert Boyd, Samuel Bowles, Colin Camerer, Ernst Fehr, Herbert Gintis, Richard McElreath, et al.** 2005. "'Economic man' in cross-cultural perspective: Behavioral experiments in 15 small-scale societies." *Behavioral and Brain Sciences* 28 (6): 795–815, discussion 815–55. <https://doi.org/10.1017/S0140525X05000142>. [37]
- Kagel, John H., and Alvin E. Roth, eds.** 2016. *The Handbook of Experimental Economics*. Edited by John H. Kagel and Alvin E. Roth. Vol. 2. Princeton, NJ, USA: Princeton University Press. [35]
- Kőszegi, Botond, and Adam Szeidl.** 2013. "A Model of Focusing in Economic Choice." *Quarterly Journal of Economics* 128 (1): 53–104. <https://doi.org/10.1093/qje/qjs049>. [6, 23, 26, 34]
- Lisi, A. Garrett.** 1995. "A solitary wave solution of the Maxwell–Dirac equations." *Journal of Physics A: Mathematical and General* 28 (18): 5385–92. <https://doi.org/10.1088/0305-4470/28/18/026>. arXiv: [hep-th/9410244](https://arxiv.org/abs/hep-th/9410244). [5]
- Luce, R. Duncan.** 1959. *Individual Choice Behavior: A Theoretical Analysis*. New York, NY, USA: John Wiley & Sons. [35]
- McClure, Samuel M., Keith M. Ericson, David Laibson, George Loewenstein, and Jonathan D. Cohen.** 2007. "Time Discounting for Primary Rewards." *Journal of Neuroscience* 27 (21): 5796–804. <https://doi.org/10.1523/jneurosci.4246-06.2007>. [23, 34]
- McClure, Samuel M., David Laibson, George Loewenstein, and Jonathan D. Cohen.** 2004. "Separate Neural Systems Value Immediate and Delayed Monetary Rewards." *Science* 306 (5695): 503–7. <https://doi.org/10.1126/science.1100907>. [23, 34]
- Samuelson, Paul.** 1937. "A Note on Measurement of Utility." *Review of Economic Studies* 4 (2): 155–61. <https://doi.org/10.2307/2967612>. [35]
- Sims, Christopher A.** 2003. "Implications of rational inattention." *Journal of Monetary Economics* 50 (3): 665–90. [https://doi.org/10.1016/S0304-3932\(03\)00029-1](https://doi.org/10.1016/S0304-3932(03)00029-1). [6]
- Sullivan, Paul.** 2016. *Fresh Thinking on Saving*. *New York Times* (2016): F2. <http://nytimes.com/2016/03/27/your-money/getting-workers-to-save-more-for-retirement.html>. [5]
- Taubinsky, Dmitry.** 2014. "From Intentions to Actions: A Model and Experimental Evidence of Inattentive Choice." Working paper. Hanover, NH, USA: Dartmouth College. <https://docs.google.com/viewer?a=v&pid=sites&srcid=ZGVmYXVsdGRvbWFpbnxkbWl0cnlwYXBlcmluN8Z3g6NmIzYWw0M0MwIiwNTc4MjkwNQ>. [6]
- Vosgerau, Joachim, Sabrina Bruyneel, Ravi Dhar, and Klaus Wertenbroch.** 2008. "Ego Depletion and Cognitive Load: Same or Different Constructs?" In *Advances in Consumer Research*, 35: 217–20. Association for Consumer Research. <http://www.acrwebsite.org/search/view-conference-proceedings.aspx?id=13549>. [35]
- Warner, John T., and Saul Pleeter.** 2001. "The Personal Discount Rate: Evidence from Military Downsizing Programs." *American Economic Review* 91 (1): 33–53. <https://doi.org/10.1257/aer.91.1.33>. [35]
- Yaari, Menahem E.** 1965. "Uncertain Lifetime, Life Insurance, and the Theory of the Consumer." *Review of Economic Studies* 32 (2): 137–50. <https://doi.org/10.2307/2296058>. [35]

Chapter 2

My Second Paper Has a Long Title That Spans Two Lines*

Joint with Adam Smith, Janet Smith, and Jeremiah Smith

2.1 Introduction

“Most people can save a few dollars a day or even \$10 a day,” she said. “That’s doable. But if you say, ‘Can you save \$300 a month or a couple of thousand dollars a year?’ people will say, ‘Whoa.’ Avoiding that ‘whoa,’ which is the hesitancy that can derail planning, is what consultants like Ms. Davidson are trying to do.”

—New York Times, March 27, 2016

This template uses the [Charter](#) typeface for the body text. Charter is a serif typeface and was designed in 1987 by [Matthew Carter](#). By contrast, all headings, tables, and captions are set in a [sans-serif typeface](#). The sans-serif typeface used in this document is [Fira Sans](#), designed by [Erik Spiekermann](#) and collaborators.

[Anonym 2]

The math settings are adjusted in the preamble to the effect that mathematical formulas are automatically typeset in the same font as the surrounding text. That is, math in a serif environment will be set in a serif font, while math in a sans-serif environment will use the sans-serif font. This is an aesthetic choice that may not please everyone given that a sans-serif font may be used in mathematical formulas to express a particular meaning. These cases are, however, very rare.

Let us cite [a couple of](#) publications (Lisi [1995](#); Andersen et al. [2008](#); Andreoni and Sprenger [2012](#); Balakrishnan, Haushofer, and Jakiela [2016](#); Bursztyn et al. [2025](#)). With the options set for BibLaTeX in the preamble, citations in the body text are [sorted chronologically](#)—irrespective of the order of the “citekeys” in

[Holger 4]

Ersetzt: some

[Lou E. 4]

Gelöscht: automatically

* This footnote can be used for acknowledgments. This is where you can express your gratitude to referees, editors, and colleagues for their valuable feedback and suggestions that helped improve your manuscript. Financial support by third parties can also be mentioned here.

your input. In the list of references, entries are sorted alphabetically by author surname. [Let's cite](#) Andersen et al. (2008) once more.

[U. R. 3]

Eingefügt

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

[Holger 5]

Eingefügt

We already included several references above.

Some [additional](#) references: See Sims (2003) and Gabaix (2014) for models of "rational inattention" or "goal-driven attention." See Bordalo, Gennaioli, and Shleifer (2012, 2013), Kőszegi and Szeidl (2013), Taubinsky (2014), and Bushong, Rabin, and Schwartzstein (2016) for models of "stimulus-driven attention."

[U. R. 4]

Check whether there are more recent publications!

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

[Lou E. 5]

Italics?

In [Section 2.3](#), we describe the [design](#) of our study. We present the data analysis and our results in [Section 2.4](#). In [Section 2.5](#), we discuss the plausibility of potential alternative explanations. [Section 2.6 concludes](#).

[Holger 6]

Gelöscht: in detail

Too wordy.

[Lou E. 6]

Ersetzt: will conclude

Let's use the present tense throughout.

2.2 Portability

2.2.1 Clean Code/Semantic Coding: Separating Content and Formatting

All written texts are structured into words and sentences. If they are long enough, texts are also structured into paragraphs. Academic texts, in addition, feature sections, subsections, and often subsubsections, as well as figures, tables, lists of references, and potentially appendices. These structural elements are associated with headings/titles and captions.

The structure of a document is conveyed to the reader through *formatting*. Effective formatting of academic documents, thus, requires that all elements of a particular type—say, section headings—be identifiable as such. Identifiability is achieved through styling that is *consistent within type* and *distinct across types*. For example, all section headings must be formatted identically, and at the same time, they must look different from subsection headings.

As a consequence, one should refrain from using manual, discretionary formatting whenever possible. In line with this principle, the source code of this template refers to the document’s structure as much as possible. This is also called “semantic coding”: using code that describes the *meaning* of the content and not *how* those elements should look.

- Examples for *semantic* commands are `\author`, `\title`, `\section`, `\emph`, `\url`, `\cite`, `\begin{figure} ... \end{figure}`, `\caption`, `\begin{quote} ... \end{quote}`.
- Examples for LaTeX commands that are *not semantic*—and should be avoided—are `\textbf`, `\textit`, `\newpage`, `\pagebreak`, `\linebreak`, `\`, `\noindent`, `\smallskip`, `\centering`, `\noindent`, `\hspace`, `\vspace`, `\cellcolor`, `\multirow`.

The reason why one should stick to using semantic commands is that they keep the code *portable*: Semantic code can be *reused* across document without any changes (or at least without major adjustments). This is not then case with nonsemantic code, that is, code that includes manual formatting instructions. Instructions like `\bigskip`, `\hspace`, `\` (even worse, `\` `\`), etc. may result in decent formatting in a particular document but are bound to lead to undesirable formatting in another document. This applies, in particular, when you face the task of combining several research papers into one, “cumulative” doctoral thesis. Moreover, manual formatting is error-prone, as it easily leads to inconsistent styling.

This entails two things: First, the formatting should be determined, as much as possible, in the *preamble* of the LaTeX document. This comes at the cost, of course, of a rather long preamble. Second, instead of defining custom commands, one should *redefine* standard LaTeX commands or use (widely adopted) packages that are available on CTAN (<https://ctan.org>).

This template follows the above principles, resulting in clean, portable code: The source code of the body of this template (apart from the boxes with examples) consists almost completely of semantic, basic LaTeX, with commands from popular packages on top.

2.2.2 Using BibLaTeX (Rather Than BibTeX)

This template uses the modern BibLaTeX framework (the *biblatex-chicago* package, <https://ctan.org/pkg/biblatex-chicago>, to be precise) instead of the vintage BibTeX framework to manage the references included in the document. The reason for this choice is that BibLaTeX is much more flexible than BibTeX and also handles multi-language references much better.

Moreover, in line with the principle describe above, BibLaTeX permits much better separation of a document’s content (in this case, the *.bib* file) and its formatting (of the citations and the bibliography) than BibTeX. In particular, suppressing output of information that is present in a *.bib* file is cumbersome with BibTeX: It amounts to editing the “bibliography style” *.bst* file, and *.bst* files have a syntax that is completely different from LaTeX and difficult to learn. It is usually easier to just remove the information from the *.bib* file. This, however, impacts the portability of the *.bib* file: In other circumstances one might want that very information to be present. BibLaTeX simplifies skipping information that is present in the *.bib* file.

An example: Researchers frequently present their research in the form of posters. Space on posters is usually quite limited. Hence, one might not want authors’ first names to be abbreviated on a poster. In the associated paper, however, the names should be printed in full to minimize ambiguity. With BibTeX, this requires time-consuming adjustments of the *.bst* file. With BibLaTeX, it is much easier: In the *.tex* file for the poster, one can include the command

```
\ExecuteBibliographyOptions{giveninits = true}
```

Similarly, in a paper, the list of references should include URLs or DOIs (digital object identifiers) whenever they exist, to make it easy for readers to locate a cited work. On a poster, by contrast, one might want to suppress DOIs and URLs. With BibTeX, this would require time-consuming adjustments of the *.bst* file. With BibLaTeX, one can simply do, for instance,

```
\AtEveryBibitem{%
  \ifentrytype{article}{\clearfield{doi}\clearfield{url}}{}%
}
```

Using BibLaTeX requires using the program *biber* instead of *bibtex* for compiling the bibliography. When using the Overleaf online service, using *biber* instead of *bibtex* is as simple as it gets: Overleaf automatically chooses *biber* to compile

the bibliography when it encounters BibLaTeX’s backend = biber option.¹ And when using the TeXstudio editor (<https://texstudio.org>), an easy way to make this happen is to include the “magic comment”

```
% !BIB program = biber
```

at the beginning of the .tex file.

2.2.3 Using the *tabularray* Package (Rather Than *tabularx*/*tabulary*/*tabu*, *booktabs*, *longtable*, *xltabular*, *multirow*, *makecell*, *colortbl*, *parnotes*, *threeparttable*, ...)

The principle of separating the content from its formatting as much as possible can also be applied to tables. The *tabularray* package (<https://ctan.org/pkg/tabularray>) allows you to do just that. *tabularray* has been around since May 2021, so it is relatively new—but it has already matured into a package that is extremely useful. *tabularray* replaces (or loads) all the other table-related LaTeX packages that you may have used so far—like *tabularx*/*tabulary*/*tabu*, *longtable*, *xltabular*, *multirow*, *makecell*, *booktabs*, *colortbl*, *parnotes*, *threeparttable*, ...

Separating the contents of a table from its formatting means that no formatting instructions whatsoever need to be placed in the table cells—neither for drawing rules between table rows nor for highlighting entries nor for increasing spacing nor for changing alignment, and so on. Instead, the placement of rules at the top and bottom of a table or between particular rows, the specification of column widths, the specification of font sizes and cell alignment, desired highlighting via boldface or a background color, etc., can all be coded as arguments to the table environment. This is beneficial in at least three aspects:

- The source code of the table’s contents is very clean. This particularly applies to animated Beamer presentations: no need to put `\pause/\only/\cellcolor/...` in table cells.
- It makes table contents *portable*: A table’s contents can be reused easily across multiple documents that are formatted in different ways.
- It also drastically simplifies the inclusion of table contents that are produced by an external program such as Python, R, or Stata.

Further additional helpful features of *tabularray* are:

- *tabularray* makes it easy to bottom-align or top-align particular table rows or individual cells entries, while keeping a different alignment for the other rows/cells.

1. This applies to the commercial version (<https://www.overleaf.com>) as well as to free versions <https://overleaf-students.astro.uni-bonn.de/> and https://uni-bonn.sciebo.de/apps/overleaf_nextcloud/launcher/launch.

- Table entries that consist of multiple lines of text become easy to produce: Just enclose them in curly braces, like in this example: ... & {Line 1 \\ Line 2} & ... \\. No need to use `\multirow` or `\makecell` any more.
- You can set document-wide formatting defaults to achieve consistent formatting of all tables. For instance, you can set defaults such that all tables feature the same top and bottom rules. Or, you could make all column heads be typeset in boldface document-wide.

When using `\UseTblrLibrary{siunitx}`, `tabularray` loads the `siunitx` package. This is another useful package which permits separating content from formatting. In particular, the `S` column type is able to round numbers to a specified precision. This means that one can include table content produced by an external program with as many decimal digits as that program produces and have LaTeX take care of the rounding.

2.3 Methods

2.3.1 Structuring Introduction and Discussion: Funnel and Reverse Funnel

A general structure that can be found in many research papers is the so-called X-shaped or “hourglass” structure.² This structure is brought about when the so-called “funnel” technique is used to write the introduction,³ and when a “reverse funnel” or “upside-down funnel” is used to write the discussion/concluding section:⁴

- (1) The paper starts with a *broad* perspective at the beginning of the introduction, which becomes *narrower* with every sentence, leading to the paper’s narrow research question.
- (2) The *narrowest* part is when the exact research question is posed, the research methodology is described in detail, and the results are derived/presented.
- (3) When the results are discussed and related to the literature in the concluding section, the perspective becomes *broad* again.

An example of the “funnel” technique can be found in [Figure 2.1](#), and of the “reverse funnel” technique in [Figure 2.2](#).

2. See https://skillsearthsciences.sites.uu.nl/writing/structure/#attachment_2387, Fig. 1, and <https://pmc.ncbi.nlm.nih.gov/articles/PMC5405644/pdf/JHRS-10-3.pdf>, Figure 1.

3. See <https://read.aupress.ca/read/read-think-write/section/9e945e9b-62cf-4dbb-b8cd-823549ede292#fig1401>, Figure 14.1, and <https://kib.ki.se/en/write-cite/academic-writing/structure-academic-texts>

4. See <https://read.aupress.ca/read/read-think-write/section/9e945e9b-62cf-4dbb-b8cd-823549ede292#fig1402>, Figure 14.2, and <https://kib.ki.se/en/write-cite/academic-writing/writing-conclusion>.

Molecular simulation methods have become an important technique in many areas of chemistry through the recent advent of effective and widespread software for molecular mechanics, molecular dynamics, and Monte Carlo simulations, and procedures for the estimation of free energy differences. In all such methods, a proper description of the electrostatic interactions within the simulated system is of key importance, ... However, in most implementations, ..., the simplest possible solution is used, ... Thus, most classical simulation methods need a point-charge parameterisation of the molecules of interest.

Unfortunately, atomic charges are not observables, i.e. they cannot unambiguously be determined by experiments or quantum chemical calculations. Therefore, a large number of methods have been suggested for the estimation of point charges [1]. Several groups have tried to derive the charges directly from experimental quantities, ... Yet, relevant data is usually missing or too scarce to allow a determination of all charges in interesting molecules. Instead, most techniques derive atomic charges from quantum chemical calculations.

The simplest way to determine quantum chemical charges is the Mulliken population analysis. ... Although, Mulliken charges are known to strongly depend on the basis set [6] and to reproduce electrostatic moments poorly [1], they are still widely used due to their simplicity. Several other ... methods have been devised to cure the problems of the Mulliken charges [1], ...

...

In this paper, we make a critical analysis of the four most popular potential-based point-charge methods, ... It is shown that these methods may give widely different results and possible explanations for this are discussed. Two alternative methods for deriving atomic charges are suggested, which avoid the arbitrariness in the selection of potential points, and their performance is judged using ...

Paragraph 1

Broad introduction with general description of the research field ("many areas of chemistry"), research topic ("molecular simulation methods"), and recent development ("through the recent advent of ..."). Followed by a more detailed description of crucial aspects in this context ("In all such methods, a proper description ... is of key importance") and of shortcomings of existing methods.

Paragraph 2

Narrowing down the topic: Description of the state of the art and of issues and limitations observed in the literature.

Paragraph 3

Further description of the state of the art: strengths and weaknesses of existing approaches.

Paragraphs 4–6

Further description of unresolved issues.

Paragraph 7

Narrow description of the specific contribution of this paper: Identification of undesirable arbitrary variation in simulation results of existing methods, which inspired development of an "alternative methods" that avoids this issue.

Figure 2.1. Illustration of the funneling approach, using the "Introduction" section from sigfridsson<empty citation>.

Note: The full text of the article by sigfridsson<empty citation> is available at https://lucris.lub.lu.se/ws/portalfiles/portal/135489685/25_charges.pdf.

Potential-based charges strongly depend on the way the potential points are selected. We have presented and tested a new method that avoids such a dependence, CHELMO. It fits the charges directly to the electrostatic moments and it performs as well, or better, as the best electrostatic potential method judging from the calculated electrostatic potential and moments. ... The major disadvantage of the method is that it cannot be used for large molecules, since there is a limited number of moments. No more than 25 independent charges in a molecule can be determined if all moments up to hexadecapole moments are used. In practice it should not be used for molecules with more than about 20 atoms since otherwise the higher moments may be poorly reproduced ... This could be remedied if moments higher than hexadecapole moments are calculated but such moments are not included in the output of normal quantum chemical packages.

In order to get a method that can be used for larger system, we constructed the CHELP-BOW method, which estimates the charges from a Boltzmann-weighted fit to the electrostatic potential. It has the advantage over the other ... methods that ... The present implementation of the CHELP-BOW method is very simplified, but in a future publication we will refine and thoroughly test the method. However, the results are conclusive enough to show that the method ... has the desired behaviour. ... Clearly, this is the method we recommend for general use.

An important advantage of the methods developed in this paper is that they are general, i.e. they can be used with any quantum chemical program and they can use experimental data (...) as well. The only thing that has to be changed is the input section of the programs. Thus any quantum chemical basis sets can be used, and any level of theory that gives a wave function may be employed. Furthermore, the methods can easily be adapted to ...

Finally, a comment on the traditional electrostatic potential methods. ... Thus, we cannot see any justification to use CHELP charges except for backward comparisons.

Paragraph 1

Narrow recap of the first contribution of the paper: brief description of the issue and how it was solved by the new method. Followed by description of the limitations of the new method. Outlook how these limitations could be overcome.

Paragraph 2

Narrow recap of the central contribution of the paper and how it relates to the literature: how existing methods were further improved upon by the second method proposed in this paper. Followed by outlook on future advances and recommendation for researchers in the field.

Paragraph 3

Broader implications due to the generality of the new method.

Paragraph 4

Other broader implications.

Figure 2.2. Illustration of the “reverse funneling” approach, using the “Concluding Remarks” subsection from sigfridsson<empty citation>.

2.3.2 Some Questions That May Help You Guide Writing Your Paper

You may find the following list of questions helpful as a guide to making sure that your manuscript covers all important aspects. The questions are primarily intended for doctoral and advanced researchers, but they may also be helpful to students writing term papers and theses.

Your introduction might answer the following questions (not necessarily in this order, but this order will give rise to a “funnel” structure):

- (1) What do we already know?
- (2) What do we not know yet (the “knowledge gap”)?
- (3) Why is it interesting/relevant to close the gap?
- (4) *What is the research question? [R]*
- (5) Why is it interesting to answer the research question?
- (6) How do we contribute to closing the knowledge gap: How do we answer the research question? (Which method do we use?)
- (7) Why is the chosen method (more) suitable (than alternative methods) for answering the research question?

And your conclusion/discussion might answer the following questions (again, not necessarily in this order, but this order will give rise to a “reverse funnel” structure):

- (8) What was the knowledge gap before the current study? [Similar to the introduction.]
- (9) *How did we contribute to closing that gap, and what did we find? [A]* (Which method/dataset did we use, and what are the results?—Recap and take-home message.)
- (10) How do our results relate to (confirm/contradict/complement/extend) the results from previous and other contemporaneous studies?
- (11) What part of the gap is still open? (Phrased a bit more negatively: What are the limitations of our approach?)
- (12) Next steps/avenues for future research: How could we go about closing the remaining knowledge gap (by removing the limitations of our current approach)?⁵

In her book *The Little Book of Research Writing*,⁶ Varanya Chaubey proposes a simpler list of only three items—the “RAP method”:

5. John Cochrane might do away with question 12. See his “Writing Tips for Ph.D. Students,” https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf.

6. For a quick summary of the book, see https://mauve-porcupine-8992.squarespace.com/s/Chaubey_Research_Writing-bxddd.pdf. See also <https://www.econscribe.org/about> and <https://arxiv.org/pdf/2012.07787>.

- The “R” stands for “research question.”
- The “A” stands for “answer” (to the research question, of course).
- The “P” stands for “positioning statement.” (How does our manuscript relate to the literature: Does it corroborate or extend or contradict others’ findings?)

Relating the “RAP” approach to the the list of questions above, question 4 is the “R,” the answer to question 9 is the “A,” and the remaining questions spell out the “P.”

2.3.3 Structuring Paragraphs in Academic Manuscripts

Obviously, there is no single “right” way of constructing paragraphs. However, a few basic rules can help you write paragraphs that are *easy to comprehend* by your readers. A common approach is to compose sentences such that they reflect the following functional sequence:⁷

- (1) *topic sentence (TS)*—which, if necessary, connects to the previous paragraph;
- (2) *supporting sentences (SS)*—which elaborate on the topic sentence; and
- (3) *concluding sentence or connecting sentence (CS)*—which refers back to the topic sentence and/or provides a link to the subsequent paragraph.

There are various ways in which the supporting sentences can be filled with content. The University of Sheffield suggests⁸ that the supporting sentences could consist of

- (1) “explanation or definitions (optional),”
- (2) “evidence and examples,” and
- (3) “comment” (an interpretation and evaluation of the evidence by you).

The University of Hull and Newcastle University (UK) consider the mnemonic “PEEL” helpful, although they define its meaning in slightly different ways: “Point, Evidence, Explanation, Link”⁹ versus “Point, Evidence, Evaluation, Link”¹⁰. One may view “Explanation”—“why the point is important and how it helps with your overall argument”—as narrower than “Evaluation”—which is a general critical reflection on the informativeness of the evidence and can include an explanation of its importance. “PEEL” relates to TS/SS/CS as follows:

- (1) “P,” the Point, corresponds to the topic sentence.

7. See <https://libguides.newcastle.edu.au/writing-paragraphs/structure>, https://www.une.edu.au/library/students/academic-writing/write-paragraphs/paragraphs/Academic-paragraphs_v2.pdf, <https://learningessentials.auckland.ac.nz/writing-effectively/paragraph/>

8. See <https://sheffield.ac.uk/study-skills/writing/academic/paragraph>.

9. See <https://libguides.hull.ac.uk/writing/paras>.

10. See <https://www.ncl.ac.uk/academic-skills-kit/writing/academic-writing/paragraphing/>

- (2) “E,” Evidence, is the first part of the supporting sentences.
- (3) “E,” Evaluation/Explanation, forms the second part of the supporting sentences.
- (4) “L,” Link, refers to the connecting sentence that links to the next paragraph.

The Wilfrid Laurier University (Canada) suggests a different mnemonic: “MEAL.”¹¹

- (1) “M: Main Point Sentence” corresponds to the topic sentence.
- (2) “E: Evidence” is the first part of the supporting sentences.
- (3) “A: Analysis/Synthesis” forms the second part of the supporting sentences.
- (4) “L: Linking-Back Sentence” refers to the concluding sentence that summarizes the paragraph and links back to the topic sentence or even the topic of the entire paper.

The first three components are largely identical according to these approaches. A difference exists in the meaning of “L”: While “PEEL” stresses the link to the *next* step in the chain of thought, “MEAL” stresses underscoring the message of the *current* paragraph. That is, the latter advocates *closure*: summarizing the paragraph and linking *back* to the topic sentence.

Yet slightly different is the approach put forth by Academic Writing UK:¹²

- (1) Topic sentence—key topic in this paragraph.
- (2) Development—the main idea/topic discussed in more detail.
- (3) Example—support/evidence/data/statistics that show your development is valid/credible.
- (4) Summary—overall main point summarized/evaluated.

It is perfectly fine to mix the different approaches. For some paragraphs, a closer explanation of the topic sentence (“development”) may be necessary, while it is unnecessary for others. And in some paragraphs, you may want underscore the central message by reiterating it in the final sentence (concluding sentence), while for other paragraphs, you will use the final sentence to set the stage for the next step in your chain of reasoning (connecting sentence).

One thing that *all* these suggestions have in common is, however, that they start with a *topic sentence* and end with a *concluding/connecting sentence*. Hence, no matter how you fill the supporting sentences, it is probably best to start a paragraph with a topic sentence and end with a concluding/connecting sentence.

11. See <https://students.wlu.ca/academics/support-and-advising/student-success/assets/resources/writing/how-to-structure-an-academic-paragraph.html>.

12. See <https://academic-englishuk.com/paragraphing/>.

Let us illustrate the “MEAL”/“PEEL” structure, with an optional explanation/definition included, through some example sentences from sigfridsson<empty citation>:

... Thus, most classical simulation methods need a point-charge parameterisation of the molecules of interest.	L: Concluding sentence
Unfortunately, atomic charges are not observables, i.e. they cannot unambiguously be determined by experiments or quantum chemical calculations. Therefore, a large number of methods have been suggested for the estimation of point charges [1]. Several groups have tried to derive the charges directly from experimental quantities, e.g. dipole moments, electrostatic potentials, or free energy differences [2–4]. Yet, relevant data is usually missing or too scarce to allow a determination of all charges in interesting molecules. Instead, most techniques derive atomic charges from quantum chemical calculations.	M/P: Topic sentence (reference to previous paragraph) Optional explanation/definition Supporting sentences: E: Evidence A/E: Analysis/Evaluation
The simplest way to determine quantum chemical charges is the Mulliken population analysis. ...	L: Connecting sentence (link to next paragraph) M/P: Topic sentence (reference to previous paragraph)

The rule to finish a paragraph with a concluding sentence entails, in particular, that a theoretical or empirical finding should never be the last thing that you mention in a paragraph. An “evidence” sentence should always be followed by an interpretation in which you tell the reader what we learn from that finding. Here is an example from sigfridsson<empty citation>:

... With these radii, the charges almost coincide with those obtained by the CHELPG method (within 0.02 e). Thus, we can conclude that the main difference between the Merz–Kollman and the CHELPG method is the 1.4 times larger effective radii of the former method, whereas the sampling schemes are almost equivalent.	Evidence: Supporting sentence with empirical finding Concluding sentence with interpretation of that finding
---	---

To reiterate, there is no single “right” way of constructing paragraphs. However, keeping the TS/SS/CS structure in mind as a guideline is bound to help you produce a text that enables readers to follow your chain of reasoning.

2.3.4 Additional Online Resources on Effective Writing in Economics

Additional useful recommendations and tips can be found in Plamen Nikolov's "Writing Tips for Crafting Effective Economics Research Papers – 2023-2024 Edition" (<https://docs.iza.org/dp16276.pdf>) and in John H. Cochrane's "Writing Tips for Ph.D. Students" (https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf).

In this section, we first present the design of the experiment (2.3.5) and derive behavioral predictions (2.3.6).

2.3.5 Design of the Main Experiment

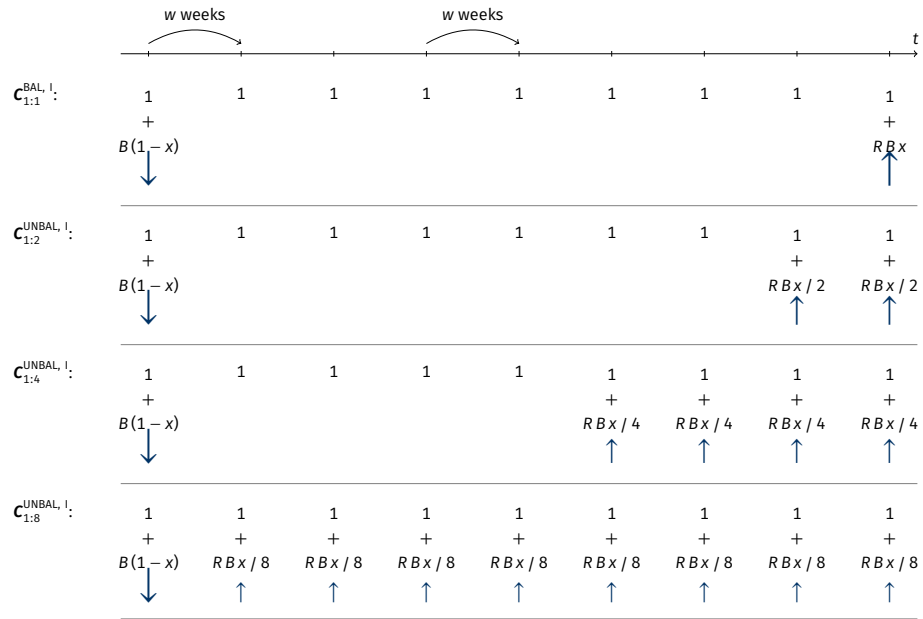
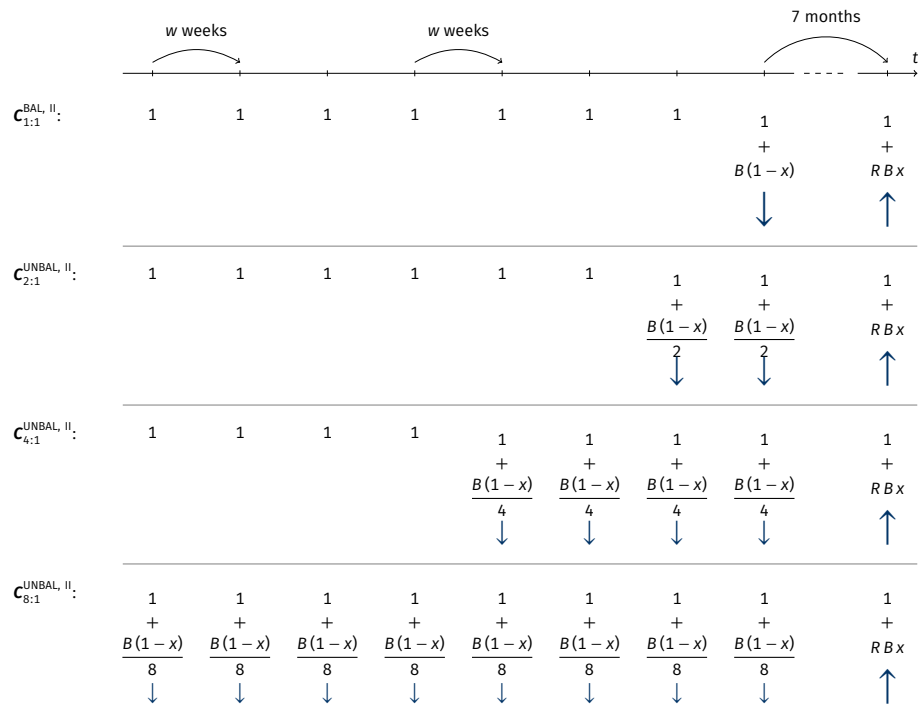
2.3.5.1 General Features

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.3.5.2 More Specific Features

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Let's test the euro symbol: €, €1,234.56, €1,234.56. Let's also test text superscripts: i^{th} and text subscripts: CO_2 and H_2O . $\sigma_\epsilon, c^\alpha$. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for

Figure 2.3. Budget Sets $C_{1:1}^{BAL, I}$ and $C_{1:n}^{UNBAL, I}$ Figure 2.4. Budget Sets $C_{1:1}^{BAL, II}$ and $C_{n:1}^{UNBAL, II}$

Notes: For the values of B , R , and w that we used, see [Section 2.3.5.4](#). The savings rate x is individuals' choice variable: they choose some $x \in X = \{0, 1/100, 2/100, \dots, 1\}$ in each trial. The arrows indicate whether and in which direction payments at the respective payment dates change if x is increased. σ_c, c^a . This figure was taken from Dertwinkel-Kalt et al. (2017).

special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Let's test the footnote settings.¹³

Figure 2.5 shows an exemplary decision screen with $B = €11$ and $r \approx 15\%$ for both $BAL_{1:1}^I$ (upper panel) and $UNBAL_{1:8}^I$ (lower panel). Through a slider, subjects choose their preferred $x \in X$.¹⁴ The slider position in Figure 2.5 indicates $x = 0.5$, i.e., the earliest payment is reduced by €5.50. Since $r \approx 15\%$ in this example, this slider position amounts to €6.30 that are paid at later payment dates. While these €6.30 are paid in a single bank transfer on the latest payment date in $BAL_{1:1}^I$, the amount is dispersed in equal parts over the last 8 payment dates in $UNBAL_{1:8}^I$ —i.e., 8 consecutive payments of €0.79.¹⁵

2.3.5.3 Some More Details

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Here's a bulleted list:

- Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

13. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

14. The slider had no initial position—it appeared only after subjects first positioned the mouse cursor over the slider bar. This was done to avoid default effects.

15. We always rounded the second decimal place up so that the sum of the payments included in a dispersed payoff was always at least as great as the respective concentrated payoff.

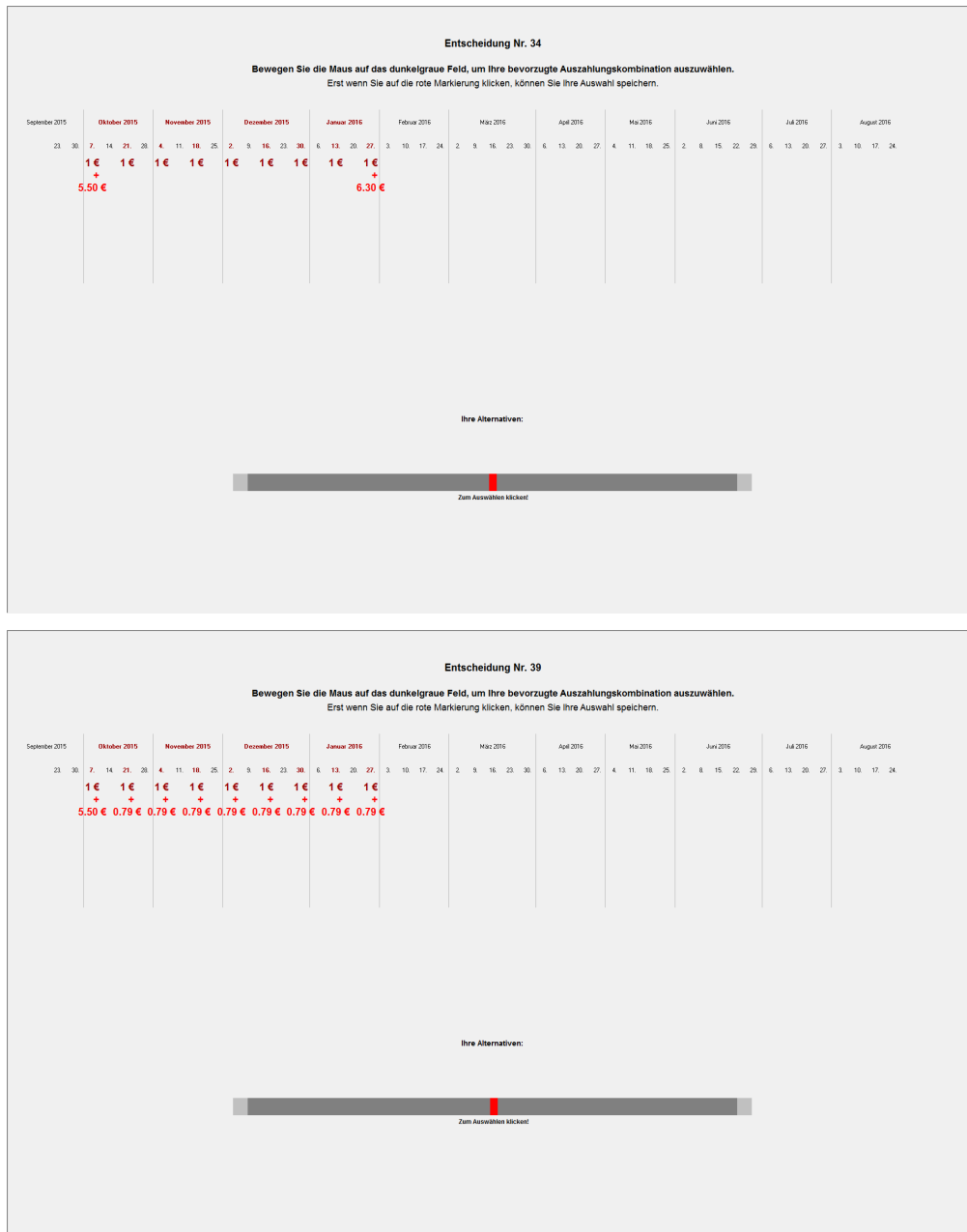


Figure 2.5. Screenshots of a $BAL_{1:1}^I$ Decision (Top) and an $UNBAL_{1:8}^I$ Decision (Bottom)

Note: This figure was taken from Dertwinkel-Kalt et al. (2017).

- Hello, here is some text without a meaning. $d\Omega = \sin\vartheta d\vartheta d\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the

letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.

- Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. If you read this text, you will get no information. $d\Omega = \sin\vartheta d\vartheta d\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

2.3.5.4 Procedure

Describe the sequence of events in your study. You could do this with the help of an enumerated list:

- (1) Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.
- (2) Hello, here is some text without a meaning. $d\Omega = \sin\vartheta d\vartheta d\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.
- (3) Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. If you read this text, you will get no information. $d\Omega = \sin\vartheta d\vartheta d\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest

gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

2.3.6 Predictions

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages.¹⁶ Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two

16. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of*

pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages.

By discounted utility we understand any intertemporal utility function that (1) is time-separable and that (2) values a payment farther in the future at most as much as an equal-sized payment closer in the future. Importantly, the predictions derived below hold for all three frequently used types of discounting—exponential, hyperbolic, and quasi-hyperbolic.

In the following, we assume that individuals base their decisions on utility derived from receiving monetary payments c_t at various dates t . This is an assumption that is frequently made in experiments on intertemporal decision making. One way to justify this assumption is that individuals anticipate to consume the payments they receive within a short period around date t . Given that the maximum payment was below €20 and that any two payment dates were separated by at least two weeks, this assumption seems reasonable (see the arguments in favor of this view in Halevy 2014). Kőszegi and Szeidl (2013) themselves make the same assumption of “money in the utility function”: “in some applications we also assume that monetary transactions induce direct utility consequences, so that for instance an agent making a payment experiences an immediate utility loss. The idea that people ex-

the alphabet and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

perience monetary transactions as immediate utility is both intuitively compelling and supported in the literature: ... some evidence on individuals' attitudes toward money, such as narrow bracketing (...) and laboratory evidence on hyperbolic discounting (...), is difficult to explain without it." Last but not least, the papers by McClure et al. (2004) and McClure et al. (2007) demonstrate that brain activation, as measured by functional magnetic resonance imaging, is similar for primary and monetary rewards. Additionally, we make the standard assumption that utility from money is increasing in its argument but not convex: $u'(c_t) \geq 0$ and $u''(c_t) \leq 0$.

2.3.6.1 Discounted Utility

Individuals make their allocation decisions by comparing the aggregated consumption utility of each earnings sequence $\mathbf{c} \in \mathcal{C}$. Discounted utility assumes that the utility of each period enters overall utility additively. That is, utility derived from the payment to be received at future date t can be expressed as $u_t(c_t) := D(t) u(c_t)$. Here, $D(t)$ denotes the individual's discount function for conversion of future utility into present utility. The discount function satisfies $0 \leq D(t)$ and $D'(t) \leq 0$, such that a payment further in the future is valued at most as much as an equal-sized payment closer in the future.¹⁷

The utility of earnings sequence $\mathbf{c}$ with payments c_t in periods $t = 1, \dots, T$ is as follows:

$\$ \$ \dots \$ \$$:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) = \sum_{t=1}^T D(t) u(c_t).$$

$\backslash[\dots \backslash]$ with manual $\backslash\tag{\dots}$:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) = \sum_{t=1}^T D(t) u(c_t). \quad (\text{II})$$

$\backslash\begin{equation} \dots \end{equation}$:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) = \sum_{t=1}^T D(t) u(c_t). \quad (2.1)$$

$\backslash\begin{equation*} \dots \end{equation*}$:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) = \sum_{t=1}^T D(t) u(c_t).$$

17. Normalization such that $D(t) \leq 1$ is not necessary in our case. Provided that t is a metric time measure, where $t = 0$ stands for the present, examples are $D(t) := \delta^t$ with some $\delta > 0$ for exponential discounting and $D(t) := (1 + \alpha t)^{-\gamma/\alpha}$ with some $\alpha, \gamma > 0$ for generalized hyperbolic discounting.

`\begin{eqnarray} ... \end{eqnarray}`:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) \quad (2.2)$$

$$= \sum_{t=1}^T D(t) u(c_t). \quad (2.3)$$

`\begin{eqnarray*} ... \end{eqnarray*}`:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t)$$

$$= \sum_{t=1}^T D(t) u(c_t).$$

`\begin{align} ... \end{align}`, equation number in the final line only:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t)$$

$$= \sum_{t=1}^T D(t) u(c_t). \quad (2.4)$$

`\begin{align} ... \end{align}`, equation number in each line:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) \quad (2.5)$$

$$= \sum_{t=1}^T D(t) u(c_t). \quad (2.6)$$

`\begin{align*} ... \end{align*}`:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t)$$

$$= \sum_{t=1}^T D(t) u(c_t).$$

`\begin{alignat}{2} ... \end{alignat}`:

$$U(\mathbf{c}) = \sum_{t=1}^T u_t(c_t) \quad (2.7)$$

$$= \sum_{t=1}^T D(t) u(c_t). \quad (2.8)$$

`\begin{alignat*}{2} ... \end{alignat*}`:

$$\begin{aligned} U(c) &= \sum_{t=1}^T u_t(c_t) \\ &= \sum_{t=1}^T D(t) u(c_t). \end{aligned}$$

Individuals choose how much to allocate to the different periods by maximizing their utility over all possible earnings sequences available within a given budget set C , see equations (II), (2.1), (2.2), (2.3), (2.4), (2.5), and (2.6). See also [Equation 2.8](#). We use the superscript ^{DU} to indicate decisions based on discounted utility.

A Subparagraph. This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Another Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.3.6.2 Focus-Weighted Utility

In this section, we extend the model of discounted utility through “focus weights,” as proposed by Kőszegi and Szeidl (2013). Period- t weights g_t scale period- t consumption utility u_t . Individuals are assumed to maximize focus-weighted utility, which is defined as follows:

$$\tilde{U}(\mathbf{c}, \mathbf{C}) := \sum_{t=1}^T g_t(\mathbf{C}) u_t(c_t). \quad (2.9)$$

In contrast to discounted utility $U(\mathbf{c})$, focus-weighted utility $\tilde{U}(\mathbf{c}, \mathbf{C})$ has two arguments: the earnings sequence $\mathbf{c}$ and the choice set $\mathbf{C}$. The latter dependence is due to the weights g_t . These are given by a strictly increasing weighting function g that takes as its argument the difference between the maximum and the minimum attainable utility in period t over all possible earnings sequences in set $\mathbf{C}$:

$$g_t(\mathbf{C}) := g[\Delta_t(\mathbf{C})] \quad \text{with} \quad \Delta_t(\mathbf{C}) := \max_{c \in \mathbf{C}} u_t(c_t) - \min_{c \in \mathbf{C}} u_t(c_t). \quad (2.10)$$

If the underlying consumption utility function is characterized by discounted utility, then $u_t(c_t) := D(t) u(c_t)$. That is, focused thinkers put more weight on period t than on period t' if the discounted-utility distance between the best and worst alternative is larger for period t than for period t' .

A Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Yet Another Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.3.6.3 Hypotheses

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. This gives rise to our first hypothesis:

Hypothesis 2.1. *This environment can be used to clearly state your hypothesis and set them apart from the body text.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Based on this, we can state our second hypothesis:

Hypothesis 2.2. *This environment can be used to clearly state your hypothesis and set them apart from the body text.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.4 Results

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference

Table 2.1. A `tabularray`-Based Example Table

Dependent variable	$\hat{d}$
Estimate	0.123*** (0.011)
Observations	750
Subjects	250

Notes: Standard errors in parentheses, clustered on the subject level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. With this, we can test our hypotheses.

2.4.1 Test of Hypothesis 2.1

Our first result supports [Hypothesis 2.1](#). Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. The analysis we conducted to obtain [Result 2.1](#) is described in detail in ???. Let’s reference a section, a subsection, and a figure from the appendices: [Section 2.C](#), [Section 2.A.2](#), [Figure 2.B.1](#).

Result 2.1. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.4.2 Test of Hypothesis 2.2

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. We thereby test [Hypothesis 2.2](#).

Result 2.2. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.*

Our second result provides evidence in support of [Hypothesis 2.2](#). Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

2.4.3 Heterogeneity

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{i=n} x_i = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\int_0^\infty e^{-\alpha x^2} dx = \frac{1}{2} \sqrt{\int_{-\infty}^\infty e^{-\alpha x^2} dx \int_{-\infty}^\infty e^{-\alpha y^2} dy} = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\sum_{k=0}^{\infty} a_0 q^k = \lim_{n \rightarrow \infty} \sum_{k=0}^n a_0 q^k = \lim_{n \rightarrow \infty} a_0 \frac{1 - q^{n+1}}{1 - q} = \frac{a_0}{1 - q}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text,

you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.4.4 Structural Estimation

Inspect the variance–covariance matrix Σ :

$$\Sigma := \text{Cov}(X) = \begin{bmatrix} \text{Var}(X_1) & \cdots & \text{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \cdots & \text{Var}(X_n) \end{bmatrix}.$$

2.5 Discussion

2.5.1 Some Limitations

Let's reference some tables: [Table 2.2](#) and ???. After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Right?

2.5.2 Utility from Money

In deriving our predictions ([Section 2.3.6](#)), we assume that subjects base their decisions on utility derived from receiving monetary payments c_t at various dates t . We

Table 2.2. Points Awarded in Our Typeface Competition—Basic Formatting That Produces Irregular Column Widths and a Suboptimal Table Width *Test Greek: $\varepsilon, \theta, \phi$*

	Utopia	Computer Modern	Charter	Times Roman	Palatino
Yoël	1	1	2	0	1
Çelik	2	0	2	1	0
Anità	1	2	1	2	0
Uğur	1	2	0	1	0
Håkan	1	0	2	0	1
Allison	2	0	1	2	1
Pía	1	0	2	1	0
David	1	0	2	1	1
Sum	10	5	12	8	4

also make the standard assumption that utility from money is increasing in its argument but not convex, i.e., $u'(c_t) > 0$ and $u''(c_t) \leq 0$. Both assumptions are frequently made in studies on intertemporal decision making.

One way to justify the assumption of utility being based on money—rather than consumption—is that individuals anticipate to consume the payments that they receive at date t within a short period around t . Given that the maximum payment was below €20 and that any two payment dates were separated by at least two weeks, this seems reasonable (see the arguments in favor of this view in Halevy 2014).

Table 2.3. Points Awarded in Our Typeface Competition—More Sophisticated Formatting with Homogeneous Column Widths and a Table Width That Equals the Text Width

	Utopia ^a	Computer Modern ^b	Charter ^c	Times Roman ^d	Palatino ^e
Yoël	1	1	2	0	1
Çelik	2	0	2	1	0
Anità	1	2	1	2	0
Uğur	1	2	0	1	0
Håkan	1	0	2	0	1
Allison	2	0	1	2	1
Pía	1	0	2	1	0
David	1	0	2	1	1
Sum	10	5	12	8	4

^a `\usepackage{fourier}`^b The $\text{\LaTeX}$ standard serif font.^c `\usepackage[charter]{mathdesign}`^d `\usepackage{newtxtext, newtxmath}`^e `\usepackage[sc]{mathpazo}`

A second justification is consistency within the discipline: Halevy (2014) points out that “in the domain of risk and uncertainty ... preferences are often defined over payments.” In line with this, Kőszegi and Szeidl (2013, p. 62) make the same assumption of “money in the utility function”:

in some applications we also assume that monetary transactions induce *direct* utility consequences, so that for instance an agent making a payment experiences an immediate utility loss. The idea that people experience monetary transactions as immediate utility is both intuitively compelling and supported in the literature: ... some evidence on individuals’ attitudes toward money, such as narrow bracketing (...) and laboratory evidence on hyperbolic discounting (...), is difficult to explain without it.

Last but not least, the papers by McClure et al. (2004) and McClure et al. (2007) demonstrate that brain activation, as measured by functional magnetic resonance imaging, is similar for primary and monetary rewards.

Let us now discuss the second assumption: that utility from money is nonconvex. We find that subjects allocate more money to the concentrated payoffs in the unbalanced than in the associated balanced budget sets—which we call concentration bias. One might argue that this relative preference for concentrated payoffs can be explained by the per-period utility function over money being convex.

Obtaining evidence on the shape of utility over money is nontrivial because it requires that at least two monetary amounts be compared with each other without the one clearly dominating the other. Thus, estimates of the curvature of the utility function over money can be obtained in two ways: the monetary amounts must be paid in different states of the world, i.e., comprise a lottery, or they have to be paid at different points in time.¹⁸ Both methods entail particular theoretical assumptions.

Andersen et al. (2008) advocate the former approach and argue that when estimating time preference parameters, one should control for the curvature of the utility function through a measure of the curvature that is based on observed choices under risk. Their study and numerous other studies on risk attitudes consistently reveal that the vast majority of subjects is risk-averse even over small stakes. Hence, for the vast majority of subjects, utility over money is concave according to this methodology (ruling out probability weighting). Others, most notably Andreoni and Sprenger (2012), have argued that the degree of curvature measured via risky choices probably overstates the degree of curvature effective in intertemporal choices, but they also find that utility is concave (albeit close to linear). Given

18. As a matter of fact, the latter was the motivation behind Samuelson (1937): “Under the following four assumptions, it is believed possible to arrive theoretically at a precise measure of the marginal utility of *money income* ...” (p. 155; emphasis in the original).

this unambiguous evidence from previous studies, it is implausible that our subjects exhibit convex utility over money.

2.6 Conclusion

Cite some more papers (Yaari 1965; Warner and Pleeter 2001; Davidoff, Brown, and Diamond 2005; Benartzi, Previtro, and Thaler 2011). Let's cite a book: Luce (1959). Let's cite a contribution to a collected volume: Harrison and Rutström (2008) and a collection (an edited volume) itself: Kagel and Roth (2016). Now let's cite presentations at conferences: Vosgerau et al. (2008) and Beute and Kort (2012). Attema et al. (2016) propose a method for “measuring discounting without measuring utility”¹⁹.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Appendix 2.A Put More Complicated Derivations and Proofs Here

2.A.1 Appendix Subsection

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

(1) Erster Listenpunkt, Stufe 1

19. The basic idea of their method is intriguingly simple: Imagine an individual who is indifferent between, say, Option A: \$10 today and Option B: \$10 in one year plus \$10 in two years. With a constant annual discount factor δ , this indifference translates to $u(\$10) = \delta u(\$10) + \delta^2 u(\$10)$, so that $u(\$10)$ cancels out, and δ can be readily calculated as the solution to $1 = \delta + \delta^2$.

- a. Erster Listenpunkt, Stufe 2
 - i. Erster Listenpunkt, Stufe 3
 - ii. Zweiter Listenpunkt, Stufe 3
 - iii. Dritter Listenpunkt, Stufe 3
 - iv. Vierter Listenpunkt, Stufe 3
 - b. Zweiter Listenpunkt, Stufe 2
 - c. Dritter Listenpunkt, Stufe 2
 - d. Vierter Listenpunkt, Stufe 2
- (2) Zweiter Listenpunkt, Stufe 1
 (3) Dritter Listenpunkt, Stufe 1
 (4) Vierter Listenpunkt, Stufe 1

The typeset math below follows the ISO recommendations that only variables be set in *italic*. Note the use of upright shapes for “d,” “e,” and “ π .” (These are entered as `\mathup{d}`, `\mathup{e}`, and `\mathup{\pi}`, respectively.)

Theorem 2.1 (Simplest form of the *Central Limit Theorem*). *Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables with mean 0 and variance 1 on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then*

$$\mathbb{P}\left(\frac{X_1 + \dots + X_n}{\sqrt{n}} \leq y\right) \rightarrow \mathfrak{N}(y) := \int_{-\infty}^y \frac{e^{-v^2/2}}{\sqrt{2\pi}} dv \quad \text{as } n \rightarrow \infty,$$

or, equivalently, letting $S_n := \sum_1^n X_k$,

$$\mathbb{E}f(S_n/\sqrt{n}) \rightarrow \int_{-\infty}^{\infty} f(v) \frac{e^{-v^2/2}}{\sqrt{2\pi}} dv \quad \text{as } n \rightarrow \infty, \text{ for every } f \in \mathfrak{b}\mathcal{C}(\mathbb{R}).$$

2.A.2 Salience

Salience theory (Bordalo, Gennaioli, and Shleifer 2012, 2013) represents a behavioral model according to which the most distinctive features of the available alternatives receive a particularly large share of attention and are therefore over-weighted. More precisely, a particular attribute out of all attributes of an alternative becomes the more salient, the more it differs from that attribute’s average level over all available alternatives.

Formally, alternatives are assumed to be uniquely characterized by the values they take in $T \geq 1$ attributes (or, “dimensions”). Utility is assumed to be additively separable in attributes, and salience attaches a decision weight to each attribute of each good which indicates how salient the respective attribute is for that good. Suppose an agent chooses one alternative from some finite choice set $\mathcal{C}$. Let t index the

T different attributes, and let k index the K available alternatives. Let $u_t(\cdot)$ denote the function which assigns utility to values in dimension t . Denote by a_t^k the level of attribute t of good k and define $u_t^k := u_t(a_t^k)$ as the utility that dimension t of good k yields. Let $\bar{u}_t$ be the average utility level, across all K goods, of dimension t . The salience of each dimension of good k is determined by a symmetric and continuous salience function $\sigma(\cdot, \cdot)$ that satisfies the following two properties:

(1) *Ordering*. Let $\mu := \text{sgn}(u_t^k - \bar{u}_t)$. Then for any $\varepsilon, \varepsilon' \geq 0$ with $\varepsilon + \varepsilon' > 0$, it holds that

$$\sigma(u_t^k + \mu \varepsilon, \bar{u}_t - \mu \varepsilon') > \sigma(u_t^k, \bar{u}_t). \quad (2.A.1)$$

(2) *Diminishing sensitivity*. For any $u_t^k, \bar{u}_t \geq 0$ and all $\varepsilon > 0$, it holds that

$$\sigma(u_t^k + \varepsilon, \bar{u}_t + \varepsilon) < \sigma(u_t^k, \bar{u}_t). \quad (2.A.2)$$

Following the smooth salience characterization proposed in Bordalo, Gennaioli, and Shleifer (2012, p. 1255), each dimension t of good k receives weight $\Delta^{-\sigma(u_t^k, \bar{u}_t)}$, where $\Delta \in (0, 1]$ is a constant that captures an agent's susceptibility to salience. $\Delta = 1$ gives rise to a rational decision maker, and the smaller Δ , the stronger is the salience bias. We call an agent with $\Delta < 1$ a salient thinker.

A reference with a large number of authors is Henrich et al. (2005).

Appendix 2.B Additional Figures

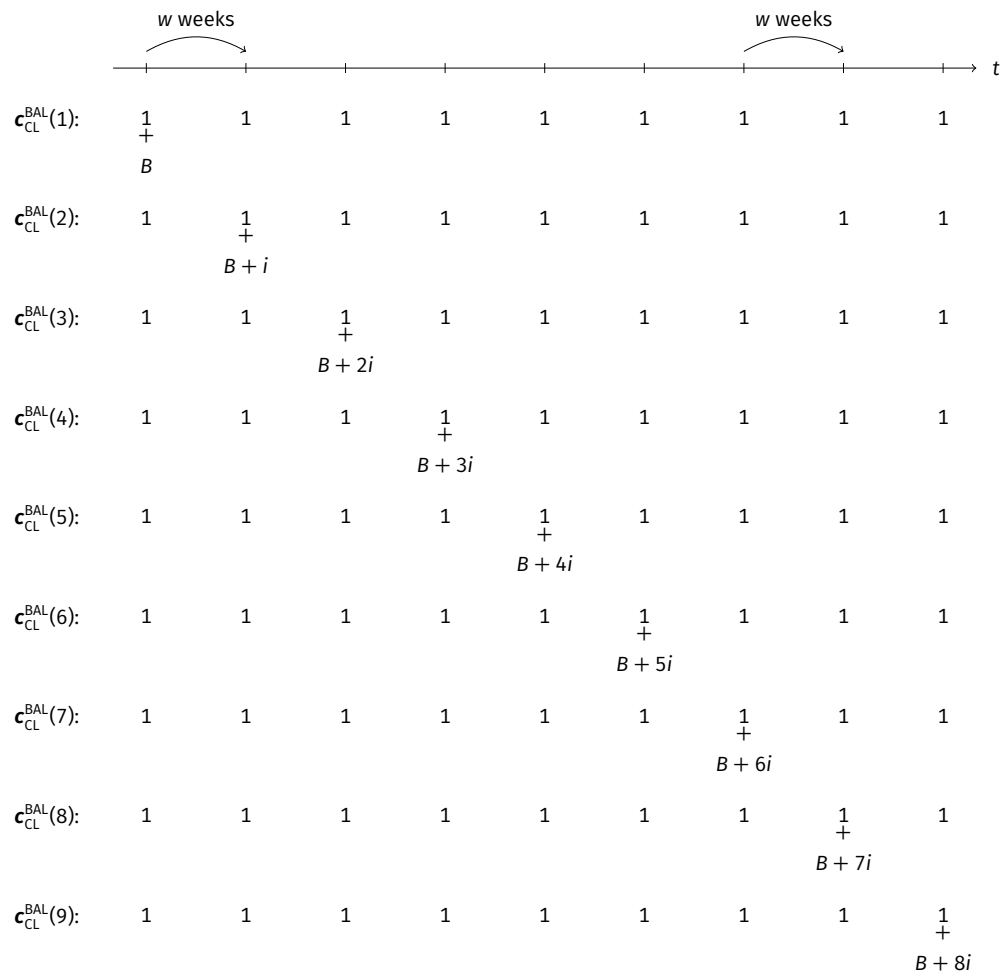


Figure 2.B.1. Earnings Sequences Included in Choice List c_{CL}^{BAL}

Notes: For the values of B , i , and w that we used see [Section 2.3](#). Figure taken from Dertwinkel-Kalt et al. (2017).

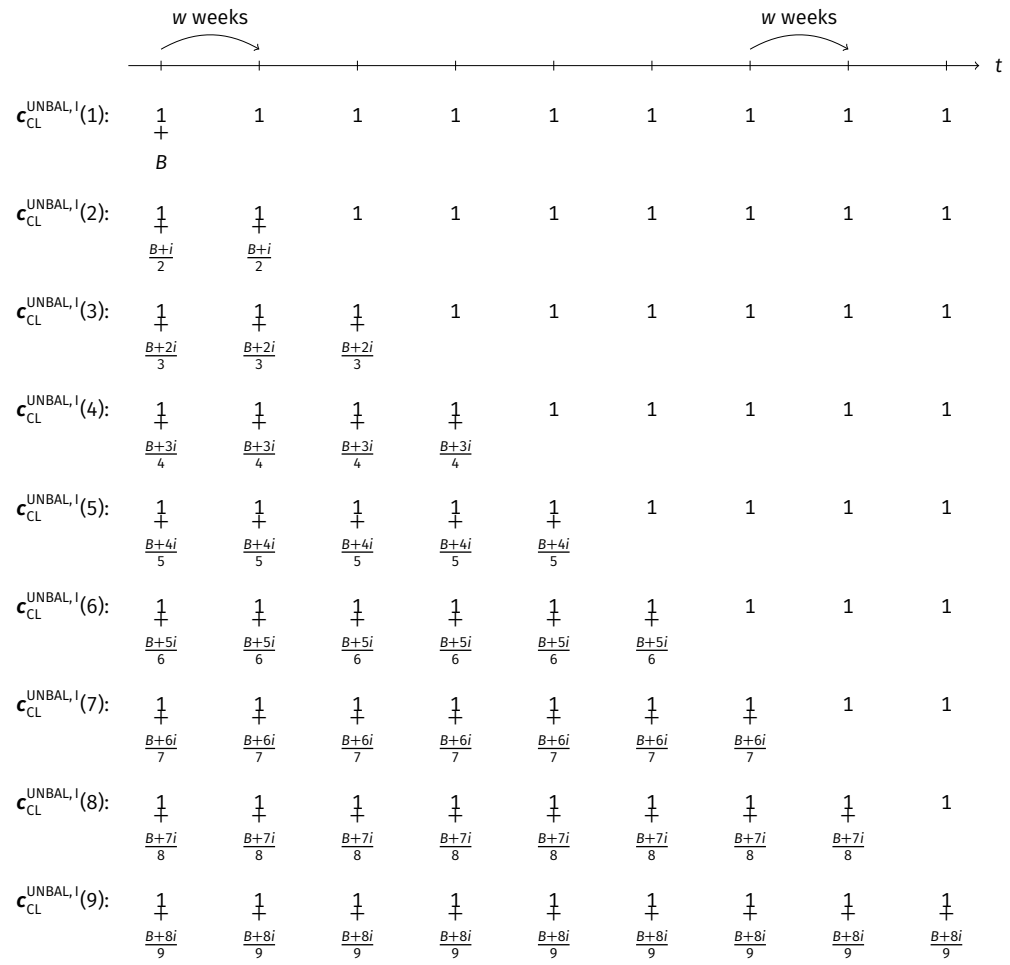


Figure 2.B.2. Earnings Sequences Included in Choice List $c_{CL}^{UNBAL, I}$

Notes: For the values of B , i , and w that we used see [Section 2.3](#). Figure taken from Dertwinkel-Kalt et al. (2017).

	w weeks					w weeks				t
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(1):$	$\frac{1}{+}$ $\frac{B}{9}$	$\frac{1}{+}$ $\frac{B}{9}$	$\frac{1}{+}$ $\frac{B}{9}$	$\frac{1}{+}$ $\frac{B}{9}$	$\frac{1}{+}$ $\frac{B}{9}$	$\frac{1}{+}$ $\frac{B}{9}$	$\frac{1}{+}$ $\frac{B}{9}$	$\frac{1}{+}$ $\frac{B}{9}$	$\frac{1}{+}$ $\frac{B}{9}$	
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(2):$	1	$\frac{1}{+}$ $\frac{B+i}{8}$	$\frac{1}{+}$ $\frac{B+i}{8}$	$\frac{1}{+}$ $\frac{B+i}{8}$	$\frac{1}{+}$ $\frac{B+i}{8}$	$\frac{1}{+}$ $\frac{B+i}{8}$	$\frac{1}{+}$ $\frac{B+i}{8}$	$\frac{1}{+}$ $\frac{B+i}{8}$	$\frac{1}{+}$ $\frac{B+i}{8}$	
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(3):$	1	1	$\frac{1}{+}$ $\frac{B+2i}{7}$	$\frac{1}{+}$ $\frac{B+2i}{7}$	$\frac{1}{+}$ $\frac{B+2i}{7}$	$\frac{1}{+}$ $\frac{B+2i}{7}$	$\frac{1}{+}$ $\frac{B+2i}{7}$	$\frac{1}{+}$ $\frac{B+2i}{7}$	$\frac{1}{+}$ $\frac{B+2i}{7}$	
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(4):$	1	1	1	$\frac{1}{+}$ $\frac{B+3i}{6}$	$\frac{1}{+}$ $\frac{B+3i}{6}$	$\frac{1}{+}$ $\frac{B+3i}{6}$	$\frac{1}{+}$ $\frac{B+3i}{6}$	$\frac{1}{+}$ $\frac{B+3i}{6}$	$\frac{1}{+}$ $\frac{B+3i}{6}$	
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(5):$	1	1	1	1	$\frac{1}{+}$ $\frac{B+4i}{5}$	$\frac{1}{+}$ $\frac{B+4i}{5}$	$\frac{1}{+}$ $\frac{B+4i}{5}$	$\frac{1}{+}$ $\frac{B+4i}{5}$	$\frac{1}{+}$ $\frac{B+4i}{5}$	
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(6):$	1	1	1	1	1	$\frac{1}{+}$ $\frac{B+5i}{4}$	$\frac{1}{+}$ $\frac{B+5i}{4}$	$\frac{1}{+}$ $\frac{B+5i}{4}$	$\frac{1}{+}$ $\frac{B+5i}{4}$	
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(7):$	1	1	1	1	1	1	$\frac{1}{+}$ $\frac{B+6i}{3}$	$\frac{1}{+}$ $\frac{B+6i}{3}$	$\frac{1}{+}$ $\frac{B+6i}{3}$	
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(8):$	1	1	1	1	1	1	1	$\frac{1}{+}$ $\frac{B+7i}{2}$	$\frac{1}{+}$ $\frac{B+7i}{2}$	
$\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}(9):$	1	1	1	1	1	1	1	1	$\frac{1}{+}$ $B + 8i$	

Figure 2.B.3. Earnings Sequences Included in Choice List $\mathbf{c}_{\text{CL}}^{\text{UNBAL, II}}$

Notes: For the values of B , i , and w that we used see [Section 2.3](#). Figure taken from Dertwinkel-Kalt et al. (2017).

Appendix 2.C Additional *siunitx* and *tabularray* Example Tables

Table 2.C.1. An Example of a Regression Table (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017). Remember to Mention the Dependent Variable!

	(1)	(2)	(3)	(4)	(5)
Treatment	-0.390 (+0.352)	-0.228 (-0.205)	-0.729* [+0.377]	-0.449* [-0.245]	-0.453** {+0.204}
Female	0.948*** (0.354)	0.0607 (0.233)	0.188 (0.372)	0.305 (0.226)	0.385* (0.222)
Female × Treatment	0.169 (0.514)	0.251 (0.325)	0.892* (0.533)	0.454 (0.341)	0.439 (0.307)
Final high school grade	-0.101 (0.198)	0.0132 (0.144)	0.0759 (0.224)	0.117 (0.146)	0.0394 (0.133)
Trait self-control	-0.0156 (0.0162)	0.00214 (0.00994)	-0.0160 (0.0147)	-0.000122 (0.0104)	-0.00678 (0.00935)
UUUU Constant	2.357*** (0.239)	1.512*** (0.144)	-0.322 (0.265)	2.158*** (0.161)	1.437*** (0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment × (1 + Female)	-0.221	0.0228	0.163	0.00435	-0.0138
p_F [Treatment × (1 + Female) = 0]	0.327	0.00767	0.192	0.000314	0.00343

Notes: Dependent variable: $m_{\cdot}$. Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

Table 2.C.2. Figure Grouping via siunitx in a Table

(1)	(2)	(3)
-0.100* (2.871)	-0.100 01* (2.871 23)	-123 456.444*** [+50 000.123]

Table 2.C.3. Overview of the Choice Lists Presented to Subjects (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017)

	Q[r] E Q[r] E Q[r] E							
	Alternative A				Alternative B			
	$c_{A,1}$	$p_{A,1}$	$c_{A,2}$	$p_{A,2}$	$c_{B,1}$	$p_{B,1}$	$c_{B,2}$	$p_{B,2}$
<i>Choice List I: risky/risky</i> ($x = €22.00$, $r = €7.50$, $k = €11.50$; 25 rows)								
Top row	€3.00	50%	€22.00	50%	€3.00	50%	€7.00	50%
Center row	€3.00	50%	€22.00	50%	€9.00	50%	€13.00	50%
Row with $m = 0$	€3.00	50%	€22.00	50%	€10.50	50%	€14.50	50%
Bottom row	€3.00	50%	€22.00	50%	€15.00	50%	€19.00	50%
<i>Choice List II: safe/risky</i> ($x = €16.00$, $r = €5.00$, $k = €5.00$; 19 rows)								
Top row	€11.00	100%			€11.00	50%	€21.00	50%
Center row	€11.00	100%			€6.50	50%	€16.50	50%
Row with $m = 0$	€11.00	100%			€6.00	50%	€16.00	50%
Bottom row	€11.00	100%			€2.00	50%	€12.00	50%
<i>Choice List III: “long shot”</i> ($x = €14.00$, $r = -€36.00$, $k = €7.00$; 21 rows)								
Top row	€7.00	90%	€50.00	10%	€7.00	90%	€10.00	10%
Row with $m = 0$	€7.00	90%	€50.00	10%	€11.00	90%	€14.00	10%
Center row	€7.00	90%	€50.00	10%	€12.00	90%	€15.00	10%
Bottom row	€7.00	90%	€50.00	10%	€17.00	90%	€20.00	10%
<i>Choice List IV: delayed payoffs</i> ($x = €18.00$, $r = €6.00$, $k = €8.50$, paid in one week; 20 rows)								
Top row	€9.50	50%	€12.00	50%	€9.50	50%	€24.00	50%
Above-center row	€9.50	50%	€12.00	50%	€5.00	50%	€19.50	50%
Below-center row	€9.50	50%	€12.00	50%	€4.50	50%	€19.00	50%
Row with $m = 0$	€9.50	50%	€12.00	50%	€3.50	50%	€18.00	50%
Bottom row	€9.50	50%	€12.00	50%	€0.00	50%	€14.50	50%

Table 2.C.4. A Really Long Table That Spans Multiple Pages

	(1)	(2)	(3)	(4)
Row 1	0.0070	0.1356	0.1560	0.8979
Row 2	0.4223	0.7311	0.4213	0.6900
Row 3	0.0767	0.5110	0.7399	0.9491
Row 4	0.5954	0.1685	0.3778	0.9960
Row 5	0.6465	0.0524	0.8895	0.1544
Row 6	0.3838	0.7069	0.1773	0.5785
Row 7	0.1537	0.5442	0.6361	0.0327
Row 8	0.0879	0.1812	0.3082	0.2942

Continued on next page.

Table 2.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 9	0.2720	0.2565	0.6214	0.8944
Row 10	0.4873	0.3064	0.9913	0.0591
Row 11	0.8387	0.1713	0.6747	0.7455
Row 12	0.0645	0.4891	0.2892	0.1013
Row 13	0.0989	0.3798	0.5795	0.3725
Row 14	0.3256	0.7080	0.0262	0.8709
Row 15	0.7867	0.8768	0.0690	0.6081
Row 16	0.2713	0.4399	0.5838	0.6107
Row 17	0.5236	0.1527	0.4402	0.8002
Row 18	0.4851	0.4619	0.4040	0.2711
Row 19	0.1742	0.8151	0.2757	0.4184
Row 20	0.0495	0.3288	0.2759	0.1452
Row 21	0.1678	0.2403	0.1993	0.3676
Row 22	0.4977	0.9472	0.2810	0.2493
Row 23	0.6777	0.6516	0.3573	0.1413
Row 24	0.3668	0.3075	0.8724	0.3945
Row 25	0.5877	0.5670	0.0417	0.5213
Row 26	0.3599	0.5485	0.2407	0.6362
Row 27	0.1029	0.9796	0.5696	0.8696
Row 28	0.3070	0.8169	0.4015	0.4386
Row 29	0.4453	0.0670	0.3726	0.3257
Row 30	0.2648	0.9977	0.8864	0.0755
Row 31	0.4085	0.2017	0.5406	0.1333
Row 32	0.4861	0.4466	0.3472	0.2486
Row 33	0.5996	0.8639	0.1837	0.7636
Row 34	0.4446	0.3755	0.6901	0.4208
Row 35	0.9616	0.3585	0.0074	0.2867
Row 36	0.5168	0.5752	0.5778	0.0060
Row 37	0.7978	0.0283	0.7998	0.9952
Row 38	0.0561	0.3133	0.1207	0.6922
Row 39	0.5237	0.1488	0.9217	0.2268
Row 40	0.0944	0.7939	0.6252	0.9836
Row 41	0.3179	0.6226	0.4493	0.4277
Row 42	0.7175	0.7267	0.8016	0.6880
Row 43	0.0192	0.4807	0.7610	0.9808
Row 44	0.9923	0.8888	0.4494	0.0645

Continued on next page.

Table 2.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 45	0.3938	0.8529	0.0496	0.0429
Row 46	0.1135	0.6166	0.5899	0.7500
Row 47	0.0654	0.1640	0.1952	0.0431
Row 48	0.8895	0.0549	0.1105	0.1284
Row 49	0.6817	0.8942	0.6597	0.3661
Row 50	0.6690	0.8817	0.2343	0.1903
Row 51	0.4091	0.0874	0.4726	0.1381
Row 52	0.9061	0.9039	0.7439	0.2061
Row 53	0.5282	0.2135	0.5223	0.7846
Row 54	0.6505	0.7404	0.8748	0.2078
Row 55	0.5824	0.8443	0.3242	0.8253
Row 56	0.0151	0.9929	0.4812	0.5010
Row 57	0.7296	0.8420	0.1535	0.4273
Row 58	0.8102	0.8068	0.1832	0.8830
Row 59	0.1650	0.5545	0.1820	0.0791
Row 60	0.5882	0.5750	0.9195	0.8993
Row 61	0.0638	0.5132	0.5994	0.0877
Row 62	0.9916	0.8032	0.0564	0.3218
Row 63	0.5555	0.4078	0.7056	0.9225
Row 64	0.8680	0.5577	0.2992	0.0941
Row 65	0.2939	0.7801	0.7039	0.7295
Row 66	0.0829	0.6756	0.5386	0.0644
Row 67	0.3868	0.4199	0.0308	0.5947
Row 68	0.0943	0.2663	0.0379	0.0887
Row 69	0.0050	0.1396	0.8348	0.2830
Row 70	0.9585	0.8018	0.4472	0.9477
Row 71	0.8153	0.2659	0.7030	0.4096
Row 72	0.7532	0.4214	0.3914	0.2360
Row 73	0.6419	0.2074	0.7386	0.0653
Row 74	0.4215	0.7004	0.3193	0.9282
Row 75	0.1307	0.8242	0.1305	0.8925
Row 76	0.5812	0.6879	0.4844	0.0464
Row 77	0.1080	0.5293	0.2700	0.4844
Row 78	0.3073	0.7945	0.8300	0.3479
Row 79	0.4777	0.5842	0.2233	0.3206
Row 80	0.7218	0.7687	0.0432	0.7268

Continued on next page.

Table 2.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 81	0.1427	0.8696	0.7573	0.1263
Row 82	0.0244	0.6493	0.6750	0.9651
Row 83	0.1925	0.4131	0.3064	0.0508
Row 84	0.8678	0.3827	0.7732	0.3896
Row 85	0.6830	0.0868	0.0773	0.1712
Row 86	0.2699	0.5507	0.1200	0.4458
Row 87	0.3873	0.8615	0.0624	0.4357
Row 88	0.0610	0.0065	0.1505	0.0287
Row 89	0.3380	0.6846	0.1305	0.8998
Row 90	0.4337	0.2892	0.9326	0.7977
Row 91	0.7618	0.7254	0.6185	0.5718
Row 92	0.2404	0.2312	0.6645	0.7351
Row 93	0.8908	0.4011	0.6728	0.4192
Row 94	0.7596	0.5054	0.3343	0.1696
Row 95	0.9736	0.2894	0.8395	0.7554
Row 96	0.2555	0.3570	0.6331	0.3460
Row 97	0.5865	0.8620	0.9528	0.8383
Row 98	0.1753	0.9843	0.5822	0.7130
Row 99	0.2085	0.7513	0.4976	0.6609
Row 100	0.8550	0.6317	0.2716	0.3482
Row 101	0.0003	0.2699	0.1657	0.9740
Row 102	0.8108	0.7631	0.4779	0.7736
Row 103	0.1700	0.7518	0.6194	0.2642
Row 104	0.9089	0.7737	0.1760	0.1838
Row 105	0.2693	0.6957	0.8645	0.7214
Row 106	0.7675	0.7649	0.1831	0.5527
Row 107	0.6605	0.6763	0.6069	0.6509
Row 108	0.9355	0.8627	0.1932	0.1369
Row 109	0.2459	0.2674	0.5147	0.3251
Row 110	0.1111	0.9926	0.6565	0.3905
Row 111	0.3883	0.7516	0.0597	0.2444
Row 112	0.3873	0.8884	0.8992	0.4628
Row 113	0.7374	0.3370	0.2922	0.8778
Row 114	0.9644	0.3383	0.7343	0.4642
Row 115	0.8793	0.1624	0.6602	0.6129
Row 116	0.7910	0.7928	0.9132	0.4582

Continued on next page.

Table 2.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 117	0.4158	0.6584	0.0655	0.3760
Row 118	0.6719	0.8505	0.2902	0.3726
Row 119	0.6456	0.6116	0.7580	0.3331
Row 120	0.9372	0.5338	0.9066	0.8391
Row 121	0.1427	0.6179	0.7094	0.5079
Row 122	0.1748	0.9789	0.1452	0.5829
Row 123	0.7514	0.2678	0.7714	0.1895
Row 124	0.4058	0.7714	0.4468	0.5559
Row 125	0.0799	0.6205	0.4477	0.3788
Row 126	0.3297	0.7600	0.5485	0.8005
Row 127	0.8873	0.3812	0.9346	0.4062
Row 128	0.5164	0.9326	0.8897	0.6300
Row 129	0.1876	0.8342	0.5704	0.9817
Row 130	0.3990	0.2170	0.8709	0.4717
Row 131	0.4454	0.3671	0.2185	0.9753
Row 132	0.8951	0.9321	0.3854	0.4805
Row 133	0.3442	0.8316	0.8667	0.6898
Row 134	0.0586	0.2090	0.3720	0.1668
Row 135	0.1312	0.5375	0.6314	0.2907
Row 136	0.5138	0.7588	0.2177	0.7461
Row 137	0.4966	0.1501	0.3993	0.0631
Row 138	0.7154	0.8785	0.8362	0.5782
Row 139	0.6265	0.2019	0.9703	0.2705
Row 140	0.5248	0.5235	0.5018	0.9854
Row 141	0.2711	0.5263	0.8829	0.8525
Row 142	0.1335	0.8354	0.0190	0.3996
Row 143	0.7644	0.3912	0.8849	0.7440
Row 144	0.4358	0.2065	0.4528	0.8955
Row 145	0.9038	0.0718	0.7912	0.5230
Row 146	0.1919	0.7559	0.2908	0.2352
Row 147	0.6801	0.3179	0.8315	0.7988
Row 148	0.7810	0.3397	0.5245	0.8478
Row 149	0.1458	0.1098	0.2659	0.2319
Row 150	0.7207	0.1931	0.2071	0.0241

Note: At the very end, you can add some notes to the table.

References

- Andersen, Steffen, Glenn W. Harrison, Morten I. Lau, and E. Elisabet Rutström.** 2008. "Eliciting Risk and Time Preferences." *Econometrica* 76 (3): 583–618. <https://doi.org/10.1111/j.1468-0262.2008.00848.x>. [49, 50, 79]
- Andreoni, James, and Charles Sprenger.** 2012. "Estimating Time Preferences from Convex Budgets." *American Economic Review* 102 (7): 3333–56. <https://doi.org/10.1257/aer.102.7.3333>. [49, 79]
- Attema, Arthur E., Han Bleichrodt, Yu Gao, Zhenxing Huang, and Peter P. Wakker.** 2016. "Measuring Discounting without Measuring Utility." *American Economic Review* 106 (6): 1476–94. <https://doi.org/10.1257/aer.20150208>. [80]
- Balakrishnan, Uttara, Johannes Haushofer, and Pamela Jakiela.** 2016. "How Soon Is Now? Evidence of Present Bias from Convex Time Budget Experiments." IZA Discussion Paper, IZA Discussion Paper Series. <http://ftp.iza.org/dp9653.pdf>. [49]
- Benartzi, Shlomo, Alessandro Previtero, and Richard H. Thaler.** 2011. "Annuitization Puzzles." *Journal of Economic Perspectives* 25 (4): 143–64. <https://doi.org/10.1257/jep.25.4.143>. [80]
- Beute, Femke, and Yvonne A. W. de Kort.** 2012. "Always Look on the Bright Side of Life: Ego-Replenishing Effects of Daylight versus Artificial Light." In *Proceedings of Experiencing Light 2012: International Conference on the Effects of Light on Wellbeing*, edited by Y. A. W. de Kort, M. P. J. Aarts, F. Beute, A. Haans, W. A. IJsselstein, D. Lakens, K. C. H. J. Smolders, and L. van Rijswijk, 1–4. Eindhoven, The Netherlands: Eindhoven University of Technology. <http://2012.experiencinglight.nl/doc/41.pdf>. [80]
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer.** 2012. "Salience Theory of Choice Under Risk." *Quarterly Journal of Economics* 127 (3): 1243–85. <https://doi.org/10.1093/qje/qjs018>. [50, 81, 82]
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer.** 2013. "Salience and Consumer Choice." *Journal of Political Economy* 121 (5): 803–43. <https://doi.org/10.1086/673885>. [50, 81]
- Bursztyn, Leonardo, Benjamin Handel, Rafael Jiménez-Durán, and Christopher Roth.** 2025. "When Product Markets Become Collective Traps: The Case of Social Media." *American Economic Review* 115 (12): 4105–36. <https://doi.org/10.1257/aer.20231468>. [49]
- Bushong, Benjamin, Matthew Rabin, and Joshua Schwartzstein.** 2016. "A Model of Relative Thinking." Working paper. Cambridge, MA, USA: Harvard University. <http://people.hbs.edu/jschwartzstein/RelativeThinking.pdf>. [50]
- Davidoff, Thomas, Jeffrey R. Brown, and Peter A. Diamond.** 2005. "Annuities and Individual Welfare." *American Economic Review* 95 (5): 1573–90. <https://doi.org/10.1257/000282805775014281>. [80]
- Dertwinkel-Kalt, Markus, Holger Gerhardt, Gerhard Riener, Frederik Schwerter, and Louis Strang.** 2017. "Concentration Bias in Intertemporal Choice." Working paper. Bonn, Germany, et al.: University of Bonn et al. https://www.dropbox.com/s/dv20mcu0qkygmjz/Concentration_Bias_in_Intertemporal_Choice.pdf. [62, 64, 83–85]
- Gabaix, Xavier.** 2014. "A Sparsity-Based Model of Bounded Rationality." *Quarterly Journal of Economics* 129 (4): 1661–710. <https://doi.org/10.1093/qje/qju024>. [50]
- Gerhardt, Holger, Hannah Schildberg-Hörisch, and Jana Willrodt.** 2017. "Does self-control depletion affect risk attitudes?" *European Economic Review* 100: 463–87. <https://doi.org/10.1016/j.eurocorev.2017.09.004>. [86, 87]

- Halevy, Yoram.** 2014. "Some Comments on the Use of Monetary and Primary Rewards in the Measurement of Time Preferences." Working paper. University of British Columbia. http://faculty.arts.ubc.ca/yhalevy/monetary_primary.pdf. [67, 78, 79]
- Harrison, Glenn W., and E. Elisabet Rutström.** 2008. "Risk Aversion in the Laboratory." Chap. 1 in *Risk Aversion in Experiments*, edited by Glenn W. Harrison and James C. Cox, 12: 41–196. Research in Experimental Economics. Bingley, UK: Emerald Group. [https://doi.org/10.1016/S0193-2306\(08\)00003-3](https://doi.org/10.1016/S0193-2306(08)00003-3). [80]
- Henrich, Joseph, Robert Boyd, Samuel Bowles, Colin Camerer, Ernst Fehr, Herbert Gintis, Richard McElreath, et al.** 2005. "'Economic man' in cross-cultural perspective: Behavioral experiments in 15 small-scale societies." *Behavioral and Brain Sciences* 28 (6): 795–815, discussion 815–55. <https://doi.org/10.1017/S0140525X05000142>. [82]
- Kagel, John H., and Alvin E. Roth, eds.** 2016. *The Handbook of Experimental Economics*. Edited by John H. Kagel and Alvin E. Roth. Vol. 2. Princeton, NJ, USA: Princeton University Press. [80]
- Kőszegi, Botond, and Adam Szeidl.** 2013. "A Model of Focusing in Economic Choice." *Quarterly Journal of Economics* 128 (1): 53–104. <https://doi.org/10.1093/qje/qjs049>. [50, 67, 71, 79]
- Lisi, A. Garrett.** 1995. "A solitary wave solution of the Maxwell–Dirac equations." *Journal of Physics A: Mathematical and General* 28 (18): 5385–92. <https://doi.org/10.1088/0305-4470/28/18/026>. arXiv: [hep-th/9410244](https://arxiv.org/abs/hep-th/9410244). [49]
- Luce, R. Duncan.** 1959. *Individual Choice Behavior: A Theoretical Analysis*. New York, NY, USA: John Wiley & Sons. [80]
- McClure, Samuel M., Keith M. Ericson, David Laibson, George Loewenstein, and Jonathan D. Cohen.** 2007. "Time Discounting for Primary Rewards." *Journal of Neuroscience* 27 (21): 5796–804. <https://doi.org/10.1523/JNEUROSCI.4246-06.2007>. [68, 79]
- McClure, Samuel M., David Laibson, George Loewenstein, and Jonathan D. Cohen.** 2004. "Separate Neural Systems Value Immediate and Delayed Monetary Rewards." *Science* 306 (5695): 503–7. <https://doi.org/10.1126/science.1100907>. [68, 79]
- Samuelson, Paul.** 1937. "A Note on Measurement of Utility." *Review of Economic Studies* 4 (2): 155–61. <https://doi.org/10.2307/2967612>. [79]
- Sims, Christopher A.** 2003. "Implications of rational inattention." *Journal of Monetary Economics* 50 (3): 665–90. [https://doi.org/10.1016/S0304-3932\(03\)00029-1](https://doi.org/10.1016/S0304-3932(03)00029-1). [50]
- Sullivan, Paul.** 2016. *Fresh Thinking on Saving*. New York Times (2016): F2. <http://nytimes.com/2016/03/27/your-money/getting-workers-to-save-more-for-retirement.html>. [49]
- Taubinsky, Dmitry.** 2014. "From Intentions to Actions: A Model and Experimental Evidence of Inattentive Choice." Working paper. Hanover, NH, USA: Dartmouth College. <https://docs.google.com/viewer?a=v&pid=sites&srcid=ZGVmYXVsdGRvbWFpbXkbWl0cnlwYXBicnN8Z3g6NmIzYWw0MwIwNTc4MjkwNQ>. [50]
- Vosgerau, Joachim, Sabrina Bruyneel, Ravi Dhar, and Klaus Wertenbroch.** 2008. "Ego Depletion and Cognitive Load: Same or Different Constructs?" In *Advances in Consumer Research*, 35: 217–20. Association for Consumer Research. <http://www.acrwebsite.org/search/view-conference-proceedings.aspx?id=13549>. [80]
- Warner, John T., and Saul Pleeter.** 2001. "The Personal Discount Rate: Evidence from Military Downsizing Programs." *American Economic Review* 91 (1): 33–53. <https://doi.org/10.1257/aer.91.1.33>. [80]
- Yaari, Menahem E.** 1965. "Uncertain Lifetime, Life Insurance, and the Theory of the Consumer." *Review of Economic Studies* 32 (2): 137–50. <https://doi.org/10.2307/2296058>. [80]

Chapter 3

Math Tests

3.1 Math Test **Serif**

3.1.1 Overview **Serif**

Default: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma; \sigma_{\epsilon}, c^{\alpha}$

mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathrm: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathup: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathit: $\alpha\alpha\alpha b\beta G\text{`}`\epsilon\epsilon\theta\vartheta P^{\circ}\sigma$

mathbf: **$\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$**

mathbfit: **$\alpha\alpha b\beta G\text{`}`\epsilon\epsilon\theta\vartheta P^{\circ}\sigma$**

mathbfup: **$\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$**

Default: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma; \sigma_{\epsilon}, c^{\alpha}$

mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathrm: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathup: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathit: $\alpha\alpha\alpha b\beta G\text{`}`\epsilon\epsilon\theta\vartheta P^{\circ}\sigma$

mathbf: **$\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$**

mathbfit: **$\alpha\alpha\alpha b\beta G\text{`}`\epsilon\epsilon\theta\vartheta P^{\circ}\sigma$**

mathbfup: **$\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$**

Default: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma; \sigma_{\epsilon}, c^{\alpha}$

mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathrm: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathup: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathit: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathbf: **$\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$**

mathbfit: **$\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$**

mathbfup: **$\alpha\alpha\alpha b\beta G\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$**

Default: $\alpha\alpha\alpha\beta\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma; \sigma_{\epsilon}, c^{\alpha}$

mathnormal: $\alpha\alpha\alpha\beta\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathrm: $\alpha\alpha\alpha\beta\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathup: $\alpha\alpha\alpha\beta\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathit: $\alpha\alpha\alpha\beta\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathbf: $\alpha\alpha\alpha\beta\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathbf{it}: $\alpha\alpha\alpha\beta\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

mathbf{fup}: $\alpha\alpha\alpha\beta\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

3.1.2 Formulas **Serif**

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega,$
 $f, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi,$
 $\omega, f, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, f, A, B, \Gamma, \Delta, E, Z, H,$
 $\Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,$

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, f, A, B, \Gamma, \Delta, E, Z, H,$
 $\Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{\nu \rightarrow \infty} v(\nu) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l I | l I /, i j, rn m, θ Θ , ϕ ψ , --

Latin vs. Greek: a α , d δ , e ϵ , i ι , k κ , n η , o σ , p ρ , β β , u υ , v ν , w ω , x χ , y γ ,
 A Δ , Λ , O Θ , Ω , T Γ , Y Υ .

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{\nu \rightarrow \infty} v(\nu) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l I | l I /, i j, rn m, θ Θ , ϕ ψ , --

Latin vs. Greek: $a \alpha, d \delta, e \epsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l l | l l /, i j, rn m, $\theta \Theta$, $\phi \psi$, --

Latin vs. Greek: $a \alpha, d \delta, e \epsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l l | l l /, i j, rn m, $\theta \Theta$, $\phi \psi$, --

Latin vs. Greek: $a \alpha, d \delta, e \epsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

3.1.3 Math Alphabets Serif

Default

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,

A, B, Γ , Δ , E, Z, H, Θ , I, K, Λ , M, N, Ξ , O, Π , P, Σ , T, Υ , Φ , X, Ψ , Ω ,

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi$,

Math Normal (`\mathnormal`)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,

A, B, Γ , Δ , E, Z, H, Θ , I, K, Λ , M, N, Ξ , O, Π , P, Σ , T, Υ , Φ , X, Ψ , Ω ,

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi$,

Math Italic (`\mathit`)

$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$
 $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,$
 $a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,$
 $A, B, \text{`}, \text{'}, E, Z, H, \text{^}, I, K, \text{~}, M, N, \text{``}, O, \text{''}, P, \text{^{\circ}}, T, \text{^{\vee}}, \text{~}, X, \text{^-}, \text{'},$
 $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$

Math Roman (`\mathrm`)

$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$
 $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,$
 $a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,$
 $A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,$
 $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$

Math Bold (`\mathbf`)

$\mathbf{0, 1, 2, 3, 4, 5, 6, 7, 8, 9,}$
 $\mathbf{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,}$
 $\mathbf{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,}$
 $\mathbf{A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,}$
 $\mathbf{\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,}$

Caligraphic (`\mathcal`)

$\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{G}, \mathcal{H}, \mathcal{I}, \mathcal{J}, \mathcal{K}, \mathcal{L}, \mathcal{M}, \mathcal{N}, \mathcal{O}, \mathcal{P}, \mathcal{Q}, \mathcal{R}, \mathcal{S}, \mathcal{T}, \mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z},$

Script (`\mathscr`)

$\mathscr{A}, \mathscr{B}, \mathscr{C}, \mathscr{D}, \mathscr{E}, \mathscr{F}, \mathscr{G}, \mathscr{H}, \mathscr{I}, \mathscr{J}, \mathscr{K}, \mathscr{L}, \mathscr{M}, \mathscr{N}, \mathscr{O}, \mathscr{P}, \mathscr{Q}, \mathscr{R}, \mathscr{S}, \mathscr{T}, \mathscr{U}, \mathscr{V}, \mathscr{W}, \mathscr{X}, \mathscr{Y}, \mathscr{Z},$

Fraktur (`\mathfrak`)

$\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{I}, \mathfrak{J}, \mathfrak{K}, \mathfrak{L}, \mathfrak{M}, \mathfrak{N}, \mathfrak{O}, \mathfrak{P}, \mathfrak{Q}, \mathfrak{R}, \mathfrak{S}, \mathfrak{T}, \mathfrak{U}, \mathfrak{V}, \mathfrak{W}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z},$
 $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{e}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n}, \mathfrak{o}, \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}, \mathfrak{t}, \mathfrak{u}, \mathfrak{v}, \mathfrak{w}, \mathfrak{x}, \mathfrak{y}, \mathfrak{z},$

Blackboard Bold (`\mathbb`)

$\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}, \mathbb{E}, \mathbb{F}, \mathbb{G}, \mathbb{H}, \mathbb{I}, \mathbb{J}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \mathbb{O}, \mathbb{P}, \mathbb{Q}, \mathbb{R}, \mathbb{S}, \mathbb{T}, \mathbb{U}, \mathbb{V}, \mathbb{W}, \mathbb{X}, \mathbb{Y}, \mathbb{Z},$

3.1.4 Character Sidebearings Serif

Default

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |\Upsilon| + |\Phi| + |X| + |\Psi| + |\Omega| +$
 $|\alpha| + |\beta| + |\gamma| + |\delta| + |\epsilon| + |\zeta| + |\eta| + |\theta| + |\iota| + |\kappa| + |\lambda| + |\mu| +$
 $|\nu| + |\xi| + |\omicron| + |\pi| + |\rho| + |\sigma| + |\tau| + |\upsilon| + |\phi| + |\chi| + |\psi| + |\omega| +$
 $|\varepsilon| + |\vartheta| + |\varpi| + |\varrho| + |\varsigma| + |\varphi| +$

Math Roman (`\mathrm`)

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |\Upsilon| + |\Phi| + |X| + |\Psi| + |\Omega| +$

Math Bold (`\mathbf`)

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |\Upsilon| + |\Phi| + |X| + |\Psi| + |\Omega| +$

Math Calligraphic (`\mathcal`)

$|\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{D}| + |\mathcal{E}| + |\mathcal{F}| + |\mathcal{G}| + |\mathcal{H}| + |\mathcal{I}| + |\mathcal{J}| + |\mathcal{K}| + |\mathcal{L}| + |\mathcal{M}| +$
 $|\mathcal{N}| + |\mathcal{O}| + |\mathcal{P}| + |\mathcal{Q}| + |\mathcal{R}| + |\mathcal{S}| + |\mathcal{T}| + |\mathcal{U}| + |\mathcal{V}| + |\mathcal{W}| + |\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}| +$

3.1.5 Superscript Positioning **Serif**

Default

$$\begin{aligned}
 &A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 + \\
 &N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 + \\
 &a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 + \\
 &n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 + \\
 &A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 + \\
 &N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 + \\
 &\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \epsilon^2 + \zeta^2 + \eta^2 + \theta^2 + \iota^2 + \kappa^2 + \lambda^2 + \mu^2 + \\
 &\nu^2 + \xi^2 + o^2 + \pi^2 + \rho^2 + \sigma^2 + \tau^2 + v^2 + \phi^2 + \chi^2 + \psi^2 + \omega^2 + \\
 &\varepsilon^2 + \vartheta^2 + \varpi^2 + \varrho^2 + \varsigma^2 + \varphi^2 +
 \end{aligned}$$

Math Roman (`\mathrm`)

$$\begin{aligned}
 &A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 + \\
 &N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 + \\
 &a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 + \\
 &n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 + \\
 &A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 + \\
 &N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 +
 \end{aligned}$$

Math Bold (`\mathbf`)

$$\begin{aligned}
 &\mathbf{A}^2 + \mathbf{B}^2 + \mathbf{C}^2 + \mathbf{D}^2 + \mathbf{E}^2 + \mathbf{F}^2 + \mathbf{G}^2 + \mathbf{H}^2 + \mathbf{I}^2 + \mathbf{J}^2 + \mathbf{K}^2 + \mathbf{L}^2 + \mathbf{M}^2 + \\
 &\mathbf{N}^2 + \mathbf{O}^2 + \mathbf{P}^2 + \mathbf{Q}^2 + \mathbf{R}^2 + \mathbf{S}^2 + \mathbf{T}^2 + \mathbf{U}^2 + \mathbf{V}^2 + \mathbf{W}^2 + \mathbf{X}^2 + \mathbf{Y}^2 + \mathbf{Z}^2 + \\
 &\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2 + \mathbf{d}^2 + \mathbf{e}^2 + \mathbf{f}^2 + \mathbf{g}^2 + \mathbf{h}^2 + \mathbf{i}^2 + \mathbf{j}^2 + \mathbf{k}^2 + \mathbf{l}^2 + \mathbf{m}^2 + \\
 &\mathbf{n}^2 + \mathbf{o}^2 + \mathbf{p}^2 + \mathbf{q}^2 + \mathbf{r}^2 + \mathbf{s}^2 + \mathbf{t}^2 + \mathbf{u}^2 + \mathbf{v}^2 + \mathbf{w}^2 + \mathbf{x}^2 + \mathbf{y}^2 + \mathbf{z}^2 + \\
 &\mathbf{A}^2 + \mathbf{B}^2 + \mathbf{\Gamma}^2 + \mathbf{\Delta}^2 + \mathbf{E}^2 + \mathbf{Z}^2 + \mathbf{H}^2 + \mathbf{\Theta}^2 + \mathbf{I}^2 + \mathbf{K}^2 + \mathbf{\Lambda}^2 + \mathbf{M}^2 + \\
 &\mathbf{N}^2 + \mathbf{\Xi}^2 + \mathbf{O}^2 + \mathbf{\Pi}^2 + \mathbf{P}^2 + \mathbf{\Sigma}^2 + \mathbf{T}^2 + \mathbf{\Upsilon}^2 + \mathbf{\Phi}^2 + \mathbf{X}^2 + \mathbf{\Psi}^2 + \mathbf{\Omega}^2 +
 \end{aligned}$$

Math Calligraphic (`\mathcal`)

$$\begin{aligned}
 &\mathcal{A}^2 + \mathcal{B}^2 + \mathcal{C}^2 + \mathcal{D}^2 + \mathcal{E}^2 + \mathcal{F}^2 + \mathcal{G}^2 + \mathcal{H}^2 + \mathcal{I}^2 + \mathcal{J}^2 + \mathcal{K}^2 + \mathcal{L}^2 + \mathcal{M}^2 + \\
 &\mathcal{N}^2 + \mathcal{O}^2 + \mathcal{P}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{S}^2 + \mathcal{T}^2 + \mathcal{U}^2 + \mathcal{V}^2 + \mathcal{W}^2 + \mathcal{X}^2 + \mathcal{Y}^2 + \mathcal{Z}^2 +
 \end{aligned}$$

3.1.6 Subscript Positioning Serif

Default

$$\begin{aligned}
 &A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i + \\
 &N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i + \\
 &a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i + \\
 &n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i + \\
 &A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i + \\
 &N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i + \\
 &\alpha_i + \beta_i + \gamma_i + \delta_i + \epsilon_i + \zeta_i + \eta_i + \theta_i + \iota_i + \kappa_i + \lambda_i + \mu_i + \\
 &\nu_i + \xi_i + o_i + \pi_i + \rho_i + \sigma_i + \tau_i + \upsilon_i + \phi_i + \chi_i + \psi_i + \omega_i + \\
 &\varepsilon_i + \vartheta_i + \varpi_i + \varrho_i + \varsigma_i + \varphi_i +
 \end{aligned}$$

Math Roman (`\mathrm`)

$$\begin{aligned}
 &A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i + \\
 &N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i + \\
 &a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i + \\
 &n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i + \\
 &A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i + \\
 &N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i +
 \end{aligned}$$

Math Bold (`\mathbf`)

$$\begin{aligned}
 &\mathbf{A_i} + \mathbf{B_i} + \mathbf{C_i} + \mathbf{D_i} + \mathbf{E_i} + \mathbf{F_i} + \mathbf{G_i} + \mathbf{H_i} + \mathbf{I_i} + \mathbf{J_i} + \mathbf{K_i} + \mathbf{L_i} + \mathbf{M_i} + \\
 &\mathbf{N_i} + \mathbf{O_i} + \mathbf{P_i} + \mathbf{Q_i} + \mathbf{R_i} + \mathbf{S_i} + \mathbf{T_i} + \mathbf{U_i} + \mathbf{V_i} + \mathbf{W_i} + \mathbf{X_i} + \mathbf{Y_i} + \mathbf{Z_i} + \\
 &\mathbf{a_i} + \mathbf{b_i} + \mathbf{c_i} + \mathbf{d_i} + \mathbf{e_i} + \mathbf{f_i} + \mathbf{g_i} + \mathbf{h_i} + \mathbf{i_i} + \mathbf{j_i} + \mathbf{k_i} + \mathbf{l_i} + \mathbf{m_i} + \\
 &\mathbf{n_i} + \mathbf{o_i} + \mathbf{p_i} + \mathbf{q_i} + \mathbf{r_i} + \mathbf{s_i} + \mathbf{t_i} + \mathbf{u_i} + \mathbf{v_i} + \mathbf{w_i} + \mathbf{x_i} + \mathbf{y_i} + \mathbf{z_i} + \\
 &\mathbf{A_i} + \mathbf{B_i} + \mathbf{\Gamma_i} + \mathbf{\Delta_i} + \mathbf{E_i} + \mathbf{Z_i} + \mathbf{H_i} + \mathbf{\Theta_i} + \mathbf{I_i} + \mathbf{K_i} + \mathbf{\Lambda_i} + \mathbf{M_i} + \\
 &\mathbf{N_i} + \mathbf{\Xi_i} + \mathbf{O_i} + \mathbf{\Pi_i} + \mathbf{P_i} + \mathbf{\Sigma_i} + \mathbf{T_i} + \mathbf{\Upsilon_i} + \mathbf{\Phi_i} + \mathbf{X_i} + \mathbf{\Psi_i} + \mathbf{\Omega_i} +
 \end{aligned}$$

Math Calligraphic (`\mathcal`)

$$\begin{aligned}
 &\mathcal{A}_i + \mathcal{B}_i + \mathcal{C}_i + \mathcal{D}_i + \mathcal{E}_i + \mathcal{F}_i + \mathcal{G}_i + \mathcal{H}_i + \mathcal{I}_i + \mathcal{J}_i + \mathcal{K}_i + \mathcal{L}_i + \mathcal{M}_i + \\
 &\mathcal{N}_i + \mathcal{O}_i + \mathcal{P}_i + \mathcal{Q}_i + \mathcal{R}_i + \mathcal{S}_i + \mathcal{T}_i + \mathcal{U}_i + \mathcal{V}_i + \mathcal{W}_i + \mathcal{X}_i + \mathcal{Y}_i + \mathcal{Z}_i +
 \end{aligned}$$

3.1.7 Accent Positioning **Serif**

Default

$\hat{O} + \hat{1} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{I} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{\Upsilon} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$
 $\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{\iota} + \hat{\kappa} + \hat{\lambda} + \hat{\mu} +$
 $\hat{\nu} + \hat{\xi} + \hat{\omicron} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{\upsilon} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$
 $\hat{\varepsilon} + \hat{\vartheta} + \hat{\varpi} + \hat{\varrho} + \hat{\varsigma} + \hat{\varphi} +$

Math Italic (`\mathit`)

$\hat{O} + \hat{1} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} + \hat{\ell} + \hat{\wp} + \hat{\imath} + \hat{j} + \hat{\tilde{i}}$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{^} + \hat{^} + \hat{E} + \hat{Z} + \hat{H} + \hat{^} + \hat{I} + \hat{K} + \hat{^} + \hat{M} +$
 $\hat{N} + \hat{^} + \hat{O} + \hat{^} + \hat{P} + \hat{^} + \hat{T} + \hat{^} + \hat{^} + \hat{X} + \hat{^} + \hat{^} +$
 $\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{\iota} + \hat{\kappa} + \hat{\lambda} + \hat{\mu} +$
 $\hat{\nu} + \hat{\xi} + \hat{\omicron} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{\upsilon} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$
 $\hat{\varepsilon} + \hat{\vartheta} + \hat{\varpi} + \hat{\varrho} + \hat{\varsigma} + \hat{\varphi} +$

Math Roman (`\mathrm`)

$\hat{O} + \hat{1} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{I} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{\Upsilon} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$

Math Bold (`\mathbf`)

$\hat{\mathbf{0}} + \hat{\mathbf{1}} + \hat{\mathbf{2}} + \hat{\mathbf{3}} + \hat{\mathbf{4}} + \hat{\mathbf{5}} + \hat{\mathbf{6}} + \hat{\mathbf{7}} + \hat{\mathbf{8}} + \hat{\mathbf{9}} +$
 $\hat{\mathbf{A}} + \hat{\mathbf{B}} + \hat{\mathbf{C}} + \hat{\mathbf{D}} + \hat{\mathbf{E}} + \hat{\mathbf{F}} + \hat{\mathbf{G}} + \hat{\mathbf{H}} + \hat{\mathbf{I}} + \hat{\mathbf{J}} + \hat{\mathbf{K}} + \hat{\mathbf{L}} + \hat{\mathbf{M}} +$
 $\hat{\mathbf{N}} + \hat{\mathbf{O}} + \hat{\mathbf{P}} + \hat{\mathbf{Q}} + \hat{\mathbf{R}} + \hat{\mathbf{S}} + \hat{\mathbf{T}} + \hat{\mathbf{U}} + \hat{\mathbf{V}} + \hat{\mathbf{W}} + \hat{\mathbf{X}} + \hat{\mathbf{Y}} + \hat{\mathbf{Z}} +$
 $\hat{\mathbf{a}} + \hat{\mathbf{b}} + \hat{\mathbf{c}} + \hat{\mathbf{d}} + \hat{\mathbf{e}} + \hat{\mathbf{f}} + \hat{\mathbf{g}} + \hat{\mathbf{h}} + \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}} + \hat{\mathbf{l}} + \hat{\mathbf{m}} +$
 $\hat{\mathbf{n}} + \hat{\mathbf{o}} + \hat{\mathbf{p}} + \hat{\mathbf{q}} + \hat{\mathbf{r}} + \hat{\mathbf{s}} + \hat{\mathbf{t}} + \hat{\mathbf{u}} + \hat{\mathbf{v}} + \hat{\mathbf{w}} + \hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}} +$
 $\hat{\mathbf{A}} + \hat{\mathbf{B}} + \hat{\mathbf{I}} + \hat{\mathbf{\Delta}} + \hat{\mathbf{E}} + \hat{\mathbf{Z}} + \hat{\mathbf{H}} + \hat{\mathbf{\Theta}} + \hat{\mathbf{I}} + \hat{\mathbf{K}} + \hat{\mathbf{\Lambda}} + \hat{\mathbf{M}} +$
 $\hat{\mathbf{N}} + \hat{\mathbf{\Xi}} + \hat{\mathbf{O}} + \hat{\mathbf{\Pi}} + \hat{\mathbf{P}} + \hat{\mathbf{\Sigma}} + \hat{\mathbf{T}} + \hat{\mathbf{\Upsilon}} + \hat{\mathbf{\Phi}} + \hat{\mathbf{X}} + \hat{\mathbf{\Psi}} + \hat{\mathbf{\Omega}} +$

Math Calligraphic (`\mathcal`)

$\hat{\mathcal{A}} + \hat{\mathcal{B}} + \hat{\mathcal{C}} + \hat{\mathcal{D}} + \hat{\mathcal{E}} + \hat{\mathcal{F}} + \hat{\mathcal{G}} + \hat{\mathcal{H}} + \hat{\mathcal{I}} + \hat{\mathcal{J}} + \hat{\mathcal{K}} + \hat{\mathcal{L}} + \hat{\mathcal{M}} +$
 $\hat{\mathcal{N}} + \hat{\mathcal{O}} + \hat{\mathcal{P}} + \hat{\mathcal{Q}} + \hat{\mathcal{R}} + \hat{\mathcal{S}} + \hat{\mathcal{T}} + \hat{\mathcal{U}} + \hat{\mathcal{V}} + \hat{\mathcal{W}} + \hat{\mathcal{X}} + \hat{\mathcal{Y}} + \hat{\mathcal{Z}} +$

3.1.8 Differentials **Serif**

$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$
 $dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$
 $da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$
 $dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$
 $d\alpha + d\beta + d\gamma + d\delta + d\epsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$
 $d\nu + d\xi + d\omicron + d\pi + d\rho + d\sigma + d\tau + d\upsilon + d\phi + d\chi + d\psi + d\omega +$
 $d\varepsilon + d\vartheta + d\varpi + d\rho + d\varsigma + d\varphi +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$

$$\begin{aligned}
& dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM + \\
& dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ + \\
& da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm + \\
& dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz + \\
& dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM + \\
& dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega + \\
& d\alpha + d\beta + d\gamma + d\delta + d\epsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu + \\
& d\nu + d\xi + do + d\pi + d\rho + d\sigma + d\tau + dv + d\phi + d\chi + d\psi + d\omega + \\
& d\varepsilon + d\vartheta + d\varpi + d\rho + d\varsigma + d\varphi + \\
& dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM + \\
& dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +
\end{aligned}$$

$$\begin{aligned}
& \partial A + \partial B + \partial C + \partial D + \partial E + \partial F + \partial G + \partial H + \partial I + \partial J + \partial K + \partial L + \partial M + \\
& \partial N + \partial O + \partial P + \partial Q + \partial R + \partial S + \partial T + \partial U + \partial V + \partial W + \partial X + \partial Y + \partial Z + \\
& \partial a + \partial b + \partial c + \partial d + \partial e + \partial f + \partial g + \partial h + \partial i + \partial j + \partial k + \partial l + \partial m + \\
& \partial n + \partial o + \partial p + \partial q + \partial r + \partial s + \partial t + \partial u + \partial v + \partial w + \partial x + \partial y + \partial z + \\
& \partial A + \partial B + \partial \Gamma + \partial \Delta + \partial E + \partial Z + \partial H + \partial \Theta + \partial I + \partial K + \partial \Lambda + \partial M + \\
& \partial N + \partial \Xi + \partial O + \partial \Pi + \partial P + \partial \Sigma + \partial T + \partial \Upsilon + \partial \Phi + \partial X + \partial \Psi + \partial \Omega + \\
& \partial \alpha + \partial \beta + \partial \gamma + \partial \delta + \partial \epsilon + \partial \zeta + \partial \eta + \partial \theta + \partial \iota + \partial \kappa + \partial \lambda + \partial \mu + \\
& \partial \nu + \partial \xi + \partial o + \partial \pi + \partial \rho + \partial \sigma + \partial \tau + \partial v + \partial \phi + \partial \chi + \partial \psi + \partial \omega + \\
& \partial \varepsilon + \partial \vartheta + \partial \varpi + \partial \rho + \partial \varsigma + \partial \varphi + \\
& \partial A + \partial B + \partial \Gamma + \partial \Delta + \partial E + \partial Z + \partial H + \partial \Theta + \partial I + \partial K + \partial \Lambda + \partial M + \\
& \partial N + \partial \Xi + \partial O + \partial \Pi + \partial P + \partial \Sigma + \partial T + \partial \Upsilon + \partial \Phi + \partial X + \partial \Psi + \partial \Omega +
\end{aligned}$$

3.1.9 Slash Kerning **Serif**

$$\begin{aligned}
& 1/A + 1/B + 1/C + 1/D + 1/E + 1/F + 1/G + 1/H + 1/I + 1/J + 1/K + 1/L + 1/M + \\
& 1/N + 1/O + 1/P + 1/Q + 1/R + 1/S + 1/T + 1/U + 1/V + 1/W + 1/X + 1/Y + 1/Z + \\
& 1/a + 1/b + 1/c + 1/d + 1/e + 1/f + 1/g + 1/h + 1/i + 1/j + 1/k + 1/l + 1/m + \\
& 1/n + 1/o + 1/p + 1/q + 1/r + 1/s + 1/t + 1/u + 1/v + 1/w + 1/x + 1/y + 1/z + \\
& 1/A + 1/B + 1/\Gamma + 1/\Delta + 1/E + 1/Z + 1/H + 1/\Theta + 1/I + 1/K + 1/\Lambda + 1/M + \\
& 1/N + 1/\Xi + 1/O + 1/\Pi + 1/P + 1/\Sigma + 1/T + 1/\Upsilon + 1/\Phi + 1/X + 1/\Psi + 1/\Omega + \\
& 1/\alpha + 1/\beta + 1/\gamma + 1/\delta + 1/\epsilon + 1/\zeta + 1/\eta + 1/\theta + 1/\iota + 1/\kappa + 1/\lambda + 1/\mu + \\
& 1/\nu + 1/\xi + 1/o + 1/\pi + 1/\rho + 1/\sigma + 1/\tau + 1/v + 1/\phi + 1/\chi + 1/\psi + 1/\omega + \\
& 1/\varepsilon + 1/\vartheta + 1/\varpi + 1/\rho + 1/\varsigma + 1/\varphi +
\end{aligned}$$

$$\begin{aligned}
& A/2 + B/2 + C/2 + D/2 + E/2 + F/2 + G/2 + H/2 + I/2 + J/2 + K/2 + L/2 + M/2 + \\
& N/2 + O/2 + P/2 + Q/2 + R/2 + S/2 + T/2 + U/2 + V/2 + W/2 + X/2 + Y/2 + Z/2 + \\
& a/2 + b/2 + c/2 + d/2 + e/2 + f/2 + g/2 + h/2 + i/2 + j/2 + k/2 + l/2 + m/2 + \\
& n/2 + o/2 + p/2 + q/2 + r/2 + s/2 + t/2 + u/2 + v/2 + w/2 + x/2 + y/2 + z/2 + \\
& A/2 + B/2 + \Gamma/2 + \Delta/2 + E/2 + Z/2 + H/2 + \Theta/2 + I/2 + K/2 + \Lambda/2 + M/2 + \\
& N/2 + \Xi/2 + O/2 + \Pi/2 + P/2 + \Sigma/2 + T/2 + \Upsilon/2 + \Phi/2 + X/2 + \Psi/2 + \Omega/2 + \\
& \alpha/2 + \beta/2 + \gamma/2 + \delta/2 + \epsilon/2 + \zeta/2 + \eta/2 + \theta/2 + \iota/2 + \kappa/2 + \lambda/2 + \mu/2 + \\
& \nu/2 + \xi/2 + o/2 + \pi/2 + \rho/2 + \sigma/2 + \tau/2 + \upsilon/2 + \phi/2 + \chi/2 + \psi/2 + \omega/2 + \\
& \varepsilon/2 + \vartheta/2 + \varpi/2 + \varrho/2 + \varsigma/2 + \varphi/2 +
\end{aligned}$$

3.1.10 (Big) Operators Serif

$$\begin{aligned}
& \sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n \quad \biguplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n \\
& \sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n \\
& \bigotimes_{i=1}^n x^n \quad \bigoplus_{i=1}^n x^n \quad \bigodot_{i=1}^n x^n \quad \bigwedge_{i=1}^n x^n \quad \bigvee_{i=1}^n x^n \quad \biguplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n
\end{aligned}$$

3.1.11 Radicals Serif

$$\begin{aligned}
& \sqrt{x+y} \quad \sqrt{x^2+y^2} \quad \sqrt{x_i^2+y_j^2} \quad \sqrt{\left(\frac{\cos x}{2}\right)} \quad \sqrt{\left(\frac{\sin x}{2}\right)} \\
& \sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{x+y}}}}}}}
\end{aligned}$$

3.1.12 Over- and Underbraces Serif

$$\overbrace{x} \quad \overbrace{x+y} \quad \overbrace{x^2+y^2} \quad \overbrace{x_i^2+y_j^2} \quad \underbrace{x} \quad \underbrace{x+y} \quad \underbrace{x_i+y_j} \quad \underbrace{x_i^2+y_j^2}$$

3.1.17 Binary Operators Serif

$x \pm y$	<code>\pm</code>	$x \cap y$	<code>\cap</code>	$x \diamond y$	<code>\diamond</code>	$x \oplus y$	<code>\oplus</code>
$x \mp y$	<code>\mp</code>	$x \cup y$	<code>\cup</code>	$x \bigtriangleup y$	<code>\bigtriangleup</code>	$x \ominus y$	<code>\ominus</code>
$x \times y$	<code>\times</code>	$x \uplus y$	<code>\uplus</code>	$x \bigtriangledown y$	<code>\bigtriangledown</code>	$x \otimes y$	<code>\otimes</code>
$x \div y$	<code>\div</code>	$x \sqcap y$	<code>\sqcap</code>	$x \triangleleft y$	<code>\triangleleft</code>	$x \oslash y$	<code>\oslash</code>
$x * y$	<code>\ast</code>	$x \sqcup y$	<code>\sqcup</code>	$x \triangleright y$	<code>\triangleright</code>	$x \odot y$	<code>\odot</code>
$x \star y$	<code>\star</code>	$x \vee y$	<code>\vee</code>	$x \lhd y$	<code>\lhd</code>	$x \bigcirc y$	<code>\bigcirc</code>
$x \circ y$	<code>\circ</code>	$x \wedge y$	<code>\wedge</code>	$x \rhd y$	<code>\rhd</code>	$x \dagger y$	<code>\dagger</code>
$x \bullet y$	<code>\bullet</code>	$x \setminus y$	<code>\setminus</code>	$x \unlhd y$	<code>\unlhd</code>	$x \ddagger y$	<code>\ddagger</code>
$x \cdot y$	<code>\cdot</code>	$x \wr y$	<code>\wr</code>	$x \unrhd y$	<code>\unrhd</code>	$x \S y$	<code>\S</code>
$x + y$	<code>+</code>	$x - y$	<code>-</code>	$x \amalg y$	<code>\amalg</code>	$x \P y$	<code>\P</code>

3.1.18 Relations Serif

$x \leq y$	<code>\leq</code>	$x \geq y$	<code>\geq</code>	$x \equiv y$	<code>\equiv</code>	$x \models y$	<code>\models</code>
$x < y$	<code><</code>	$x > y$	<code>></code>	$x \sim y$	<code>\sim</code>	$x \perp y$	<code>\perp</code>
$x \preceq y$	<code>\preceq</code>	$x \succeq y$	<code>\succeq</code>	$x \simeq y$	<code>\simeq</code>	$x y$	<code> </code>
$x \ll y$	<code>\ll</code>	$x \gg y$	<code>\gg</code>	$x \asymp y$	<code>\asymp</code>	$x \parallel y$	<code>\parallel</code>
$x \subset y$	<code>\subset</code>	$x \supset y$	<code>\supset</code>	$x \approx y$	<code>\approx</code>	$x \bowtie y$	<code>\bowtie</code>
$x \subseteq y$	<code>\subseteq</code>	$x \supseteq y$	<code>\supseteq</code>	$x \cong y$	<code>\cong</code>	$x \Join y$	<code>\Join</code>
$x \sqsubset y$	<code>\sqsubset</code>	$x \sqsupset y$	<code>\sqsupset</code>	$x \neq y$	<code>\neq</code>	$x \smile y$	<code>\smile</code>
$x \sqsubseteq y$	<code>\sqsubseteq</code>	$x \sqsupseteq y$	<code>\sqsupseteq</code>	$x \doteq y$	<code>\doteq</code>	$x \frown y$	<code>\frown</code>
$x \in y$	<code>\in</code>	$x \ni y$	<code>\ni</code>	$x \propto y$	<code>\propto</code>	$x = y$	<code>=</code>
$x \vdash y$	<code>\vdash</code>	$x \dashv y$	<code>\dashv</code>	$x < y$	<code><</code>	$x > y$	<code>></code>
$x : y$	<code>:</code>						

3.1.19 Punctuation Serif

x, y	<code>,</code>	$x; y$	<code>;</code>	$x : y$	<code>\colon</code>	$x \cdot y$	<code>\ldotp</code>	$x \cdot y$	<code>\cdot</code>
--------	----------------	--------	----------------	---------	---------------------	-------------	---------------------	-------------	--------------------

3.1.20 Arrows Serif

$x \leftarrow y$	<code>\leftarrow</code>	$x \longleftarrow y$	<code>\longleftarrow</code>	$x \uparrow y$	<code>\uparrow</code>
$x \Leftarrow y$	<code>\Leftarrow</code>	$x \Longleftarrow y$	<code>\Longleftarrow</code>	$x \Uparrow y$	<code>\Uparrow</code>
$x \rightarrow y$	<code>\rightarrow</code>	$x \longrightarrow y$	<code>\longrightarrow</code>	$x \downarrow y$	<code>\downarrow</code>
$x \Rightarrow y$	<code>\Rightarrow</code>	$x \Longrightarrow y$	<code>\Longrightarrow</code>	$x \Downarrow y$	<code>\Downarrow</code>
$x \leftrightarrow y$	<code>\leftrightarrow</code>	$x \longleftrightarrow y$	<code>\longleftrightarrow</code>	$x \Updownarrow y$	<code>\Updownarrow</code>
$x \Leftrightarrow y$	<code>\Leftrightarrow</code>	$x \Longleftrightarrow y$	<code>\Longleftrightarrow</code>	$x \Updownarrow y$	<code>\Updownarrow</code>
$x \mapsto y$	<code>\mapsto</code>	$x \longmapsto y$	<code>\longmapsto</code>	$x \nearrow y$	<code>\nearrow</code>
$x \hookrightarrow y$	<code>\hookrightarrow</code>	$x \hookrightarrow y$	<code>\hookrightarrow</code>	$x \searrow y$	<code>\searrow</code>
$x \leftharpoonup y$	<code>\leftharpoonup</code>	$x \rightharpoonup y$	<code>\rightharpoonup</code>	$x \swarrow y$	<code>\swarrow</code>
$x \leftharpoonupdown y$	<code>\leftharpoonupdown</code>	$x \rightharpoonupdown y$	<code>\rightharpoonupdown</code>	$x \nwarrow y$	<code>\nwarrow</code>
$x \Rightarrow y$	<code>\rightharpoonup</code>	$x \rightsquigarrow y$	<code>\leadsto</code>		

3.1.21 Miscellaneous Symbols Serif

$x \dots y$	<code>\ldots</code>	$x \cdots y$	<code>\cdots</code>	$x \vdots y$	<code>\vdots</code>	$x \ddots y$	<code>\ddots</code>
$x \aleph y$	<code>\aleph</code>	$x \prime y$	<code>\prime</code>	$x \forall y$	<code>\forall</code>	$x \infty y$	<code>\infty</code>
$x \hbar y$	<code>\hbar</code>	$x \emptyset y$	<code>\emptyset</code>	$x \exists y$	<code>\exists</code>	$x \Box y$	<code>\Box</code>
$x \imath y$	<code>\imath</code>	$x \nabla y$	<code>\nabla</code>	$x \neg y$	<code>\neg</code>	$x \Diamond y$	<code>\Diamond</code>
$x \jmath y$	<code>\jmath</code>	$x \sqrt{y}$	<code>\sqrt</code>	$x \flat y$	<code>\flat</code>	$x \triangle y$	<code>\triangle</code>
$x \ell y$	<code>\ell</code>	$x \top y$	<code>\top</code>	$x \natural y$	<code>\natural</code>	$x \clubsuit y$	<code>\clubsuit</code>
$x \wp y$	<code>\wp</code>	$x \bot y$	<code>\bot</code>	$x \sharp y$	<code>\sharp</code>	$x \diamondsuit y$	<code>\diamondsuit</code>
$x \Re y$	<code>\Re</code>	$x \parallel y$	<code>\parallel</code>	$x \backslash y$	<code>\backslash</code>	$x \heartsuit y$	<code>\heartsuit</code>
$x \Im y$	<code>\Im</code>	$x \angle y$	<code>\angle</code>	$x \partial y$	<code>\partial</code>	$x \spadesuit y$	<code>\spadesuit</code>
$x \mho y$	<code>\mho</code>	$x \cdot y$	<code>\cdot</code>	$x y$	<code> </code>	$x ! y$	<code>!</code>

3.1.22 Variable-Sized Operators Serif

$x \sum y$	<code>\sum</code>	$x \bigcap y$	<code>\bigcap</code>	$x \odot y$	<code>\bigodot</code>
$x \prod y$	<code>\prod</code>	$x \bigcup y$	<code>\bigcup</code>	$x \otimes y$	<code>\bigotimes</code>
$x \coprod y$	<code>\coprod</code>	$x \bigsqcup y$	<code>\bigsqcup</code>	$x \oplus y$	<code>\bigoplus</code>
$x \int y$	<code>\int</code>	$x \bigvee y$	<code>\bigvee</code>	$x \biguplus y$	<code>\biguplus</code>
$x \oint y$	<code>\oint</code>	$x \bigwedge y$	<code>\bigwedge</code>		

3.1.23 Log-Like Operators Serif

$x \arccos y$	$x \cos y$	$x \csc y$	$x \exp y$	$x \ker y$	$x \limsup y$	$x \min y$	$x \sinh y$
$x \arcsin y$	$x \cosh y$	$x \deg y$	$x \gcd y$	$x \lg y$	$x \ln y$	$x \Pr y$	$x \sup y$
$x \arctan y$	$x \cot y$	$x \det y$	$x \hom y$	$x \lim y$	$x \log y$	$x \sec y$	$x \tan y$
$x \arg y$	$x \coth y$	$x \dim y$	$x \inf y$	$x \liminf y$	$x \max y$	$x \sin y$	$x \tanh y$

3.1.24 Delimiters **Serif**

$x(y$	$($	$x)y$	$)$	$x \uparrow y$	$\uparrow$	$x \Uparrow y$	$\Uparrow$
$x[y$	$[$	$x]y$	$]$	$x \downarrow y$	$\downarrow$	$x \Downarrow y$	$\Downarrow$
$x\{y$	$\{$	$x\}y$	$\}$	$x \updownarrow y$	$\updownarrow$	$x \Updownarrow y$	$\Updownarrow$
$x\lfloor y$	$\lfloor$	$x\rfloor y$	$\rfloor$	$x\lceil y$	$\lceil$	$x\rceil y$	$\rceil$
$x\langle y$	$\langle$	$x\rangle y$	$\rangle$	x/y	$/$	$x\backslash y$	$\backslash$
$x y$	$ $	$x y$	$ $				

3.1.25 Large Delimiters **Serif**

$\left($	$\rmoustache$	$\int$	$\lmoustache$	$\right)$	$\rgroup$	$\left($	$\lgroup$
$\downarrow$	$\arrowvert$	$\parallel$	$\Arrowvert$	$\downarrow$	$\bracevert$		

3.1.26 Math Mode Accents **Serif**

$\hat{a}$	$\hat{a}$	$\acute{a}$	$\acute{a}$	$\bar{a}$	$\bar{a}$	$\dot{a}$	$\dot{a}$	$\breve{a}$	$\breve{a}$
$\check{a}$	$\check{a}$	$\grave{a}$	$\grave{a}$	$\vec{a}$	$\vec{a}$	$\ddot{a}$	$\ddot{a}$	$\tilde{a}$	$\tilde{a}$

3.1.27 Miscellaneous Constructions **Serif**

$\widetilde{abc}$	$\widetilde{abc}$	$\widehat{abc}$	$\widehat{abc}$
$\overleftarrow{abc}$	$\overleftarrow{abc}$	$\overrightarrow{abc}$	$\overrightarrow{abc}$
$\overline{abc}$	$\overline{abc}$	$\underline{abc}$	$\underline{abc}$
$\overbrace{abc}$	$\overbrace{abc}$	$\underbrace{abc}$	$\underbrace{abc}$
$\sqrt{abc}$	$\sqrt{abc}$	$\sqrt[n]{abc}$	$\sqrt[n]{abc}$
f'	f'	$\frac{abc}{xyz}$	$\frac{abc}{xyz}$

3.1.28 AMS Delimiters **Serif**

$\ulcorner$	$\ulcorner$	$\urcorner$	$\urcorner$	$\llcorner$	$\llcorner$	$\lrcorner$	$\lrcorner$
-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------

3.1.29 AMS Arrows Serif

$x \dashrightarrow y$	<code>\dashrightarrow</code>	$x \dashleftarrow y$	<code>\dashleftarrow</code>
$x \leftrightsquigarrow y$	<code>\leftrightsquigarrow</code>	$x \rightleftarrows y$	<code>\leftrightsquigarrow</code>
$x \Lleftarrow y$	<code>\Lleftarrow</code>	$x \twoheadleftarrow y$	<code>\twoheadleftarrow</code>
$x \leftarrowtail y$	<code>\leftarrowtail</code>	$x \looparrowleft y$	<code>\looparrowleft</code>
$x \leftrightharpoons y$	<code>\leftrightharpoons</code>	$x \curvearrowleft y$	<code>\curvearrowleft</code>
$x \circlearrowleft y$	<code>\circlearrowleft</code>	$x \lsh y$	<code>\lsh</code>
$x \upuparrows y$	<code>\upuparrows</code>	$x \upharpoonleft y$	<code>\upharpoonleft</code>
$x \downharpoonleft y$	<code>\downharpoonleft</code>	$x \multimap y$	<code>\multimap</code>
$x \leftrightsquigarrow y$	<code>\leftrightsquigarrow</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \rightleftarrows y$	<code>\rightleftarrows</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \rightleftarrows y$	<code>\rightleftarrows</code>	$x \twoheadrightarrow y$	<code>\twoheadrightarrow</code>
$x \rightarrowtail y$	<code>\rightarrowtail</code>	$x \looparrowright y$	<code>\looparrowright</code>
$x \rightharpoons y$	<code>\rightharpoons</code>	$x \curvearrowright y$	<code>\curvearrowright</code>
$x \circlearrowright y$	<code>\circlearrowright</code>	$x \Rsh y$	<code>\Rsh</code>
$x \downharpoonright y$	<code>\downharpoonright</code>	$x \upharpoonright y$	<code>\upharpoonright</code>
$x \downharpoonright y$	<code>\downharpoonright</code>	$x \rightsquigarrow y$	<code>\rightsquigarrow</code>

3.1.30 AMS Negated Arrows Serif

$x \nleftarrow y$	<code>\nleftarrow</code>	$x \nrightarrow y$	<code>\nrightarrow</code>
$x \nLleftarrow y$	<code>\nLleftarrow</code>	$x \nRrightarrow y$	<code>\nRrightarrow</code>
$x \nleftrightarrow y$	<code>\nleftrightarrow</code>	$x \nleftrightarrow y$	<code>\nleftrightarrow</code>

3.1.31 AMS Greek Serif

$x \digamma y$ `\digamma` $x \varkappa y$ `\varkappa`

3.1.32 AMS Hebrew Serif

$x \beth y$ `\beth` $x \daleth y$ `\daleth` $x \gimel y$ `\gimel`

3.1.33 AMS Miscellaneous **Serif**

$x\hbar y$	<code>\hbar</code>	$x\hslash y$	<code>\hslash</code>
$x\triangle y$	<code>\vartriangle</code>	$x\nabla y$	<code>\triangledown</code>
$x\square y$	<code>\square</code>	$x\lozenge y$	<code>\lozenge</code>
$x\circledS y$	<code>\circledS</code>	$x\angle y$	<code>\angle</code>
$x\measuredangle y$	<code>\measuredangle</code>	$x\nexists y$	<code>\nexists</code>
$x\mho y$	<code>\mho</code>	$x\Finv y$	<code>\Finv</code> ^u
$x\Game y$	<code>\Game</code> ^u	$x\Bbbk y$	<code>\Bbbk</code> ^u
$x\backprime y$	<code>\backprime</code>	$x\varnothing y$	<code>\varnothing</code>
$x\blacktriangle y$	<code>\blacktriangle</code>	$x\blacktriangledown y$	<code>\blacktriangledown</code>
$x\blacksquare y$	<code>\blacksquare</code>	$x\blacklozenge y$	<code>\blacklozenge</code>
$x\bigstar y$	<code>\bigstar</code>	$x\sphericalangle y$	<code>\sphericalangle</code>
$x\complement y$	<code>\complement</code>	$x\eth y$	<code>\eth</code>
$x\diagup y$	<code>\diagup</code> ^u	$x\diagdown y$	<code>\diagdown</code> ^u

^u Not defined in `amssymb.sty`, define using the `\newsymbol` command.

3.1.34 AMS Binary Operators **Serif**

$x\dotplus y$	<code>\dotplus</code>	$x\smallsetminus y$	<code>\smallsetminus</code>
$x\Cap y$	<code>\Cap</code>	$x\cup y$	<code>\Cup</code>
$x\barwedge y$	<code>\barwedge</code>	$x\veebar y$	<code>\veebar</code>
$x\doublebarwedge y$	<code>\doublebarwedge</code>	$x\boxminus y$	<code>\boxminus</code>
$x\boxtimes y$	<code>\boxtimes</code>	$x\boxdot y$	<code>\boxdot</code>
$x\boxplus y$	<code>\boxplus</code>	$x\divideontimes y$	<code>\divideontimes</code>
$x\ltimes y$	<code>\ltimes</code>	$x\rtimes y$	<code>\rtimes</code>
$x\leftthreetimes y$	<code>\leftthreetimes</code>	$x\rightthreetimes y$	<code>\rightthreetimes</code>
$x\curlywedge y$	<code>\curlywedge</code>	$x\curlyvee y$	<code>\curlyvee</code>
$x\circledast y$	<code>\circledast</code>	$x\circledcirc y$	<code>\circledcirc</code>
$x\circledcirc y$	<code>\circledcirc</code>	$x\centerdot y$	<code>\centerdot</code>
$x\intercal y$	<code>\intercal</code>		

3.1.35 AMS Relations **Serif**

$x \leqslant y$	<code>\leqslant</code>
$x \lesssim y$	<code>\lesssim</code>
$x \approx y$	<code>\approx</code>
$x \lll y$	<code>\lll</code>
$x \lesseqgtr y$	<code>\lesseqgtr</code>
$x \doteqdot y$	<code>\doteqdot</code>
$x \fallingdotseq y$	<code>\fallingdotseq</code>
$x \backsimeq y$	<code>\backsimeq</code>
$x \Subset y$	<code>\Subset</code>
$x \preccurlyeq y$	<code>\preccurlyeq</code>
$x \prec\sim y$	<code>\prec\sim</code>
$x \vartriangleleft y$	<code>\vartriangleleft</code>
$x \vDash y$	<code>\vDash</code>
$x \smallsmile y$	<code>\smallsmile</code>
$x \bumpeq y$	<code>\bumpeq</code>
$x \geqq y$	<code>\geqq</code>
$x \gtrless y$	<code>\gtrless</code>
$x \gtrapprox y$	<code>\gtrapprox</code>
$x \ggg y$	<code>\ggg</code>
$x \gtreqless y$	<code>\gtreqless</code>
$x \eqcirc y$	<code>\eqcirc</code>
$x \triangleq y$	<code>\triangleq</code>
$x \thickapprox y$	<code>\thickapprox</code>
$x \supseteq y$	<code>\supseteq</code>
$x \succcurlyeq y$	<code>\succcurlyeq</code>
$x \succsim y$	<code>\succsim</code>
$x \triangleright y$	<code>\triangleright</code>
$x \Vdash y$	<code>\Vdash</code>
$x \parallel y$	<code>\shortparallel</code>
$x \pitchfork y$	<code>\pitchfork</code>
$x \blacktriangleleft y$	<code>\blacktriangleleft</code>
$x \backepsilon y$	<code>\backepsilon</code>
$x \because y$	<code>\because</code>

3.1.36 AMS Negated Relations Serif

$x \nless y$	<code>\nless</code>	$x \nleq y$	<code>\nleq</code>
$x \nleqslant y$	<code>\nleqslant</code>	$x \nleqq y$	<code>\nleqq</code>
$x \lesseqgtr y$	<code>\lesseqgtr</code>	$x \lesseqgtr y$	<code>\lesseqgtr</code>
$x \lvertneqq y$	<code>\lvertneqq</code>	$x \lvertneqq y$	<code>\lvertneqq</code>
$x \napprox y$	<code>\napprox</code>	$x \nprec y$	<code>\nprec</code>
$x \npreceq y$	<code>\npreceq</code>	$x \nprecnsim y$	<code>\nprecnsim</code>
$x \nprecnapprox y$	<code>\nprecnapprox</code>	$x \nsim y$	<code>\nsim</code>
$x \nshortmid y$	<code>\nshortmid</code>	$x \nmid y$	<code>\nmid</code>
$x \nvDash y$	<code>\nvDash</code>	$x \nvDash y$	<code>\nvDash</code>
$x \ntriangleleft y$	<code>\ntriangleleft</code>	$x \ntrianglelefteq y$	<code>\ntrianglelefteq</code>
$x \nsubseteq y$	<code>\nsubseteq</code>	$x \subsetneq y$	<code>\subsetneq</code>
$x \varsubsetneq y$	<code>\varsubsetneq</code>	$x \subsetneqq y$	<code>\subsetneqq</code>
$x \varsubsetneqq y$	<code>\varsubsetneqq</code>	$x \ntriangleright y$	<code>\ntriangleright</code>
$x \ngeq y$	<code>\ngeq</code>	$x \ngeqslant y$	<code>\ngeqslant</code>
$x \ngeqq y$	<code>\ngeqq</code>	$x \gneq y$	<code>\gneq</code>
$x \gtrless y$	<code>\gtrless</code>	$x \gtrless y$	<code>\gtrless</code>
$x \gnsim y$	<code>\gnsim</code>	$x \gnsim y$	<code>\gnsim</code>
$x \nsucc y$	<code>\nsucc</code>	$x \nsucceq y$	<code>\nsucceq</code>
$x \nsucceqq y$	<code>\nsucceqq</code>	$x \succnsim y$	<code>\succnsim</code>
$x \succnapprox y$	<code>\succnapprox</code>	$x \ncong y$	<code>\ncong</code>
$x \nshortparallel y$	<code>\nshortparallel</code>	$x \nparallel y$	<code>\nparallel</code>
$x \nVDash y$	<code>\nVDash</code>	$x \nVDash y$	<code>\nVDash</code>
$x \ntriangleright y$	<code>\ntriangleright</code>	$x \ntrianglerighteq y$	<code>\ntrianglerighteq</code>
$x \nsupseteq y$	<code>\nsupseteq</code>	$x \supsetneqq y$	<code>\supsetneqq</code>
$x \supsetneq y$	<code>\supsetneq</code>	$x \varsupsetneq y$	<code>\varsupsetneq</code>
$x \supsetneqq y$	<code>\supsetneqq</code>	$x \varsupsetneqq y$	<code>\varsupsetneqq</code>

3.1.37 Math “Torture” Test Serif

Most of the following examples are taken from *The T_EXbook* (Knuth 1984, see <https://ctan.org/pkg/texbook>) and were adapted for L^AT_EX from Karl Berry’s torture test for plain T_EX math fonts.

$x + y - z, \quad x + y * z, \quad z * y / z, \quad (x + y)(x - y) = x^2 - y^2,$
 $x \times y \cdot z = [xyz], \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,$
 $x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y / z, \quad x := y, \quad x \leq y \neq z, \quad x \sim y \simeq z \quad x \equiv y \neq z, \quad x \subset y \subseteq z$
 $\sin 2\theta = 2 \sin \theta \cos \theta, \quad O(n \log n \log n), \quad \Pr(X > x) = \exp(-x/\mu),$
 $(x \in A(n) \mid x \in B(n)), \quad \bigcup_n X_n \parallel \bigcap_n Y_n$
 In-text matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} a & b & c \\ 1 & m & n \end{pmatrix}$.

$$a_0+\frac{1}{a_1+\frac{1}{a_2+\frac{1}{a_3+\frac{1}{a_4}}}}$$

$$\binom{p}{2}x^2y^{p-2}-\frac{1}{1-x}\frac{1}{1-x^2}=\frac{a+1}{b}\Big/\frac{c+1}{d}.$$

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|\varphi(x+\mathrm{i}y)|^2=0$$

$$\pi(n)=\sum_{m=2}^n\left[\left(\sum_{k=1}^{m-1}\lfloor(m/k)/\lceil m/k\rceil\right)^{-1}\right].$$

$$\int_0^\infty \frac{t-\mathrm{i}b}{t^2+b^2}e^{\mathrm{i}at}\,\mathrm{d}t=e^{ab}E_1(ab),\quad a,b>0.$$

$$\boldsymbol{A}:=\begin{pmatrix}x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda\end{pmatrix}.$$

$$\begin{pmatrix}a & b & c \\ d & e & f\end{pmatrix}\begin{pmatrix}u & x \\ v & y \\ w & z\end{pmatrix}$$

$$\boldsymbol{A}=\begin{pmatrix}a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn}\end{pmatrix}$$

$$\begin{array}{c}C\\C\\C'\end{array}\begin{array}{c}I\\ \left(\begin{array}{ccc}1&0&0\\b&1-b&0\\0&a&1-a\end{array}\right)\\ \end{array}$$

$$\sum_{n=0}^\infty a_n z^n \quad \text{converges if} \quad |z| < \Big(\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}\Big)^{-1}.$$

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \rightarrow f'(x) \quad \text{as } \Delta x \rightarrow 0.$$

$$\|u_i\| = 1, \quad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of } \left\{ \begin{array}{l} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{array} \right\} \text{ is } \left\{ \begin{array}{l} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{array} \right\}.$$

$$\begin{aligned} T(n) &\leq T(2^{\lceil \lg n \rceil}) \leq c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\ &< 3c \cdot 3^{\lg n} \\ &= 3cn^{\lg 3}. \end{aligned}$$

$$\begin{aligned} (x+y)(x-y) &= x^2 - xy + yx - y^2 \\ &= x^2 - y^2 \\ (x+y)^2 &= x^2 + 2xy + y^2. \end{aligned}$$

$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} dr d\theta \\ &= \int_0^{2\pi} \left(e^{-\frac{r^2}{2}} \Big|_{r=0}^{r=\infty} \right) d\theta \\ &= \pi. \end{aligned}$$

$$\prod_{k \geq 0} \frac{1}{(1 - q^k z)} = \sum_{n \geq 0} z^n / \prod_{1 \leq k \leq n} (1 - q^k).$$

$$\sum_{\substack{0 < i \leq m \\ 0 < j \leq n}} p(i,j) \neq \sum_{i=1}^p \sum_{j=1}^q \sum_{k=1}^r a_{ij} b_{jk} c_{ki} \neq \sum_{\substack{1 \leq i \leq p \\ 1 \leq j \leq q \\ 1 \leq k \leq r}} a_{ij} b_{jk} c_{ki}$$

$$\max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\text{Inline math: } \max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$p_1(n) = \lim_{m \rightarrow \infty} \sum_{\nu=0}^{\infty} (1 - \cos^{2m}(\nu!^n \pi/n))$$

$$\text{Inline math: } p_1(n) = \lim_{m \rightarrow \infty} \sum_{\nu=0}^{\infty} (1 - \cos^{2m}(\nu!^n \pi/n))$$

3.2 Math Test **Serif Bold**

3.2.1 Overview **Serif Bold**

Default: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma; \sigma_{\epsilon}, c^{\alpha}$

mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathrm: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathup: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathit: $\alpha\alpha\alpha b\beta G\text{``}\epsilon\epsilon\theta\vartheta P^{\circ}\sigma$

mathbf: $\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathbfit: $\alpha\alpha b\beta G\text{``}\epsilon\epsilon\theta\vartheta P^{\circ}\sigma$

mathbfup: $\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

Default: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma; \sigma_{\epsilon}, c^{\alpha}$

mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathrm: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathup: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathit: $\alpha\alpha\alpha b\beta G\text{``}\epsilon\epsilon\theta\vartheta P^{\circ}\sigma$

mathbf: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathbfit: $\alpha\alpha\alpha b\beta G\text{``}\epsilon\epsilon\theta\vartheta P^{\circ}\sigma$

mathbfup: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

Default: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma; \sigma_{\epsilon}, c^{\alpha}$

mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathrm: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathup: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathit: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathbf: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathbfit: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathbfup: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

Default: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma; \sigma_{\epsilon}, c^{\alpha}$

mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathrm: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathup: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathit: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathbf: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathbfit: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

mathbfup: $\alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\vartheta P\Pi\Sigma\sigma$

3.2.2 Formulas **Serif Bold**

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi,$
 $\omega, \mathfrak{f}, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi,$
 $\omega, \mathfrak{f}, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$
 $\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathfrak{f}, A, B, \Gamma, \Delta, E, Z, H,$
 $\Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$
 $\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathfrak{f}, A, B, \Gamma, \Delta, E, Z, H,$
 $\Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$
 $\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
 $\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
 $\lim_{\nu \rightarrow \infty} v(\nu) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$
 $\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l I | l I /, i j, rn m, θ Θ , ϕ ψ , --

Latin vs. Greek: a α , d δ , e ϵ , i ι , k κ , n η , o σ , p ρ , β β , u υ , v ν , w ω , x χ , y γ ,
 A Δ Λ , O Θ Ω , T Γ , Y Υ .

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{\nu \rightarrow \infty} v(\nu) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l I | l I /, i j, rn m, θ Θ , ϕ ψ , --

Latin vs. Greek: a α , d δ , e ϵ , i ι , k κ , n η , o σ , p ρ , β β , u υ , v ν , w ω , x χ , y γ ,
 A Δ Λ , O Θ Ω , T Γ , Y Υ .

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{\nu \rightarrow \infty} v(\nu) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l I | l I /, i j, rn m, θ Θ , ϕ ψ , --

Latin vs. Greek: $a \alpha, d \delta, e \varepsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$$

$$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: $0 \ O \ O, 1 \ l \ l \ | \ l \ l \ / , i \ j, r n \ m, \theta \ \Theta, \phi \ \psi, - \ -$

Latin vs. Greek: $a \alpha, d \delta, e \varepsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

3.2.3 Math Alphabets **Serif Bold**

Default

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,

A, B, Γ , Δ , E, Z, H, Θ , I, K, Λ , M, N, Ξ , O, Π , P, Σ , T, Υ , Φ , X, Ψ , Ω ,

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$

Math Normal (`\mathnormal`)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,

A, B, Γ , Δ , E, Z, H, Θ , I, K, Λ , M, N, Ξ , O, Π , P, Σ , T, Υ , Φ , X, Ψ , Ω ,

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$

Math Italic (`\mathit`)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,

A, B, $\grave{}$, $\acute{}$, E, Z, H, $\hat{}$, I, K, $\tilde{}$, M, N, $\ddot{}$, O, $\acute{}$, P, $\circ$, T, $\grave{}$, $\tilde{}$, X, $\bar{}$, $\dot{}$,

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$

Math Roman (\mathrm)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
 A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
 a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
 A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
 α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, ϑ, Ϙ, ρ, ζ, φ,

Math Bold (\mathbf)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
 A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
 a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
 A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
 α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, ϑ, Ϙ, ρ, ζ, φ,

Caligraphic (\mathcal)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

Script (\mathscr)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

Fraktur (\mathfrak)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,

Blackboard Bold (\mathbb)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

3.2.4 Character Sidebearings **Serif Bold**

Default

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |\Upsilon| + |\Phi| + |X| + |\Psi| + |\Omega| +$
 $|\alpha| + |\beta| + |\gamma| + |\delta| + |\epsilon| + |\zeta| + |\eta| + |\theta| + |\iota| + |\kappa| + |\lambda| + |\mu| +$
 $|\nu| + |\xi| + |\omicron| + |\pi| + |\rho| + |\sigma| + |\tau| + |\upsilon| + |\phi| + |\chi| + |\psi| + |\omega| +$
 $|\varepsilon| + |\vartheta| + |\varpi| + |\varrho| + |\varsigma| + |\varphi| +$

Math Roman (`\mathrm`)

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |\Upsilon| + |\Phi| + |X| + |\Psi| + |\Omega| +$

Math Bold (`\mathbf`)

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |\Upsilon| + |\Phi| + |X| + |\Psi| + |\Omega| +$

Math Calligraphic (`\mathcal`)

$|\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{D}| + |\mathcal{E}| + |\mathcal{F}| + |\mathcal{G}| + |\mathcal{H}| + |\mathcal{I}| + |\mathcal{J}| + |\mathcal{K}| + |\mathcal{L}| + |\mathcal{M}| +$
 $|\mathcal{N}| + |\mathcal{O}| + |\mathcal{P}| + |\mathcal{Q}| + |\mathcal{R}| + |\mathcal{S}| + |\mathcal{T}| + |\mathcal{U}| + |\mathcal{V}| + |\mathcal{W}| + |\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}| +$

3.2.5 Superscript Positioning **Serif Bold**

Default

$$\begin{aligned}
 &A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 + \\
 &N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 + \\
 &a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 + \\
 &n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 + \\
 &A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 + \\
 &N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 + \\
 &\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \epsilon^2 + \zeta^2 + \eta^2 + \theta^2 + \iota^2 + \kappa^2 + \lambda^2 + \mu^2 + \\
 &\nu^2 + \xi^2 + \omicron^2 + \pi^2 + \rho^2 + \sigma^2 + \tau^2 + \upsilon^2 + \phi^2 + \chi^2 + \psi^2 + \omega^2 + \\
 &\varepsilon^2 + \vartheta^2 + \varpi^2 + \varrho^2 + \varsigma^2 + \varphi^2 +
 \end{aligned}$$

Math Roman (`\mathrm`)

$$\begin{aligned}
 &A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 + \\
 &N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 + \\
 &a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 + \\
 &n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 + \\
 &A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 + \\
 &N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 +
 \end{aligned}$$

Math Bold (`\mathbf`)

$$\begin{aligned}
 &A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 + \\
 &N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 + \\
 &a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 + \\
 &n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 + \\
 &A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 + \\
 &N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 +
 \end{aligned}$$

Math Calligraphic (`\mathcal`)

$$\begin{aligned}
 &\mathcal{A}^2 + \mathcal{B}^2 + \mathcal{C}^2 + \mathcal{D}^2 + \mathcal{E}^2 + \mathcal{F}^2 + \mathcal{G}^2 + \mathcal{H}^2 + \mathcal{I}^2 + \mathcal{J}^2 + \mathcal{K}^2 + \mathcal{L}^2 + \mathcal{M}^2 + \\
 &\mathcal{N}^2 + \mathcal{O}^2 + \mathcal{P}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{S}^2 + \mathcal{T}^2 + \mathcal{U}^2 + \mathcal{V}^2 + \mathcal{W}^2 + \mathcal{X}^2 + \mathcal{Y}^2 + \mathcal{Z}^2 +
 \end{aligned}$$

3.2.6 Subscript Positioning **Serif Bold**

Default

$$\begin{aligned}
 &A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i + \\
 &N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i + \\
 &a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i + \\
 &n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i + \\
 &A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i + \\
 &N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i + \\
 &\alpha_i + \beta_i + \gamma_i + \delta_i + \epsilon_i + \zeta_i + \eta_i + \theta_i + \iota_i + \kappa_i + \lambda_i + \mu_i + \\
 &\nu_i + \xi_i + o_i + \pi_i + \rho_i + \sigma_i + \tau_i + v_i + \phi_i + \chi_i + \psi_i + \omega_i + \\
 &\varepsilon_i + \vartheta_i + \varpi_i + \varrho_i + \varsigma_i + \varphi_i +
 \end{aligned}$$

Math Roman (`\mathrm`)

$$\begin{aligned}
 &A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i + \\
 &N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i + \\
 &a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i + \\
 &n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i + \\
 &A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i + \\
 &N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i +
 \end{aligned}$$

Math Bold (`\mathbf`)

$$\begin{aligned}
 &A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i + \\
 &N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i + \\
 &a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i + \\
 &n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i + \\
 &A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i + \\
 &N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i +
 \end{aligned}$$

Math Calligraphic (`\mathcal`)

$$\begin{aligned}
 &\mathcal{A}_i + \mathcal{B}_i + \mathcal{C}_i + \mathcal{D}_i + \mathcal{E}_i + \mathcal{F}_i + \mathcal{G}_i + \mathcal{H}_i + \mathcal{I}_i + \mathcal{J}_i + \mathcal{K}_i + \mathcal{L}_i + \mathcal{M}_i + \\
 &\mathcal{N}_i + \mathcal{O}_i + \mathcal{P}_i + \mathcal{Q}_i + \mathcal{R}_i + \mathcal{S}_i + \mathcal{T}_i + \mathcal{U}_i + \mathcal{V}_i + \mathcal{W}_i + \mathcal{X}_i + \mathcal{Y}_i + \mathcal{Z}_i +
 \end{aligned}$$

3.2.7 Accent Positioning **Serif Bold**

Default

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{I} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{\Upsilon} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$
 $\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{i} + \hat{\kappa} + \hat{\lambda} + \hat{\mu} +$
 $\hat{\nu} + \hat{\xi} + \hat{o} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{v} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$
 $\hat{\varepsilon} + \hat{\vartheta} + \hat{\varpi} + \hat{\varrho} + \hat{\varsigma} + \hat{\varphi} +$

Math Italic (`\mathit`)

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} + \hat{\ell} + \hat{\wp} + \hat{i} + \hat{j} + \hat{\tilde{i}}$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{\text{^}} + \hat{\text{^}} + \hat{E} + \hat{Z} + \hat{H} + \hat{\text{^}} + \hat{I} + \hat{K} + \hat{\text{^}} + \hat{M} +$
 $\hat{N} + \hat{\text{^}} + \hat{O} + \hat{\text{^}} + \hat{P} + \hat{\text{^}} + \hat{T} + \hat{\text{^}} + \hat{\text{^}} + \hat{X} + \hat{\text{^}} + \hat{\text{^}} +$
 $\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{i} + \hat{\kappa} + \hat{\lambda} + \hat{\mu} +$
 $\hat{\nu} + \hat{\xi} + \hat{o} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{v} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$
 $\hat{\varepsilon} + \hat{\vartheta} + \hat{\varpi} + \hat{\varrho} + \hat{\varsigma} + \hat{\varphi} +$

Math Roman (`\mathrm`)

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{I} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{\Upsilon} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$

Math Bold ($\mathbf{\text{}}\backslash\mathbf{mathbf{b}}\mathbf{f}$)

$\hat{O} + \hat{1} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{I} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{\Upsilon} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$

Math Calligraphic ($\mathcal{\text{}}\backslash\mathbf{mathcal{a}}\mathbf{l}$)

$\mathcal{A} + \mathcal{B} + \mathcal{C} + \mathcal{D} + \mathcal{E} + \mathcal{F} + \mathcal{G} + \mathcal{H} + \mathcal{I} + \mathcal{J} + \mathcal{K} + \mathcal{L} + \mathcal{M} +$
 $\mathcal{N} + \mathcal{O} + \mathcal{P} + \mathcal{Q} + \mathcal{R} + \mathcal{S} + \mathcal{T} + \mathcal{U} + \mathcal{V} + \mathcal{W} + \mathcal{X} + \mathcal{Y} + \mathcal{Z} +$

3.2.8 Differentials Serif Bold

$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$
 $dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$
 $da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$
 $dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$
 $d\alpha + d\beta + d\gamma + d\delta + d\epsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$
 $d\nu + d\xi + do + d\pi + d\rho + d\sigma + d\tau + dv + d\phi + d\chi + d\psi + d\omega +$
 $d\epsilon + d\vartheta + d\varpi + d\rho + d\zeta + d\varphi +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$

$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$
 $dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$
 $da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$
 $dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$
 $d\alpha + d\beta + d\gamma + d\delta + d\epsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$
 $d\nu + d\xi + do + d\pi + d\rho + d\sigma + d\tau + dv + d\phi + d\chi + d\psi + d\omega +$
 $d\varepsilon + d\vartheta + d\varpi + d\rho + d\zeta + d\varphi +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$

$\partial A + \partial B + \partial C + \partial D + \partial E + \partial F + \partial G + \partial H + \partial I + \partial J + \partial K + \partial L + \partial M +$
 $\partial N + \partial O + \partial P + \partial Q + \partial R + \partial S + \partial T + \partial U + \partial V + \partial W + \partial X + \partial Y + \partial Z +$
 $\partial a + \partial b + \partial c + \partial d + \partial e + \partial f + \partial g + \partial h + \partial i + \partial j + \partial k + \partial l + \partial m +$
 $\partial n + \partial o + \partial p + \partial q + \partial r + \partial s + \partial t + \partial u + \partial v + \partial w + \partial x + \partial y + \partial z +$
 $\partial A + \partial B + \partial \Gamma + \partial \Delta + \partial E + \partial Z + \partial H + \partial \Theta + \partial I + \partial K + \partial \Lambda + \partial M +$
 $\partial N + \partial \Xi + \partial O + \partial \Pi + \partial P + \partial \Sigma + \partial T + \partial \Upsilon + \partial \Phi + \partial X + \partial \Psi + \partial \Omega +$
 $\partial \alpha + \partial \beta + \partial \gamma + \partial \delta + \partial \epsilon + \partial \zeta + \partial \eta + \partial \theta + \partial \iota + \partial \kappa + \partial \lambda + \partial \mu +$
 $\partial \nu + \partial \xi + \partial o + \partial \pi + \partial \rho + \partial \sigma + \partial \tau + \partial v + \partial \phi + \partial \chi + \partial \psi + \partial \omega +$
 $\partial \varepsilon + \partial \vartheta + \partial \varpi + \partial \rho + \partial \zeta + \partial \varphi +$
 $\partial A + \partial B + \partial \Gamma + \partial \Delta + \partial E + \partial Z + \partial H + \partial \Theta + \partial I + \partial K + \partial \Lambda + \partial M +$
 $\partial N + \partial \Xi + \partial O + \partial \Pi + \partial P + \partial \Sigma + \partial T + \partial \Upsilon + \partial \Phi + \partial X + \partial \Psi + \partial \Omega +$

3.2.9 Slash Kerning **Serif Bold**

$1/A + 1/B + 1/C + 1/D + 1/E + 1/F + 1/G + 1/H + 1/I + 1/J + 1/K + 1/L + 1/M +$
 $1/N + 1/O + 1/P + 1/Q + 1/R + 1/S + 1/T + 1/U + 1/V + 1/W + 1/X + 1/Y + 1/Z +$
 $1/a + 1/b + 1/c + 1/d + 1/e + 1/f + 1/g + 1/h + 1/i + 1/j + 1/k + 1/l + 1/m +$
 $1/n + 1/o + 1/p + 1/q + 1/r + 1/s + 1/t + 1/u + 1/v + 1/w + 1/x + 1/y + 1/z +$
 $1/A + 1/B + 1/\Gamma + 1/\Delta + 1/E + 1/Z + 1/H + 1/\Theta + 1/I + 1/K + 1/\Lambda + 1/M +$
 $1/N + 1/\Xi + 1/O + 1/\Pi + 1/P + 1/\Sigma + 1/T + 1/\Upsilon + 1/\Phi + 1/X + 1/\Psi + 1/\Omega +$
 $1/\alpha + 1/\beta + 1/\gamma + 1/\delta + 1/\epsilon + 1/\zeta + 1/\eta + 1/\theta + 1/\iota + 1/\kappa + 1/\lambda + 1/\mu +$
 $1/\nu + 1/\xi + 1/o + 1/\pi + 1/\rho + 1/\sigma + 1/\tau + 1/v + 1/\phi + 1/\chi + 1/\psi + 1/\omega +$
 $1/\varepsilon + 1/\vartheta + 1/\varpi + 1/\rho + 1/\zeta + 1/\varphi +$

$$\begin{aligned}
& A/2 + B/2 + C/2 + D/2 + E/2 + F/2 + G/2 + H/2 + I/2 + J/2 + K/2 + L/2 + M/2 + \\
& N/2 + O/2 + P/2 + Q/2 + R/2 + S/2 + T/2 + U/2 + V/2 + W/2 + X/2 + Y/2 + Z/2 + \\
& a/2 + b/2 + c/2 + d/2 + e/2 + f/2 + g/2 + h/2 + i/2 + j/2 + k/2 + l/2 + m/2 + \\
& n/2 + o/2 + p/2 + q/2 + r/2 + s/2 + t/2 + u/2 + v/2 + w/2 + x/2 + y/2 + z/2 + \\
& A/2 + B/2 + \Gamma/2 + \Delta/2 + E/2 + Z/2 + H/2 + \Theta/2 + I/2 + K/2 + \Lambda/2 + M/2 + \\
& N/2 + \Xi/2 + O/2 + \Pi/2 + P/2 + \Sigma/2 + T/2 + \Upsilon/2 + \Phi/2 + X/2 + \Psi/2 + \Omega/2 + \\
& \alpha/2 + \beta/2 + \gamma/2 + \delta/2 + \epsilon/2 + \zeta/2 + \eta/2 + \theta/2 + \iota/2 + \kappa/2 + \lambda/2 + \mu/2 + \\
& \nu/2 + \xi/2 + o/2 + \pi/2 + \rho/2 + \sigma/2 + \tau/2 + v/2 + \phi/2 + \chi/2 + \psi/2 + \omega/2 + \\
& \varepsilon/2 + \vartheta/2 + \varpi/2 + \varrho/2 + \varsigma/2 + \varphi/2 +
\end{aligned}$$

3.2.10 (Big) Operators **Serif Bold**

$$\begin{aligned}
& \sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n \quad \biguplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n \\
& \sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n \\
& \bigotimes_{i=1}^n x^n \quad \bigoplus_{i=1}^n x^n \quad \bigodot_{i=1}^n x^n \quad \bigwedge_{i=1}^n x^n \quad \bigvee_{i=1}^n x^n \quad \biguplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n
\end{aligned}$$

3.2.11 Radicals **Serif Bold**

$$\begin{aligned}
& \sqrt{x+y} \quad \sqrt{x^2+y^2} \quad \sqrt{x_i^2+y_j^2} \quad \sqrt{\left(\frac{\cos x}{2}\right)} \quad \sqrt{\left(\frac{\sin x}{2}\right)} \\
& \sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{x+y}}}}}}}
\end{aligned}$$

3.2.12 Over- and Underbraces **Serif Bold**

$$\overbrace{x} \quad \overbrace{x+y} \quad \overbrace{x^2+y^2} \quad \overbrace{x_i^2+y_j^2} \quad \underbrace{x} \quad \underbrace{x+y} \quad \underbrace{x_i+y_j} \quad \underbrace{x_i^2+y_j^2}$$

3.2.13 Normal and Wide Accents **Serif Bold**

$$\dot{x} \quad \ddot{x} \quad \vec{x} \quad \bar{x} \quad \overline{x} \quad \overline{xx} \quad \tilde{x} \quad \widetilde{x} \quad \widetilde{xx} \quad \widetilde{xxx} \quad \hat{x} \quad \widehat{x} \quad \widehat{xx} \quad \widehat{xxx}$$
$$\hat{x} \quad \check{x} \quad \tilde{x} \quad \acute{x} \quad \grave{x} \quad \dot{x} \quad \ddot{x} \quad \breve{x} \quad \bar{x} \quad \vec{x}$$

3.2.14 Long Arrows **Serif Bold**

3.2.15 Left and Right Delimiters **Serif Bold**

$$-(f) - -[f] - -[f] - -[f] - -\langle f \rangle - -\{f\} -$$

Using `\left` and `\right`.

$$-(f) \quad -[f] \quad -[f] \quad -[f] \quad -\langle f \rangle \quad -\{f\} \quad -$$
$$-) f (- -] f [- - / f / - - \setminus f \setminus - - / f \setminus - - \setminus f / -$$

3.2.16 Big-g-g Delimiters **Serif Bold**

$$-\left|\begin{array}{ccccccccccc} & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \end{array}\right| - \quad -\left(\left(\left(\left(\left(\left(\left(-\right)\right)\right)\right)\right)\right)\right)\right) -$$
[illegible]
$$- \left[\left[\left[\left[\left[\left[\left[- \right] \right] \right] \right] \right] \right] \right] - \quad - \left(\left(\left(\left(\left(\left((-) \right) \right) \right) \right) \right) \right) \right) -$$
$$-\langle\langle\langle\langle\langle\langle\langle\langle\rangle\rangle\rangle\rangle\rangle\rangle\rangle- \quad -\left(\left(\left(\left(\left(\left(\left(\left(-\right)\right)\right)\right)\right)\right)\right)\right)-$$

3.2.17 Binary Operators **Serif Bold**

$x \pm y$	<code>\pm</code>	$x \cap y$	<code>\cap</code>	$x \diamond y$	<code>\diamond</code>	$x \oplus y$	<code>\oplus</code>
$x \mp y$	<code>\mp</code>	$x \cup y$	<code>\cup</code>	$x \bigtriangleup y$	<code>\bigtriangleup</code>	$x \ominus y$	<code>\ominus</code>
$x \times y$	<code>\times</code>	$x \uplus y$	<code>\uplus</code>	$x \bigtriangledown y$	<code>\bigtriangledown</code>	$x \otimes y$	<code>\otimes</code>
$x \div y$	<code>\div</code>	$x \sqcap y$	<code>\sqcap</code>	$x \triangleleft y$	<code>\triangleleft</code>	$x \oslash y$	<code>\oslash</code>
$x * y$	<code>\ast</code>	$x \sqcup y$	<code>\sqcup</code>	$x \triangleright y$	<code>\triangleright</code>	$x \odot y$	<code>\odot</code>
$x \star y$	<code>\star</code>	$x \vee y$	<code>\vee</code>	$x \lhd y$	<code>\lhd</code>	$x \bigcirc y$	<code>\bigcirc</code>
$x \circ y$	<code>\circ</code>	$x \wedge y$	<code>\wedge</code>	$x \rhd y$	<code>\rhd</code>	$x \dagger y$	<code>\dagger</code>
$x \bullet y$	<code>\bullet</code>	$x \setminus y$	<code>\setminus</code>	$x \unlhd y$	<code>\unlhd</code>	$x \ddagger y$	<code>\ddagger</code>
$x \cdot y$	<code>\cdot</code>	$x \wr y$	<code>\wr</code>	$x \unrhd y$	<code>\unrhd</code>	$x \S y$	<code>\S</code>
$x + y$	<code>+</code>	$x - y$	<code>-</code>	$x \amalg y$	<code>\amalg</code>	$x \P y$	<code>\P</code>

3.2.18 Relations **Serif Bold**

$x \leq y$	<code>\leq</code>	$x \geq y$	<code>\geq</code>	$x \equiv y$	<code>\equiv</code>	$x \models y$	<code>\models</code>
$x < y$	<code><</code>	$x > y$	<code>></code>	$x \sim y$	<code>\sim</code>	$x \perp y$	<code>\perp</code>
$x \preceq y$	<code>\preceq</code>	$x \succeq y$	<code>\succeq</code>	$x \simeq y$	<code>\simeq</code>	$x y$	<code> </code>
$x \ll y$	<code>\ll</code>	$x \gg y$	<code>\gg</code>	$x \asymp y$	<code>\asymp</code>	$x \parallel y$	<code>\parallel</code>
$x \subset y$	<code>\subset</code>	$x \supset y$	<code>\supset</code>	$x \approx y$	<code>\approx</code>	$x \bowtie y$	<code>\bowtie</code>
$x \subseteq y$	<code>\subseteq</code>	$x \supseteq y$	<code>\supseteq</code>	$x \cong y$	<code>\cong</code>	$x \Join y$	<code>\Join</code>
$x \sqsubset y$	<code>\sqsubset</code>	$x \sqsupset y$	<code>\sqsupset</code>	$x \neq y$	<code>\neq</code>	$x \smile y$	<code>\smile</code>
$x \sqsubseteq y$	<code>\sqsubseteq</code>	$x \sqsupseteq y$	<code>\sqsupseteq</code>	$x \doteq y$	<code>\doteq</code>	$x \frown y$	<code>\frown</code>
$x \in y$	<code>\in</code>	$x \ni y$	<code>\ni</code>	$x \propto y$	<code>\propto</code>	$x = y$	<code>=</code>
$x \vdash y$	<code>\vdash</code>	$x \dashv y$	<code>\dashv</code>	$x < y$	<code><</code>	$x > y$	<code>></code>
$x : y$	<code>:</code>						

3.2.19 Punctuation **Serif Bold**

x, y	<code>,</code>	$x; y$	<code>;</code>	$x : y$	<code>\colon</code>	$x \cdot y$	<code>\ldotp</code>	$x \cdot y$	<code>\cdot</code>
--------	----------------	--------	----------------	---------	---------------------	-------------	---------------------	-------------	--------------------

3.2.20 Arrows **Serif Bold**

$x \leftarrow y$	<code>\leftarrow</code>	$x \longleftarrow y$	<code>\longleftarrow</code>	$x \uparrow y$	<code>\uparrowarrow</code>
$x \Leftarrow y$	<code>\Leftarrow</code>	$x \Longleftarrow y$	<code>\Longleftarrow</code>	$x \Uparrow y$	<code>\Uparrow</code>
$x \rightarrow y$	<code>\rightarrow</code>	$x \longrightarrow y$	<code>\longrightarrow</code>	$x \downarrow y$	<code>\downarrowarrow</code>
$x \Rightarrow y$	<code>\Rightarrow</code>	$x \Longrightarrow y$	<code>\Longrightarrow</code>	$x \Downarrow y$	<code>\Downarrow</code>
$x \leftrightarrow y$	<code>\leftrightharpoonup</code>	$x \longleftrightarrow y$	<code>\longleftrightarrow</code>	$x \updownarrow y$	<code>\updownarrowarrow</code>
$x \Leftrightarrow y$	<code>\Leftrightharpoonup</code>	$x \Longleftrightarrow y$	<code>\Longleftrightarrow</code>	$x \Updownarrow y$	<code>\Updownarrow</code>
$x \mapsto y$	<code>\mapsto</code>	$x \longmapsto y$	<code>\longmapsto</code>	$x \nearrow y$	<code>\nearrow</code>
$x \hookrightarrow y$	<code>\hookrightarrow</code>	$x \hookrightarrow y$	<code>\hookrightarrow</code>	$x \searrow y$	<code>\searrow</code>
$x \leftharpoonup y$	<code>\leftharpoonup</code>	$x \rightharpoonup y$	<code>\rightharpoonup</code>	$x \swarrow y$	<code>\swarrow</code>
$x \leftharpoondown y$	<code>\leftharpoondown</code>	$x \rightharpoondown y$	<code>\rightharpoondown</code>	$x \nwarrow y$	<code>\nwarrow</code>
$x \rightleftharpoons y$	<code>\rightleftharpoons</code>	$x \leadsto y$	<code>\leadsto</code>		

3.2.21 Miscellaneous Symbols **Serif Bold**

$x\ldots y$	<code>\ldots</code>	$x\cdots y$	<code>\cdots</code>	$x\dot{y}$	<code>\vdots</code>	$x\ddot{y}$	<code>\ddots</code>
$x\aleph y$	<code>\aleph</code>	$x'y$	<code>\prime</code>	$x\forall y$	<code>\forall</code>	$x\infty y$	<code>\infty</code>
$x\hbar y$	<code>\hbar</code>	$x\emptyset y$	<code>\emptyset</code>	$x\exists y$	<code>\exists</code>	$x\Box y$	<code>\Box</code>
$x\imath y$	<code>\imath</code>	$x\nabla y$	<code>\nabla</code>	$x\neg y$	<code>\neg</code>	$x\Diamond y$	<code>\Diamond</code>
$x\jmath y$	<code>\jmath</code>	$x\sqrt{y}$	<code>\surd</code>	$x\flat y$	<code>\flat</code>	$x\triangle y$	<code>\triangle</code>
$x\ell y$	<code>\ell</code>	$x\top y$	<code>\top</code>	$x\natural y$	<code>\natural</code>	$x\clubsuit y$	<code>\clubsuit</code>
$x\wp y$	<code>\wp</code>	$x\bot y$	<code>\bot</code>	$x\sharp y$	<code>\sharp</code>	$x\diamond y$	<code>\diamond</code>
$x\Re y$	<code>\Re</code>	$x\ y$	<code>\ </code>	$x\backslash y$	<code>\backslash</code>	$x\heartsuit y$	<code>\heartsuit</code>
$x\Im y$	<code>\Im</code>	$x\angle y$	<code>\angle</code>	$x\partial y$	<code>\partial</code>	$x\spadesuit y$	<code>\spadesuit</code>
$x\Upsilon y$	<code>\Upsilon</code>	$x.y$	<code>.</code>	$x y$	<code> </code>	$x!y$	<code>!</code>

3.2.22 Variable-Sized Operators **Serif Bold**

$x\sum y$	<code>\sum</code>	$x\bigcap y$	<code>\bigcap</code>	$x\bigodot y$	<code>\bigodot</code>
$x\prod y$	<code>\prod</code>	$x\bigcup y$	<code>\bigcup</code>	$x\bigotimes y$	<code>\bigotimes</code>
$x\coprod y$	<code>\coprod</code>	$x\bigsqcup y$	<code>\bigsqcup</code>	$x\bigoplus y$	<code>\bigoplus</code>
$x\int y$	<code>\int</code>	$x\bigvee y$	<code>\bigvee</code>	$x\biguplus y$	<code>\biguplus</code>
$x\oint y$	<code>\oint</code>	$x\bigwedge y$	<code>\bigwedge</code>		

3.2.23 Log-Like Operators **Serif Bold**

xarccosy	xcosy	xcscy	xexpy	xkery	xlimsupy	xminy	xsinyh
xarcsiny	xcoshy	xdegy	xgedy	xlg y	xlny	xPry	xsupy
xarctany	xcoty	xdety	xhom y	xlimy	xlogy	xsecy	xtany
xargy	xcothy	xdim y	xinfy	xlim inf y	xmaxy	xsiny	xtanh y

3.2.24 Delimiters Serif Bold

$x(y$	$($	$x)y$	$)$	$x \uparrow y$	<code>\uparrow</code>	$x \Uparrow y$	<code>\Uparrow</code>
$x[y$	$[$	$x]y$	$]$	$x \downarrow y$	<code>\downarrow</code>	$x \Downarrow y$	<code>\Downarrow</code>
$x\{y$	<code>\{</code>	$x\}y$	<code>\}</code>	$x \updownarrow y$	<code>\updownarrow</code>	$x \Updownarrow y$	<code>\Updownarrow</code>
$x\lfloor y$	<code>\lfloor</code>	$x\rfloor y$	<code>\rfloor</code>	$x\lceil y$	<code>\lceil</code>	$x\rceil y$	<code>\rceil</code>
$x\langle y$	<code>\langle</code>	$x\rangle y$	<code>\rangle</code>	x/y	<code>/</code>	$x\backslash y$	<code>\backslash</code>
$x y$	<code> </code>	$x y$	<code>\ </code>				

3.2.25 Large Delimiters Serif Bold

$\left($	<code>\rmoustache</code>	$\int$	<code>\lmoustache</code>	$\right)$	<code>\rgroup</code>	$\left($	<code>\lgroup</code>
$\downarrow$	<code>\arrowvert</code>	$\Uparrow$	<code>\Arrowvert</code>	$\left\{$	<code>\bracevert</code>		

3.2.26 Math Mode Accents Serif Bold

$\hat{a}$	<code>\hat{a}</code>	$\acute{a}$	<code>\acute{a}</code>	$\bar{a}$	<code>\bar{a}</code>	$\dot{a}$	<code>\dot{a}</code>	$\breve{a}$	<code>\breve{a}</code>
$\check{a}$	<code>\check{a}</code>	$\grave{a}$	<code>\grave{a}</code>	$\vec{a}$	<code>\vec{a}</code>	$\ddot{a}$	<code>\ddot{a}</code>	$\tilde{a}$	<code>\tilde{a}</code>

3.2.27 Miscellaneous Constructions Serif Bold

$\widetilde{abc}$	<code>\widetilde{abc}</code>	$\widehat{abc}$	<code>\widehat{abc}</code>
$\overleftarrow{abc}$	<code>\overleftarrow{abc}</code>	$\overrightarrow{abc}$	<code>\overrightarrow{abc}</code>
$\overline{abc}$	<code>\overline{abc}</code>	$\underline{abc}$	<code>\underline{abc}</code>
$\overbrace{abc}$	<code>\overbrace{abc}</code>	$\underbrace{abc}$	<code>\underbrace{abc}</code>
$\sqrt{abc}$	<code>\sqrt{abc}</code>	$\sqrt[n]{abc}$	<code>\sqrt[n]{abc}</code>
f'	<code>f'</code>	$\frac{abc}{xyz}$	<code>\frac{abc}{xyz}</code>

3.2.28 AMS Delimiters Serif Bold

$\ulcorner$	<code>\ulcorner</code>	$\urcorner$	<code>\urcorner</code>	$\llcorner$	<code>\llcorner</code>	$\lrcorner$	<code>\lrcorner</code>
-------------	------------------------	-------------	------------------------	-------------	------------------------	-------------	------------------------

3.2.29 AMS Arrows **Serif Bold**

$x \dashrightarrow y$	<code>\dashrightarrow</code>	$x \dashleftarrow y$	<code>\dashleftarrow</code>
$x \leftrightsquigarrow y$	<code>\leftrightsquigarrow</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \Lleftarrow y$	<code>\Lleftarrow</code>	$x \twoheadleftarrow y$	<code>\twoheadleftarrow</code>
$x \leftarrowtail y$	<code>\leftarrowtail</code>	$x \looparrowleft y$	<code>\looparrowleft</code>
$x \leftrightharpoons y$	<code>\leftrightharpoons</code>	$x \curvearrowleft y$	<code>\curvearrowleft</code>
$x \circlearrowleft y$	<code>\circlearrowleft</code>	$x \lsh y$	<code>\Lsh</code>
$x \upuparrows y$	<code>\upuparrows</code>	$x \upharpoonleft y$	<code>\upharpoonleft</code>
$x \downharpoonleft y$	<code>\downharpoonleft</code>	$x \multimap y$	<code>\multimap</code>
$x \leftrightsquigarrow y$	<code>\leftrightsquigarrow</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \rightleftarrows y$	<code>\rightleftarrows</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \rightleftarrows y$	<code>\rightleftarrows</code>	$x \twoheadrightarrow y$	<code>\twoheadrightarrow</code>
$x \rightarrowtail y$	<code>\rightarrowtail</code>	$x \looparrowright y$	<code>\looparrowright</code>
$x \rightleftharpoons y$	<code>\rightleftharpoons</code>	$x \curvearrowright y$	<code>\curvearrowright</code>
$x \circlearrowright y$	<code>\circlearrowright</code>	$x \Rsh y$	<code>\Rsh</code>
$x \downdownarrows y$	<code>\downdownarrows</code>	$x \upharpoonright y$	<code>\upharpoonright</code>
$x \downharpoonright y$	<code>\downharpoonright</code>	$x \rightsquigarrow y$	<code>\rightsquigarrow</code>

3.2.30 AMS Negated Arrows **Serif Bold**

$x \nleftarrow y$	<code>\nleftarrow</code>	$x \nrightarrow y$	<code>\nrightarrow</code>
$x \nLleftarrow y$	<code>\nLleftarrow</code>	$x \nRrightarrow y$	<code>\nRrightarrow</code>
$x \nleftrightarrow y$	<code>\nleftrightarrow</code>	$x \nLeftrightarrow y$	<code>\nLeftrightarrow</code>

3.2.31 AMS Greek **Serif Bold**

$x \digamma y$ `\digamma` $x \varkappa y$ `\varkappa`

3.2.32 AMS Hebrew **Serif Bold**

$x \beth y$ `\beth` $x \daleth y$ `\daleth` $x \gimel y$ `\gimel`

3.2.33 AMS Miscellaneous **Serif Bold**

$x\hbar y$	<code>\hbar</code>	$x\hslash y$	<code>\hslash</code>
$x\Delta y$	<code>\vartriangle</code>	$x\nabla y$	<code>\triangledown</code>
$x\square y$	<code>\square</code>	$x\lozenge y$	<code>\lozenge</code>
$x\circledcirc y$	<code>\circledcirc</code>	$x\angle y$	<code>\angle</code>
$x\measuredangle y$	<code>\measuredangle</code>	$x\nexists y$	<code>\nexists</code>
$x\mho y$	<code>\mho</code>	$x\Finv y$	<code>\Finv</code>
$x\Game y$	<code>\Game</code>	$x\Bbbk y$	<code>\Bbbk</code>
$x\backprime y$	<code>\backprime</code>	$x\varnothing y$	<code>\varnothing</code>
$x\blacktriangle y$	<code>\blacktriangle</code>	$x\blacktriangledown y$	<code>\blacktriangledown</code>
$x\blacksquare y$	<code>\blacksquare</code>	$x\blacklozenge y$	<code>\blacklozenge</code>
$x\bigstar y$	<code>\bigstar</code>	$x\sphericalangle y$	<code>\sphericalangle</code>
$x\complement y$	<code>\complement</code>	$x\eth y$	<code>\eth</code>
x/y	<code>\diagup</code>	$x\diagdown y$	<code>\diagdown</code>

^u Not defined in `amssymb.sty`, define using the `\newsymbol` command.

3.2.34 AMS Binary Operators **Serif Bold**

$x\dotplus y$	<code>\dotplus</code>	$x\smallsetminus y$	<code>\smallsetminus</code>
$x\Cap y$	<code>\Cap</code>	$x\Cup y$	<code>\Cup</code>
$x\barwedge y$	<code>\barwedge</code>	$x\veebar y$	<code>\veebar</code>
$x\doublebarwedge y$	<code>\doublebarwedge</code>	$x\boxminus y$	<code>\boxminus</code>
$x\boxtimes y$	<code>\boxtimes</code>	$x\boxdot y$	<code>\boxdot</code>
$x\boxplus y$	<code>\boxplus</code>	$x\divideontimes y$	<code>\divideontimes</code>
$x\ltimes y$	<code>\ltimes</code>	$x\rtimes y$	<code>\rtimes</code>
$x\leftthreetimes y$	<code>\leftthreetimes</code>	$x\rightthreetimes y$	<code>\rightthreetimes</code>
$x\curlywedge y$	<code>\curlywedge</code>	$x\curlyvee y$	<code>\curlyvee</code>
$x\circleddash y$	<code>\circleddash</code>	$x\circledast y$	<code>\circledast</code>
$x\circledcirc y$	<code>\circledcirc</code>	$x\centerdot y$	<code>\centerdot</code>
$x\intercal y$	<code>\intercal</code>		

3.2.35 AMS Relations **Serif Bold**

$x \leqslant y$	<code>\leqslant</code>
$x \lesssim y$	<code>\lesssim</code>
$x \approx y$	<code>\approx</code>
$x \lll y$	<code>\lll</code>
$x \lesseqgtr y$	<code>\lesseqgtr</code>
$x \doteqdot y$	<code>\doteqdot</code>
$x \fallingdotseq y$	<code>\fallingdotseq</code>
$x \backsimeq y$	<code>\backsimeq</code>
$x \Subset y$	<code>\Subset</code>
$x \preccurlyeq y$	<code>\preccurlyeq</code>
$x \precapprox y$	<code>\precapprox</code>
$x \vartriangleleft y$	<code>\vartriangleleft</code>
$x \vDash y$	<code>\vDash</code>
$x \smallsmile y$	<code>\smallsmile</code>
$x \bumpeq y$	<code>\bumpeq</code>
$x \geqq y$	<code>\geqq</code>
$x \gtrless y$	<code>\gtrless</code>
$x \gtrapprox y$	<code>\gtrapprox</code>
$x \ggg y$	<code>\ggg</code>
$x \gtreqless y$	<code>\gtreqless</code>
$x \eqcirc y$	<code>\eqcirc</code>
$x \triangleq y$	<code>\triangleq</code>
$x \thickapprox y$	<code>\thickapprox</code>
$x \supset y$	<code>\supset</code>
$x \succcurlyeq y$	<code>\succcurlyeq</code>
$x \succapprox y$	<code>\succapprox</code>
$x \triangleright y$	<code>\triangleright</code>
$x \Vdash y$	<code>\Vdash</code>
$x \parallel y$	<code>\parallel</code>
$x \pitchfork y$	<code>\pitchfork</code>
$x \blacktriangleleft y$	<code>\blacktriangleleft</code>
$x \backepsilon y$	<code>\backepsilon</code>
$x \because y$	<code>\because</code>

3.2.36 AMS Negated Relations **Serif Bold**

$x \nless y$	<code>\nless</code>	$x \nleq y$	<code>\nleq</code>
$x \nleqslant y$	<code>\nleqslant</code>	$x \nleqq y$	<code>\nleqq</code>
$x \lneq y$	<code>\lneq</code>	$x \lneqq y$	<code>\lneqq</code>
$x \lvertneqq y$	<code>\lvertneqq</code>	$x \nlsim y$	<code>\nlsim</code>
$x \napprox y$	<code>\napprox</code>	$x \nprec y$	<code>\nprec</code>
$x \npreceq y$	<code>\npreceq</code>	$x \nprecnsim y$	<code>\nprecnsim</code>
$x \nprecnapprox y$	<code>\nprecnapprox</code>	$x \nsim y$	<code>\nsim</code>
$x \nshortmid y$	<code>\nshortmid</code>	$x \nmid y$	<code>\nmid</code>
$x \nvDash y$	<code>\nvDash</code>	$x \nVDash y$	<code>\nVDash</code>
$x \ntriangleleft y$	<code>\ntriangleleft</code>	$x \ntrianglelefteq y$	<code>\ntrianglelefteq</code>
$x \nsubseteq y$	<code>\nsubseteq</code>	$x \subsetneq y$	<code>\subsetneq</code>
$x \subsetneqq y$	<code>\subsetneqq</code>	$x \subsetneqq y$	<code>\subsetneqq</code>
$x \varsubsetneqq y$	<code>\varsubsetneqq</code>	$x \ntriangleright y$	<code>\ntriangleright</code>
$x \ngeq y$	<code>\ngeq</code>	$x \ngeqslant y$	<code>\ngeqslant</code>
$x \ngeqq y$	<code>\ngeqq</code>	$x \gneq y$	<code>\gneq</code>
$x \gneqq y$	<code>\gneqq</code>	$x \gvertneqq y$	<code>\gvertneqq</code>
$x \gnsim y$	<code>\gnsim</code>	$x \gnapprox y$	<code>\gnapprox</code>
$x \nsucc y$	<code>\nsucc</code>	$x \nsucceq y$	<code>\nsucceq</code>
$x \nsucceqq y$	<code>\nsucceqq</code>	$x \succnsim y$	<code>\succnsim</code>
$x \succapprox y$	<code>\succapprox</code>	$x \ncong y$	<code>\ncong</code>
$x \nshortparallel y$	<code>\nshortparallel</code>	$x \nparallel y$	<code>\nparallel</code>
$x \nVdash y$	<code>\nVdash</code>	$x \nVDash y$	<code>\nVDash</code>
$x \ntriangleright y$	<code>\ntriangleright</code>	$x \ntrianglerighteq y$	<code>\ntrianglerighteq</code>
$x \nsupseteq y$	<code>\nsupseteq</code>	$x \supsetneqq y$	<code>\supsetneqq</code>
$x \supsetneqq y$	<code>\supsetneqq</code>	$x \varsupsetneqq y$	<code>\varsupsetneqq</code>
$x \supseteq y$	<code>\supseteq</code>	$x \varsupseteq y$	<code>\varsupseteq</code>

3.2.37 Math “Torture” Test **Serif Bold**

Most of the following examples are taken from *The TeXbook* (Knuth 1984, see <https://ctan.org/pkg/texbook>) and were adapted for L^AT_EX from Karl Berry’s torture test for plain TeX math fonts.

$x + y - z$, $x + y * z$, $z * y / z$, $(x + y)(x - y) = x^2 - y^2$,
 $x \times y \cdot z = [xyz]$, $x \circ y \bullet z$, $x \cup y \cap z$, $x \sqcup y \sqcap z$,
 $x \vee y \wedge z$, $x \pm y \mp z$, $x = y / z$, $x := y$, $x \leq y \neq z$, $x \sim y \simeq z \equiv y \not\equiv z$, $x \subset y \subseteq z$
 $\sin 2\theta = 2 \sin \theta \cos \theta$, $O(n \log n \log n)$, $\Pr(X > x) = \exp(-x/\mu)$,
 $(x \in A(n) \mid x \in B(n))$, $\bigcup_n X_n \parallel \bigcap_n Y_n$
 In-text matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} a & b & c \\ 1 & m & n \end{pmatrix}$.

$$a_0+\frac{1}{a_1+\frac{1}{a_2+\frac{1}{a_3+\frac{1}{a_4}}}}$$

$$\binom{p}{2}x^2y^{p-2}-\frac{1}{1-x}\frac{1}{1-x^2}=\frac{a+1}{b}\Big/\frac{c+1}{d}.$$

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|\varphi(x+\mathrm{i}y)|^2=0$$

$$\pi(n)=\sum_{m=2}^n\left[\left(\sum_{k=1}^{m-1}\lfloor(m/k)/\lceil m/k\rceil\rfloor\right)^{-1}\right].$$

$$\int_0^\infty \frac{t-\mathrm{i}b}{t^2+b^2}e^{\mathrm{i}at}\,\mathrm{d}t=e^{ab}E_1(ab),\quad a,b>0.$$

$$A:=\begin{pmatrix}x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda\end{pmatrix}.$$

$$\begin{pmatrix}a & b & c \\ d & e & f\end{pmatrix}\begin{pmatrix}u & x \\ v & y \\ w & z\end{pmatrix}$$

$$A=\begin{pmatrix}a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn}\end{pmatrix}$$

$$M=\begin{matrix} & C & I & C' \\ \begin{matrix} C \\ I \\ C' \end{matrix} & \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{matrix}$$

$$\sum_{n=0}^\infty a_n z^n \quad \text{converges if} \quad |z|<\Big(\limsup_{n\rightarrow\infty}\sqrt[n]{|a_n|}\Big)^{-1}.$$

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \rightarrow f'(x) \quad \text{as } \Delta x \rightarrow 0.$$

$$\|u_i\| = 1, \quad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of } \left\{ \begin{array}{l} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{array} \right\} \text{ is } \left\{ \begin{array}{l} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{array} \right\}.$$

$$\begin{aligned} T(n) &\leq T(2^{\lceil \lg n \rceil}) \leq c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\ &< 3c \cdot 3^{\lg n} \\ &= 3cn^{\lg 3}. \end{aligned}$$

$$\begin{aligned} (x+y)(x-y) &= x^2 - xy + yx - y^2 \\ &= x^2 - y^2 \\ (x+y)^2 &= x^2 + 2xy + y^2. \end{aligned}$$

$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} dr d\theta \\ &= \int_0^{2\pi} \left(e^{-\frac{r^2}{2}} \Big|_{r=0}^{r=\infty} \right) d\theta \\ &= \pi. \end{aligned}$$

$$\prod_{k \geq 0} \frac{1}{(1 - q^k z)} = \sum_{n \geq 0} z^n / \prod_{1 \leq k \leq n} (1 - q^k).$$

$$\sum_{\substack{0 < i \leq m \\ 0 < j \leq n}} p(i,j) \neq \sum_{i=1}^p \sum_{j=1}^q \sum_{k=1}^r a_{ij} b_{jk} c_{ki} \neq \sum_{\substack{1 \leq i \leq p \\ 1 \leq j \leq q \\ 1 \leq k \leq r}} a_{ij} b_{jk} c_{ki}$$

$$\max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\text{Inline math: } \max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$p_1(n) = \lim_{m \rightarrow \infty} \sum_{\nu=0}^{\infty} (1 - \cos^{2m}(\nu!^n \pi/n))$$

$$\text{Inline math: } p_1(n) = \lim_{m \rightarrow \infty} \sum_{\nu=0}^{\infty} (1 - \cos^{2m}(\nu!^n \pi/n))$$

3.3 Math Test Sans Serif

3.3.1 Overview Sans Serif

Default: $\alpha\alpha\alpha\beta\beta\mathit{G}\mathit{G}\mathit{T}\varepsilon\varepsilon\theta\theta\mathit{P}\mathit{P}\mathit{\Sigma}\sigma$; $\sigma_{\varepsilon}, c^{\alpha}$
mathnormal: $\alpha\alpha\alpha\beta\beta\mathit{G}\mathit{G}\mathit{T}\varepsilon\varepsilon\theta\theta\mathit{P}\mathit{P}\mathit{\Sigma}\sigma$
mathrm: $\alpha\alpha\alpha\beta\beta\mathit{G}\mathit{G}\mathit{T}\varepsilon\varepsilon\theta\theta\mathit{P}\mathit{P}\mathit{\Sigma}\sigma$
mathup: $\alpha\alpha\alpha\beta\beta\mathit{G}\mathit{G}\mathit{T}\varepsilon\varepsilon\theta\theta\mathit{P}\mathit{P}\mathit{\Sigma}\sigma$
mathit: $\alpha\alpha\alpha\beta\beta\mathit{G}\mathit{G}\mathit{T}\varepsilon\varepsilon\theta\theta\mathit{P}\mathit{P}\mathit{\Sigma}\sigma$
mathbf: **$\alpha\alpha\alpha\beta\beta\mathit{G}\mathit{G}\mathit{T}\varepsilon\varepsilon\theta\theta\mathit{P}\mathit{P}\mathit{\Sigma}\sigma$**
mathbfit: **$\alpha\alpha\beta\mathit{G}\mathit{G}\mathit{T}\varepsilon\varepsilon\theta\theta\mathit{P}\mathit{P}\mathit{\Sigma}\sigma$**
mathbfup: **$\alpha\alpha\beta\mathit{G}\mathit{G}\mathit{T}\varepsilon\varepsilon\theta\theta\mathit{P}\mathit{P}\mathit{\Sigma}\sigma$**

Default: $\alpha\alpha\alpha\beta\beta\mathbf{G}\mathbf{T}\varepsilon\varepsilon\theta\theta\mathbf{P}\mathbf{\Pi}\mathbf{\Sigma}\sigma$; $\sigma_\varepsilon, c^\alpha$
 $\mathrm{mathnormal}$: $\alpha\alpha\alpha\beta\beta\mathbf{G}\mathbf{T}\varepsilon\varepsilon\theta\theta\mathbf{P}\mathbf{\Pi}\mathbf{\Sigma}\sigma$
 mathrm : $\alpha\alpha\alpha\beta\beta\mathbf{G}\mathbf{T}\varepsilon\varepsilon\theta\theta\mathbf{P}\mathbf{\Pi}\mathbf{\Sigma}\sigma$
 mathup : $\alpha\alpha\alpha\beta\beta\mathbf{G}\mathbf{T}\varepsilon\varepsilon\theta\theta\mathbf{P}\mathbf{\Pi}\mathbf{\Sigma}\sigma$
 mathit : $\alpha\alpha\alpha\beta\beta\mathbf{G}\mathbf{T}\varepsilon\varepsilon\theta\theta\mathbf{P}\mathbf{\Pi}\mathbf{\Sigma}\sigma$
 mathbf : $\alpha\alpha\alpha\beta\beta\mathbf{G}\mathbf{T}\varepsilon\varepsilon\theta\theta\mathbf{P}\mathbf{\Pi}\mathbf{\Sigma}\sigma$
 mathbf fit: $\alpha\alpha\alpha\beta\beta\mathbf{G}\mathbf{T}\varepsilon\varepsilon\theta\theta\mathbf{P}\mathbf{\Pi}\mathbf{\Sigma}\sigma$
 mathbf fup: $\alpha\alpha\alpha\beta\beta\mathbf{G}\mathbf{T}\varepsilon\varepsilon\theta\theta\mathbf{P}\mathbf{\Pi}\mathbf{\Sigma}\sigma$

Default: $\alpha\alpha\alpha\beta\mathit{G}\Gamma\epsilon\epsilon\theta\theta\mathit{P}\Pi\mathit{\Sigma}\sigma$; $\sigma_\epsilon, c^\alpha$
mathnormal: $\alpha\alpha\alpha\beta\mathit{G}\Gamma\epsilon\epsilon\theta\theta\mathit{P}\Pi\mathit{\Sigma}\sigma$
mathrm: $\alpha\alpha\alpha\beta\mathit{G}\Gamma\epsilon\epsilon\theta\theta\mathit{P}\Pi\mathit{\Sigma}\sigma$
mathup: $\alpha\alpha\alpha\beta\mathit{G}\Gamma\epsilon\epsilon\theta\theta\mathit{P}\Pi\mathit{\Sigma}\sigma$
mathit: $\alpha\alpha\alpha\beta\mathit{G}\Gamma\epsilon\epsilon\theta\theta\mathit{P}\Pi\mathit{\Sigma}\sigma$
mathbf: $\alpha\alpha\alpha\beta\mathit{G}\Gamma\epsilon\epsilon\theta\theta\mathit{P}\Pi\mathit{\Sigma}\sigma$
mathbfit: $\alpha\alpha\alpha\beta\mathit{G}\Gamma\epsilon\epsilon\theta\theta\mathit{P}\Pi\mathit{\Sigma}\sigma$
mathbfup: $\alpha\alpha\alpha\beta\mathit{G}\Gamma\epsilon\epsilon\theta\theta\mathit{P}\Pi\mathit{\Sigma}\sigma$

Default: $\alpha\alpha\alpha\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta\Pi\Pi\sigma$; σ_ϵ , c^α
 $\mathrm{mathnormal}$: $\alpha\alpha\alpha\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta\Pi\Pi\sigma$
 mathrm : $\alpha\alpha\alpha\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta\Pi\Pi\sigma$
 mathup : $\alpha\alpha\alpha\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta\Pi\Pi\sigma$
 mathit : $\alpha\alpha\alpha\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta\Pi\Pi\sigma$
 mathbf : $\alpha\alpha\alpha\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta\Pi\Pi\sigma$
 mathbf fit: $\alpha\alpha\alpha\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta\Pi\Pi\sigma$
 mathbf fup: $\alpha\alpha\alpha\beta\beta\Gamma\Gamma\epsilon\epsilon\theta\theta\Pi\Pi\sigma$

3.3.2 Formulas Sans Serif

$$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \text{Г}, \text{Δ}, \text{E}, \text{Z}, \text{H},$$

$$\Theta, \text{I}, \text{K}, \text{Λ}, \text{M}, \text{N}, \Xi, \text{O}, \text{Π}, \text{P}, \Sigma, \text{T}, \Upsilon, \Phi, \text{X}, \Psi, \Omega, \text{F},$$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \Gamma, \Delta, \text{E}, \text{Z}, \text{H}, \Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F},$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \Gamma, \Delta, \text{E}, \text{Z}, \text{H}, \Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F},$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \Gamma, \Delta, \text{E}, \text{Z}, \text{H}, \Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F},$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l l | l l /, i j, r n m, $\theta \Theta$, $\phi \psi$, --

Latin vs. Greek: $a \alpha, d \delta, e \varepsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l l | l l /, i j, r n m, $\theta \Theta$, $\phi \psi$, --

Latin vs. Greek: $a \alpha, d \delta, e \varepsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l l | l l /, i j, r n m, $\theta \Theta$, $\phi \psi$, --

Latin vs. Greek: $a \alpha, d \delta, e \varepsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$$

$$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 O O, 1 l l | l l /, i j, rn m, θ Θ, ϕ ψ, – –

Latin vs. Greek: a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, β β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.

3.3.3 Math Alphabets Sans Serif

Default

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, θ, π, ρ, ζ, φ,

Math Normal (\mathnormal)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, θ, π, ρ, ζ, φ,

Math Italic (\mathit)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, θ, π, ρ, ζ, φ,

Math Roman (`\mathrm`)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
 A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
 a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
 A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
 α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, ϑ, Ϙ, ϙ, Ϛ, ϛ, Ϝ,

Math Bold (`\mathbf`)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, θ, π, ρ, Ϛ, ϛ,

Caligraphic (`\mathcal`)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

Script (`\mathscr`)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

Fraktur (`\mathfrak`)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
 a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,

Blackboard Bold (`\mathbb`)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

3.3.4 Character Sidebearings Sans Serif

Default

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |Y| + |\Phi| + |X| + |\Psi| + |\Omega| +$
 $|\alpha| + |\beta| + |\gamma| + |\delta| + |\epsilon| + |\zeta| + |\eta| + |\theta| + |\iota| + |\kappa| + |\lambda| + |\mu| +$
 $|\nu| + |\xi| + |\omicron| + |\pi| + |\rho| + |\sigma| + |\tau| + |\upsilon| + |\phi| + |\chi| + |\psi| + |\omega| +$
 $|\varepsilon| + |\vartheta| + |\varpi| + |\rho| + |\varsigma| + |\phi| +$

Math Roman ($\backslash\mathrm{rm}$)

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |\Upsilon| + |\Phi| + |X| + |\Psi| + |\Omega| +$

Math Bold ($\backslash\mathbf{bf}$)

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |Y| + |\Phi| + |X| + |\Psi| + |\Omega| +$

Math Calligraphic ($\backslash\mathrm{cal}$)

$|\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{D}| + |\mathcal{E}| + |\mathcal{F}| + |\mathcal{G}| + |\mathcal{H}| + |\mathcal{I}| + |\mathcal{J}| + |\mathcal{K}| + |\mathcal{L}| + |\mathcal{M}| +$
 $|\mathcal{N}| + |\mathcal{O}| + |\mathcal{P}| + |\mathcal{Q}| + |\mathcal{R}| + |\mathcal{S}| + |\mathcal{T}| + |\mathcal{U}| + |\mathcal{V}| + |\mathcal{W}| + |\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}| +$

3.3.5 Superscript Positioning Sans Serif

Default

$$\begin{aligned}
 &A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 + \\
 &N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 + \\
 &a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 + \\
 &n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 + \\
 &A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 + \\
 &N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + Y^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 + \\
 &\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \varepsilon^2 + \zeta^2 + \eta^2 + \theta^2 + \iota^2 + \kappa^2 + \lambda^2 + \mu^2 + \\
 &\nu^2 + \xi^2 + o^2 + \pi^2 + \rho^2 + \sigma^2 + \tau^2 + \upsilon^2 + \phi^2 + \chi^2 + \psi^2 + \omega^2 + \\
 &\varepsilon^2 + \theta^2 + \pi^2 + \rho^2 + \varsigma^2 + \phi^2 +
 \end{aligned}$$

Math Roman (`\mathrm`)

$$\begin{aligned}
 &A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 + \\
 &N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 + \\
 &a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 + \\
 &n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 + \\
 &A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 + \\
 &N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 +
 \end{aligned}$$

Math Bold (`\mathbf`)

$$\begin{aligned}
 &\mathbf{A}^2 + \mathbf{B}^2 + \mathbf{C}^2 + \mathbf{D}^2 + \mathbf{E}^2 + \mathbf{F}^2 + \mathbf{G}^2 + \mathbf{H}^2 + \mathbf{I}^2 + \mathbf{J}^2 + \mathbf{K}^2 + \mathbf{L}^2 + \mathbf{M}^2 + \\
 &\mathbf{N}^2 + \mathbf{O}^2 + \mathbf{P}^2 + \mathbf{Q}^2 + \mathbf{R}^2 + \mathbf{S}^2 + \mathbf{T}^2 + \mathbf{U}^2 + \mathbf{V}^2 + \mathbf{W}^2 + \mathbf{X}^2 + \mathbf{Y}^2 + \mathbf{Z}^2 + \\
 &\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2 + \mathbf{d}^2 + \mathbf{e}^2 + \mathbf{f}^2 + \mathbf{g}^2 + \mathbf{h}^2 + \mathbf{i}^2 + \mathbf{j}^2 + \mathbf{k}^2 + \mathbf{l}^2 + \mathbf{m}^2 + \\
 &\mathbf{n}^2 + \mathbf{o}^2 + \mathbf{p}^2 + \mathbf{q}^2 + \mathbf{r}^2 + \mathbf{s}^2 + \mathbf{t}^2 + \mathbf{u}^2 + \mathbf{v}^2 + \mathbf{w}^2 + \mathbf{x}^2 + \mathbf{y}^2 + \mathbf{z}^2 + \\
 &\mathbf{A}^2 + \mathbf{B}^2 + \mathbf{\Gamma}^2 + \mathbf{\Delta}^2 + \mathbf{E}^2 + \mathbf{Z}^2 + \mathbf{H}^2 + \mathbf{\Theta}^2 + \mathbf{I}^2 + \mathbf{K}^2 + \mathbf{\Lambda}^2 + \mathbf{M}^2 + \\
 &\mathbf{N}^2 + \mathbf{\Xi}^2 + \mathbf{O}^2 + \mathbf{\Pi}^2 + \mathbf{P}^2 + \mathbf{\Sigma}^2 + \mathbf{T}^2 + \mathbf{Y}^2 + \mathbf{\Phi}^2 + \mathbf{X}^2 + \mathbf{\Psi}^2 + \mathbf{\Omega}^2 +
 \end{aligned}$$

Math Calligraphic (`\mathcal`)

$$\begin{aligned}
 &\mathcal{A}^2 + \mathcal{B}^2 + \mathcal{C}^2 + \mathcal{D}^2 + \mathcal{E}^2 + \mathcal{F}^2 + \mathcal{G}^2 + \mathcal{H}^2 + \mathcal{I}^2 + \mathcal{J}^2 + \mathcal{K}^2 + \mathcal{L}^2 + \mathcal{M}^2 + \\
 &\mathcal{N}^2 + \mathcal{O}^2 + \mathcal{P}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{S}^2 + \mathcal{T}^2 + \mathcal{U}^2 + \mathcal{V}^2 + \mathcal{W}^2 + \mathcal{X}^2 + \mathcal{Y}^2 + \mathcal{Z}^2 +
 \end{aligned}$$

3.3.6 Subscript Positioning Sans Serif

Default

$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$
 $N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$
 $a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$
 $n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$
 $A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i +$
 $N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + Y_i + \Phi_i + X_i + \Psi_i + \Omega_i +$
 $\alpha_i + \beta_i + \gamma_i + \delta_i + \varepsilon_i + \zeta_i + \eta_i + \theta_i + \iota_i + \kappa_i + \lambda_i + \mu_i +$
 $\nu_i + \xi_i + o_i + \pi_i + \rho_i + \sigma_i + \tau_i + \upsilon_i + \phi_i + \chi_i + \psi_i + \omega_i +$
 $\varepsilon_i + \theta_i + \pi_i + \rho_i + \varsigma_i + \phi_i +$

Math Roman ($\backslash\mathrm{rm}$)

$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$
 $N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$
 $a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$
 $n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$
 $A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i +$
 $N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i +$

Math Bold ($\backslash\mathbf{bf}$)

$\mathbf{A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +}$
 $\mathbf{N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +}$
 $\mathbf{a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +}$
 $\mathbf{n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +}$
 $\mathbf{A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i +}$
 $\mathbf{N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + Y_i + \Phi_i + X_i + \Psi_i + \Omega_i +}$

Math Calligraphic ($\backslash\mathrm{cal}$)

$\mathcal{A}_i + \mathcal{B}_i + \mathcal{C}_i + \mathcal{D}_i + \mathcal{E}_i + \mathcal{F}_i + \mathcal{G}_i + \mathcal{H}_i + \mathcal{I}_i + \mathcal{J}_i + \mathcal{K}_i + \mathcal{L}_i + \mathcal{M}_i +$
 $\mathcal{N}_i + \mathcal{O}_i + \mathcal{P}_i + \mathcal{Q}_i + \mathcal{R}_i + \mathcal{S}_i + \mathcal{T}_i + \mathcal{U}_i + \mathcal{V}_i + \mathcal{W}_i + \mathcal{X}_i + \mathcal{Y}_i + \mathcal{Z}_i +$

3.3.7 Accent Positioning Sans Serif

Default

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{T} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{Y} + \hat{\Phi} + \hat{\chi} + \hat{\Psi} + \hat{\Omega} +$
 $\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{\iota} + \hat{\kappa} + \hat{\lambda} + \hat{\mu} +$
 $\hat{\nu} + \hat{\xi} + \hat{o} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{u} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$
 $\hat{\epsilon} + \hat{\theta} + \hat{\pi} + \hat{\rho} + \hat{\zeta} + \hat{\phi} +$

Math Italic (`\mathit`)

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} + \hat{\ell} + \hat{\wp} + \hat{\imath} + \hat{j} + \hat{\tilde{i}}$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{T} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{Y} + \hat{\Phi} + \hat{\chi} + \hat{\Psi} + \hat{\Omega} +$
 $\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{\iota} + \hat{\kappa} + \hat{\lambda} + \hat{\mu} +$
 $\hat{\nu} + \hat{\xi} + \hat{o} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{u} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$
 $\hat{\epsilon} + \hat{\theta} + \hat{\pi} + \hat{\rho} + \hat{\zeta} + \hat{\phi} +$

Math Roman (`\mathrm`)

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{T} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{Y} + \hat{\Phi} + \hat{\chi} + \hat{\Psi} + \hat{\Omega} +$

Math Bold ($\backslash\mathrm{bf}$)

$\hat{\mathbf{O}} + \hat{\mathbf{1}} + \hat{\mathbf{2}} + \hat{\mathbf{3}} + \hat{\mathbf{4}} + \hat{\mathbf{5}} + \hat{\mathbf{6}} + \hat{\mathbf{7}} + \hat{\mathbf{8}} + \hat{\mathbf{9}} +$
 $\hat{\mathbf{A}} + \hat{\mathbf{B}} + \hat{\mathbf{C}} + \hat{\mathbf{D}} + \hat{\mathbf{E}} + \hat{\mathbf{F}} + \hat{\mathbf{G}} + \hat{\mathbf{H}} + \hat{\mathbf{I}} + \hat{\mathbf{J}} + \hat{\mathbf{K}} + \hat{\mathbf{L}} + \hat{\mathbf{M}} +$
 $\hat{\mathbf{N}} + \hat{\mathbf{O}} + \hat{\mathbf{P}} + \hat{\mathbf{Q}} + \hat{\mathbf{R}} + \hat{\mathbf{S}} + \hat{\mathbf{T}} + \hat{\mathbf{U}} + \hat{\mathbf{V}} + \hat{\mathbf{W}} + \hat{\mathbf{X}} + \hat{\mathbf{Y}} + \hat{\mathbf{Z}} +$
 $\hat{\mathbf{a}} + \hat{\mathbf{b}} + \hat{\mathbf{c}} + \hat{\mathbf{d}} + \hat{\mathbf{e}} + \hat{\mathbf{f}} + \hat{\mathbf{g}} + \hat{\mathbf{h}} + \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}} + \hat{\mathbf{l}} + \hat{\mathbf{m}} +$
 $\hat{\mathbf{n}} + \hat{\mathbf{o}} + \hat{\mathbf{p}} + \hat{\mathbf{q}} + \hat{\mathbf{r}} + \hat{\mathbf{s}} + \hat{\mathbf{t}} + \hat{\mathbf{u}} + \hat{\mathbf{v}} + \hat{\mathbf{w}} + \hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}} +$
 $\hat{\mathbf{A}} + \hat{\mathbf{B}} + \hat{\mathbf{r}} + \hat{\mathbf{\Delta}} + \hat{\mathbf{E}} + \hat{\mathbf{Z}} + \hat{\mathbf{H}} + \hat{\mathbf{\Theta}} + \hat{\mathbf{I}} + \hat{\mathbf{K}} + \hat{\mathbf{\Lambda}} + \hat{\mathbf{M}} +$
 $\hat{\mathbf{N}} + \hat{\mathbf{\Xi}} + \hat{\mathbf{O}} + \hat{\mathbf{\Pi}} + \hat{\mathbf{P}} + \hat{\mathbf{\Sigma}} + \hat{\mathbf{T}} + \hat{\mathbf{Y}} + \hat{\mathbf{\Phi}} + \hat{\mathbf{X}} + \hat{\mathbf{\Psi}} + \hat{\mathbf{\Omega}} +$

Math Calligraphic ($\backslash\mathrm{cal}$)

$\hat{\mathcal{A}} + \hat{\mathcal{B}} + \hat{\mathcal{C}} + \hat{\mathcal{D}} + \hat{\mathcal{E}} + \hat{\mathcal{F}} + \hat{\mathcal{G}} + \hat{\mathcal{H}} + \hat{\mathcal{I}} + \hat{\mathcal{J}} + \hat{\mathcal{K}} + \hat{\mathcal{L}} + \hat{\mathcal{M}} +$
 $\hat{\mathcal{N}} + \hat{\mathcal{O}} + \hat{\mathcal{P}} + \hat{\mathcal{Q}} + \hat{\mathcal{R}} + \hat{\mathcal{S}} + \hat{\mathcal{T}} + \hat{\mathcal{U}} + \hat{\mathcal{V}} + \hat{\mathcal{W}} + \hat{\mathcal{X}} + \hat{\mathcal{Y}} + \hat{\mathcal{Z}} +$

3.3.8 Differentials Sans Serif

$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$
 $dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$
 $da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$
 $dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + dY + d\Phi + dX + d\Psi + d\Omega +$
 $d\alpha + d\beta + d\gamma + d\delta + d\varepsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$
 $dv + d\xi + do + d\pi + dp + d\sigma + d\tau + du + d\phi + d\chi + d\psi + d\omega +$
 $d\varepsilon + d\theta + d\pi + dp + d\varsigma + d\phi +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + dY + d\Phi + dX + d\Psi + d\Omega +$

$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$
 $dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$
 $da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$
 $dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + dY + d\Phi + dX + d\Psi + d\Omega +$
 $d\alpha + d\beta + d\gamma + d\delta + d\varepsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$
 $dv + d\xi + do + d\pi + d\rho + d\sigma + d\tau + d\nu + d\phi + d\chi + d\psi + d\omega +$
 $d\varepsilon + d\theta + d\pi + d\rho + d\varsigma + d\phi +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + dY + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$

$\partial A + \partial B + \partial C + \partial D + \partial E + \partial F + \partial G + \partial H + \partial I + \partial J + \partial K + \partial L + \partial M +$
 $\partial N + \partial O + \partial P + \partial Q + \partial R + \partial S + \partial T + \partial U + \partial V + \partial W + \partial X + \partial Y + \partial Z +$
 $\partial a + \partial b + \partial c + \partial d + \partial e + \partial f + \partial g + \partial h + \partial i + \partial j + \partial k + \partial l + \partial m +$
 $\partial n + \partial o + \partial p + \partial q + \partial r + \partial s + \partial t + \partial u + \partial v + \partial w + \partial x + \partial y + \partial z +$
 $\partial A + \partial B + \partial \Gamma + \partial \Delta + \partial E + \partial Z + \partial H + \partial \Theta + \partial I + \partial K + \partial \Lambda + \partial M +$
 $\partial N + \partial \Xi + \partial O + \partial \Pi + \partial P + \partial \Sigma + \partial T + \partial Y + \partial \Phi + \partial X + \partial \Psi + \partial \Omega +$
 $\partial \alpha + \partial \beta + \partial \gamma + \partial \delta + \partial \varepsilon + \partial \zeta + \partial \eta + \partial \theta + \partial \iota + \partial \kappa + \partial \lambda + \partial \mu +$
 $\partial \nu + \partial \xi + \partial o + \partial \pi + \partial \rho + \partial \sigma + \partial \tau + \partial \nu + \partial \phi + \partial \chi + \partial \psi + \partial \omega +$
 $\partial \varepsilon + \partial \theta + \partial \pi + \partial \rho + \partial \varsigma + \partial \phi +$
 $\partial A + \partial B + \partial \Gamma + \partial \Delta + \partial E + \partial Z + \partial H + \partial \Theta + \partial I + \partial K + \partial \Lambda + \partial M +$
 $\partial N + \partial \Xi + \partial O + \partial \Pi + \partial P + \partial \Sigma + \partial T + \partial Y + \partial \Upsilon + \partial \Phi + \partial X + \partial \Psi + \partial \Omega +$

3.3.9 Slash Kerning **Sans Serif**

$1/A + 1/B + 1/C + 1/D + 1/E + 1/F + 1/G + 1/H + 1/I + 1/J + 1/K + 1/L + 1/M +$
 $1/N + 1/O + 1/P + 1/Q + 1/R + 1/S + 1/T + 1/U + 1/V + 1/W + 1/X + 1/Y + 1/Z +$
 $1/a + 1/b + 1/c + 1/d + 1/e + 1/f + 1/g + 1/h + 1/i + 1/j + 1/k + 1/l + 1/m +$
 $1/n + 1/o + 1/p + 1/q + 1/r + 1/s + 1/t + 1/u + 1/v + 1/w + 1/x + 1/y + 1/z +$
 $1/A + 1/B + 1/\Gamma + 1/\Delta + 1/E + 1/Z + 1/H + 1/\Theta + 1/I + 1/K + 1/\Lambda + 1/M +$
 $1/N + 1/\Xi + 1/O + 1/\Pi + 1/P + 1/\Sigma + 1/T + 1/Y + 1/\Phi + 1/X + 1/\Psi + 1/\Omega +$
 $1/\alpha + 1/\beta + 1/\gamma + 1/\delta + 1/\varepsilon + 1/\zeta + 1/\eta + 1/\theta + 1/\iota + 1/\kappa + 1/\lambda + 1/\mu +$
 $1/\nu + 1/\xi + 1/o + 1/\pi + 1/\rho + 1/\sigma + 1/\tau + 1/\upsilon + 1/\phi + 1/\chi + 1/\psi + 1/\omega +$
 $1/\varepsilon + 1/\theta + 1/\pi + 1/\rho + 1/\varsigma + 1/\phi +$

$$\begin{aligned} &A/2 + B/2 + C/2 + D/2 + E/2 + F/2 + G/2 + H/2 + I/2 + J/2 + K/2 + L/2 + M/2 + \\ &N/2 + O/2 + P/2 + Q/2 + R/2 + S/2 + T/2 + U/2 + V/2 + W/2 + X/2 + Y/2 + Z/2 + \\ &a/2 + b/2 + c/2 + d/2 + e/2 + f/2 + g/2 + h/2 + i/2 + j/2 + k/2 + l/2 + m/2 + \\ &n/2 + o/2 + p/2 + q/2 + r/2 + s/2 + t/2 + u/2 + v/2 + w/2 + x/2 + y/2 + z/2 + \\ &A/2 + B/2 + \Gamma/2 + \Delta/2 + E/2 + Z/2 + H/2 + \Theta/2 + I/2 + K/2 + \Lambda/2 + M/2 + \\ &N/2 + \Xi/2 + O/2 + \Pi/2 + P/2 + \Sigma/2 + T/2 + Y/2 + \Phi/2 + X/2 + \Psi/2 + \Omega/2 + \\ &\alpha/2 + \beta/2 + \gamma/2 + \delta/2 + \varepsilon/2 + \zeta/2 + \eta/2 + \theta/2 + \iota/2 + \kappa/2 + \lambda/2 + \mu/2 + \\ &\nu/2 + \xi/2 + o/2 + \pi/2 + \rho/2 + \sigma/2 + \tau/2 + \upsilon/2 + \phi/2 + \chi/2 + \psi/2 + \omega/2 + \\ &\varepsilon/2 + \theta/2 + \pi/2 + \rho/2 + \varsigma/2 + \phi/2 + \end{aligned}$$

3.3.10 (Big) Operators Sans Serif

$$\begin{array}{ccccccc} \sum_{i=1}^n x^n & \prod_{i=1}^n x^n & \coprod_{i=1}^n x^n & \int_{i=1}^n x^n & \oint_{i=1}^n x^n & \uplus_{i=1}^n x^n & \cup_{i=1}^n x^n \\ \otimes_{i=1}^n x^n & \oplus_{i=1}^n x^n & \odot_{i=1}^n x^n & \bigwedge_{i=1}^n x^n & \bigvee_{i=1}^n x^n & \cap_{i=1}^n x^n & \sqcup_{i=1}^n x^n \end{array}$$

$$\sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n$$

$$\bigotimes_{i=1}^n x^n \quad \bigoplus_{i=1}^n x^n \quad \bigodot_{i=1}^n x^n \quad \bigwedge_{i=1}^n x^n \quad \bigvee_{i=1}^n x^n \quad \uplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n$$

3.3.11 Radicals Sans Serif

$$\sqrt{x+y} \quad \sqrt{x^2+y^2} \quad \sqrt{x_i^2+y_j^2} \quad \sqrt{\left(\frac{\cos x}{2}\right)} \quad \sqrt{\left(\frac{\sin x}{2}\right)}$$

$$\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{x+y}}}}}}}}$$

3.3.12 Over- and Underbraces Sans Serif

$$\overbrace{x} \quad \overbrace{x+y} \quad \overbrace{x^2+y^2} \quad \overbrace{x_i^2+y_j^2} \quad \underbrace{x} \quad \underbrace{x+y} \quad \underbrace{x_i+y_j} \quad \underbrace{x_i^2+y_j^2}$$

3.3.17 Binary Operators Sans Serif

$x \pm y$	<code>\pm</code>	$x \cap y$	<code>\cap</code>	$x \diamond y$	<code>\diamond</code>	$x \oplus y$	<code>\oplus</code>
$x \mp y$	<code>\mp</code>	$x \cup y$	<code>\cup</code>	$x \triangle y$	<code>\bigtriangleup</code>	$x \ominus y$	<code>\ominus</code>
$x \times y$	<code>\times</code>	$x \uplus y$	<code>\uplus</code>	$x \nabla y$	<code>\bigtriangledown</code>	$x \otimes y$	<code>\otimes</code>
$x \div y$	<code>\div</code>	$x \sqcap y$	<code>\sqcap</code>	$x \triangleleft y$	<code>\triangleleft</code>	$x \oslash y$	<code>\oslash</code>
$x * y$	<code>\ast</code>	$x \sqcup y$	<code>\sqcup</code>	$x \triangleright y$	<code>\triangleright</code>	$x \odot y$	<code>\odot</code>
$x \star y$	<code>\star</code>	$x \vee y$	<code>\vee</code>	$x \lhd y$	<code>\lhd</code>	$x \bigcirc y$	<code>\bigcirc</code>
$x \circ y$	<code>\circ</code>	$x \wedge y$	<code>\wedge</code>	$x \rhd y$	<code>\rhd</code>	$x \dagger y$	<code>\dagger</code>
$x \bullet y$	<code>\bullet</code>	$x \setminus y$	<code>\setminus</code>	$x \unlhd y$	<code>\unlhd</code>	$x \ddagger y$	<code>\ddagger</code>
$x \cdot y$	<code>\cdot</code>	$x \wr y$	<code>\wr</code>	$x \unrhd y$	<code>\unrhd</code>	$x \S y$	<code>\S</code>
$x + y$	<code>+</code>	$x - y$	<code>-</code>	$x \amalg y$	<code>\amalg</code>	$x \P y$	<code>\P</code>

3.3.18 Relations Sans Serif

$x \leq y$	<code>\leq</code>	$x \geq y$	<code>\geq</code>	$x \equiv y$	<code>\equiv</code>	$x \models y$	<code>\models</code>
$x < y$	<code>\prec</code>	$x > y$	<code>\succ</code>	$x \sim y$	<code>\sim</code>	$x \perp y$	<code>\perp</code>
$x \leq y$	<code>\preceq</code>	$x \geq y$	<code>\succeq</code>	$x \simeq y$	<code>\simeq</code>	$x y$	<code>\mid</code>
$x \ll y$	<code>\ll</code>	$x \gg y$	<code>\gg</code>	$x \asymp y$	<code>\asymp</code>	$x \parallel y$	<code>\parallel</code>
$x \subset y$	<code>\subset</code>	$x \supset y$	<code>\supset</code>	$x \approx y$	<code>\approx</code>	$x \bowtie y$	<code>\bowtie</code>
$x \subseteq y$	<code>\subseteq</code>	$x \supseteq y$	<code>\supseteq</code>	$x \cong y$	<code>\cong</code>	$x \Join y$	<code>\Join</code>
$x \sqsubset y$	<code>\sqsubset</code>	$x \sqsupset y$	<code>\sqsupset</code>	$x \neq y$	<code>\neq</code>	$x \smile y$	<code>\smile</code>
$x \sqsubseteq y$	<code>\sqsubseteq</code>	$x \sqsupseteq y$	<code>\sqsupseteq</code>	$x \doteq y$	<code>\doteq</code>	$x \frown y$	<code>\frown</code>
$x \in y$	<code>\in</code>	$x \ni y$	<code>\ni</code>	$x \propto y$	<code>\propto</code>	$x = y$	<code>=</code>
$x \vdash y$	<code>\vdash</code>	$x \dashv y$	<code>\dashv</code>	$x < y$	<code><</code>	$x > y$	<code>></code>
$x : y$	<code>:</code>						

3.3.19 Punctuation Sans Serif

x, y	<code>,</code>	$x; y$	<code>;</code>	$x : y$	<code>\colon</code>	$x \cdot y$	<code>\ldotp</code>	$x \cdot y$	<code>\cdotp</code>
--------	----------------	--------	----------------	---------	---------------------	-------------	---------------------	-------------	---------------------

$x \arccos y$	$x \cos y$	$x \csc y$	$x \exp y$	$x \ker y$	$x \limsup y$	$x \min y$	$x \sinh y$
$x \arcsin y$	$x \cosh y$	$x \deg y$	$x \gcd y$	$x \lg y$	$x \ln y$	$x \Pr y$	$x \sup y$
$x \arctan y$	$x \cot y$	$x \det y$	$x \hom y$	$x \lim y$	$x \log y$	$x \sec y$	$x \tan y$
$x \arg y$	$x \coth y$	$x \dim y$	$x \inf y$	$x \liminf y$	$x \max y$	$x \sin y$	$x \tanh y$

3.3.24 Delimiters Sans Serif

$x(y$	$($	$x)y$	$)$	$x \uparrow y$	<code>\uparrow</code>	$x \Uparrow y$	<code>\Uparrow</code>
$x[y$	$[$	$x]y$	$]$	$x \downarrow y$	<code>\downarrow</code>	$x \Downarrow y$	<code>\Downarrow</code>
$x\{y$	<code>\{</code>	$x\}y$	<code>\}</code>	$x \updownarrow y$	<code>\updownarrow</code>	$x \Updownarrow y$	<code>\Updownarrow</code>
$x\lfloor y$	<code>\lfloor</code>	$x\rfloor y$	<code>\rfloor</code>	$x\lceil y$	<code>\lceil</code>	$x\rceil y$	<code>\rceil</code>
$x\langle y$	<code>\langle</code>	$x\rangle y$	<code>\rangle</code>	x/y	<code>/</code>	$x\backslash y$	<code>\backslash</code>
$x y$	<code> </code>	$x y$	<code>\ </code>				

3.3.25 Large Delimiters Sans Serif

$\left($	<code>\rmoustache</code>	$\int$	<code>\lmoustache</code>	$\right)$	<code>\rgroup</code>	$\left($	<code>\lgroup</code>
$\Big $	<code>\arrowvert</code>	$\Big\ $	<code>\Arrowvert</code>	$\Big $	<code>\bracevert</code>		

3.3.26 Math Mode Accents Sans Serif

$\hat{a}$	<code>\hat{a}</code>	$\acute{a}$	<code>\acute{a}</code>	$\bar{a}$	<code>\bar{a}</code>	$\dot{a}$	<code>\dot{a}</code>	$\breve{a}$	<code>\breve{a}</code>
$\check{a}$	<code>\check{a}</code>	$\grave{a}$	<code>\grave{a}</code>	$\vec{a}$	<code>\vec{a}</code>	$\ddot{a}$	<code>\ddot{a}</code>	$\tilde{a}$	<code>\tilde{a}</code>

3.3.27 Miscellaneous Constructions Sans Serif

$\widetilde{abc}$	<code>\widetilde{abc}</code>	$\widehat{abc}$	<code>\widehat{abc}</code>
$\overleftarrow{abc}$	<code>\overleftarrow{abc}</code>	$\overrightarrow{abc}$	<code>\overrightarrow{abc}</code>
$\overline{abc}$	<code>\overline{abc}</code>	$\underline{abc}$	<code>\underline{abc}</code>
$\overbrace{abc}$	<code>\overbrace{abc}</code>	$\underbrace{abc}$	<code>\underbrace{abc}</code>
$\sqrt{abc}$	<code>\sqrt{abc}</code>	$\sqrt[n]{abc}$	<code>\sqrt[n]{abc}</code>
f'	<code>f'</code>	$\frac{abc}{xyz}$	<code>\frac{abc}{xyz}</code>

3.3.28 AMS Delimiters Sans Serif

$x\ulcorner y$	<code>\ulcorner</code>	$x\urcorner y$	<code>\urcorner</code>	$x\llcorner y$	<code>\llcorner</code>	$x\lrcorner y$	<code>\lrcorner</code>
----------------	------------------------	----------------	------------------------	----------------	------------------------	----------------	------------------------

3.3.29 AMS Arrows Sans Serif

$x \dashrightarrow y$	<code>\dashrightarrow</code>	$x \dashleftarrow y$	<code>\dashleftarrow</code>
$x \leftrightsquigarrow y$	<code>\leftrightsquigarrow</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \Lleftarrow y$	<code>\Lleftarrow</code>	$x \twoheadleftarrow y$	<code>\twoheadleftarrow</code>
$x \leftarrowtail y$	<code>\leftarrowtail</code>	$x \looparrowleft y$	<code>\looparrowleft</code>
$x \leftrightharpoons y$	<code>\leftrightharpoons</code>	$x \curvearrowleft y$	<code>\curvearrowleft</code>
$x \circlearrowleft y$	<code>\circlearrowleft</code>	$x \lsh y$	<code>\lsh</code>
$x \upuparrows y$	<code>\upuparrows</code>	$x \upharpoonleft y$	<code>\upharpoonleft</code>
$x \downharpoonleft y$	<code>\downharpoonleft</code>	$x \multimap y$	<code>\multimap</code>
$x \leftrightsquigarrow y$	<code>\leftrightsquigarrow</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \rightleftarrows y$	<code>\rightleftarrows</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \rightleftarrows y$	<code>\rightleftarrows</code>	$x \twoheadrightarrow y$	<code>\twoheadrightarrow</code>
$x \rightarrowtail y$	<code>\rightarrowtail</code>	$x \looparrowright y$	<code>\looparrowright</code>
$x \rightleftharpoons y$	<code>\rightleftharpoons</code>	$x \curvearrowright y$	<code>\curvearrowright</code>
$x \circlearrowright y$	<code>\circlearrowright</code>	$x \Rsh y$	<code>\Rsh</code>
$x \downdownarrows y$	<code>\downdownarrows</code>	$x \upharpoonright y$	<code>\upharpoonright</code>
$x \downharpoonright y$	<code>\downharpoonright</code>	$x \rightsquigarrow y$	<code>\rightsquigarrow</code>

3.3.30 AMS Negated Arrows Sans Serif

$x \nleftarrow y$	<code>\nleftarrow</code>	$x \nrightarrow y$	<code>\nrightarrow</code>
$x \nLleftarrow y$	<code>\nLleftarrow</code>	$x \nRrightarrow y$	<code>\nRrightarrow</code>
$x \nleftrightarrow y$	<code>\nleftrightarrow</code>	$x \nLeftrightarrow y$	<code>\nLeftrightarrow</code>

3.3.31 AMS Greek Sans Serif

$x \digamma y$ `\digamma` $x \varkappa y$ `\varkappa`

3.3.32 AMS Hebrew Sans Serif

$x \beth y$ `\beth` $x \daleth y$ `\daleth` $x \gimel y$ `\gimel`

3.3.33 AMS Miscellaneous Sans Serif

$x\hbar y$	<code>\hbar</code>	$x\hslash y$	<code>\hslash</code>
$x\triangle y$	<code>\vartriangle</code>	$x\nabla y$	<code>\triangledown</code>
$x\square y$	<code>\square</code>	$x\lozenge y$	<code>\lozenge</code>
$x\circledcirc y$	<code>\circledcirc</code>	$x\angle y$	<code>\angle</code>
$x\measuredangle y$	<code>\measuredangle</code>	$x\nexists y$	<code>\nexists</code>
$x\mho y$	<code>\mho</code>	$x\Finv y$	<code>\Finv</code> ^u
$x\Game y$	<code>\Game</code> ^u	$x\Bbbk y$	<code>\Bbbk</code> ^u
$x\backprime y$	<code>\backprime</code>	$x\varnothing y$	<code>\varnothing</code>
$x\blacktriangle y$	<code>\blacktriangle</code>	$x\blacktriangledown y$	<code>\blacktriangledown</code>
$x\blacksquare y$	<code>\blacksquare</code>	$x\blacklozenge y$	<code>\blacklozenge</code>
$x\bigstar y$	<code>\bigstar</code>	$x\sphericalangle y$	<code>\sphericalangle</code>
$x\complement y$	<code>\complement</code>	$x\eth y$	<code>\eth</code>
$x\diagup y$	<code>\diagup</code> ^u	$x\diagdown y$	<code>\diagdown</code> ^u

^u Not defined in `amssymb.sty`, define using the `\newsymbol` command.

3.3.34 AMS Binary Operators Sans Serif

$x\dotplus y$	<code>\dotplus</code>	$x\smallsetminus y$	<code>\smallsetminus</code>
$x\cap y$	<code>\Cap</code>	$x\cup y$	<code>\Cup</code>
$x\bar{\wedge} y$	<code>\barwedge</code>	$x\veebar y$	<code>\veebar</code>
$x\overline{\wedge} y$	<code>\doublebarwedge</code>	$x\boxminus y$	<code>\boxminus</code>
$x\boxtimes y$	<code>\boxtimes</code>	$x\boxdot y$	<code>\boxdot</code>
$x\boxplus y$	<code>\boxplus</code>	$x\divideontimes y$	<code>\divideontimes</code>
$x\ltimes y$	<code>\ltimes</code>	$x\rtimes y$	<code>\rtimes</code>
$x\leftthreetimes y$	<code>\leftthreetimes</code>	$x\rightthreetimes y$	<code>\rightthreetimes</code>
$x\curlywedge y$	<code>\curlywedge</code>	$x\curlyvee y$	<code>\curlyvee</code>
$x\ominus y$	<code>\circleddash</code>	$x\circledast y$	<code>\circledast</code>
$x\odot y$	<code>\circledcirc</code>	$x\centerdot y$	<code>\centerdot</code>
$x\intercal y$	<code>\intercal</code>		

3.3.35 AMS Relations Sans Serif

$x \leqslant y$	<code>\leqslant</code>
$x \lesssim y$	<code>\lesssim</code>
$x \approx y$	<code>\approx</code>
$x \lll y$	<code>\lll</code>
$x \lesseqgtr y$	<code>\lesseqgtr</code>
$x \doteqdot y$	<code>\doteqdot</code>
$x \fallingdotseq y$	<code>\fallingdotseq</code>
$x \subseteq y$	<code>\subseteq</code>
$x \Subset y$	<code>\Subset</code>
$x \preccurlyeq y$	<code>\preccurlyeq</code>
$x \prec\sim y$	<code>\prec\sim</code>
$x \triangleleft y$	<code>\triangleleft</code>
$x \vDash y$	<code>\vDash</code>
$x \smallsmile y$	<code>\smallsmile</code>
$x \bumpeq y$	<code>\bumpeq</code>
$x \geqq y$	<code>\geqq</code>
$x \gtrless y$	<code>\gtrless</code>
$x \gtrapprox y$	<code>\gtrapprox</code>
$x \ggg y$	<code>\ggg</code>
$x \gtreqless y$	<code>\gtreqless</code>
$x \eqcirc y$	<code>\eqcirc</code>
$x \triangleq y$	<code>\triangleq</code>
$x \thickapprox y$	<code>\thickapprox</code>
$x \supseteq y$	<code>\supseteq</code>
$x \succcurlyeq y$	<code>\succcurlyeq</code>
$x \succsim y$	<code>\succsim</code>
$x \triangleright y$	<code>\triangleright</code>
$x \Vdash y$	<code>\Vdash</code>
$x \parallel y$	<code>\parallel</code>
$x \pitchfork y$	<code>\pitchfork</code>
$x \blacktriangleleft y$	<code>\blacktriangleleft</code>
$x \backepsilon y$	<code>\backepsilon</code>
$x \because y$	<code>\because</code>

3.3.36 AMS Negated Relations Sans Serif

$x \nless y$	<code>\nless</code>	$x \nleq y$	<code>\nleq</code>
$x \nleqslant y$	<code>\nleqslant</code>	$x \nleqq y$	<code>\nleqq</code>
$x \lneq y$	<code>\lneq</code>	$x \lesseqgtr y$	<code>\lesseqgtr</code>
$x \lvertneqq y$	<code>\lvertneqq</code>	$x \lnsim y$	<code>\lnsim</code>
$x \lnapprox y$	<code>\lnapprox</code>	$x \nprec y$	<code>\nprec</code>
$x \npreceq y$	<code>\npreceq</code>	$x \precnsim y$	<code>\precnsim</code>
$x \precnapprox y$	<code>\precnapprox</code>	$x \nsim y$	<code>\nsim</code>
$x \nshortmid y$	<code>\nshortmid</code>	$x \nmid y$	<code>\nmid</code>
$x \nvDash y$	<code>\nvDash</code>	$x \nVDash y$	<code>\nVDash</code>
$x \ntriangleleft y$	<code>\ntriangleleft</code>	$x \ntrianglelefteq y$	<code>\ntrianglelefteq</code>
$x \nsubseteq y$	<code>\nsubseteq</code>	$x \subsetneq y$	<code>\subsetneq</code>
$x \varsubsetneq y$	<code>\varsubsetneq</code>	$x \subsetneqq y$	<code>\subsetneqq</code>
$x \varsubsetneqq y$	<code>\varsubsetneqq</code>	$x \ngtr y$	<code>\ngtr</code>
$x \ngeq y$	<code>\ngeq</code>	$x \ngeqslant y$	<code>\ngeqslant</code>
$x \ngeqq y$	<code>\ngeqq</code>	$x \gneq y$	<code>\gneq</code>
$x \gneqq y$	<code>\gneqq</code>	$x \gvertneqq y$	<code>\gvertneqq</code>
$x \gnsim y$	<code>\gnsim</code>	$x \gnapprox y$	<code>\gnapprox</code>
$x \nsucc y$	<code>\nsucc</code>	$x \nsucceq y$	<code>\nsucceq</code>
$x \nsucceqq y$	<code>\nsucceqq</code>	$x \succnsim y$	<code>\succnsim</code>
$x \succnapprox y$	<code>\succnapprox</code>	$x \ncong y$	<code>\ncong</code>
$x \nshortparallel y$	<code>\nshortparallel</code>	$x \nparallel y$	<code>\nparallel</code>
$x \nVDash y$	<code>\nVDash</code>	$x \nVdash y$	<code>\nVdash</code>
$x \ntriangleright y$	<code>\ntriangleright</code>	$x \ntrianglerighteq y$	<code>\ntrianglerighteq</code>
$x \nsupseteq y$	<code>\nsupseteq</code>	$x \nsupseteqq y$	<code>\nsupseteqq</code>
$x \supsetneq y$	<code>\supsetneq</code>	$x \supsetneqq y$	<code>\supsetneqq</code>
$x \supseteqq y$	<code>\supseteqq</code>	$x \varsupseteq y$	<code>\varsupseteq</code>

3.3.37 Math “Torture” Test Sans Serif

Most of the following examples are taken from *The T_EXbook* (Knuth 1984, see <https://ctan.org/pkg/texbook>) and were adapted for L^AT_EX from Karl Berry’s torture test for plain T_EX math fonts.

$x + y - z, \quad x + y * z, \quad z * y / z, \quad (x + y)(x - y) = x^2 - y^2,$
 $x \times y \cdot z = [xyz], \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,$
 $x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y / z, \quad x := y, \quad x \leq y \neq z, \quad x \sim y \simeq z \equiv y \not\equiv z, \quad x \subset y \subseteq z$
 $\sin 2\theta = 2 \sin \theta \cos \theta, \quad O(n \log n \log n), \quad \Pr(X > x) = \exp(-x/\mu),$
 $(x \in A(n) \mid x \in B(n)), \quad \bigcup_n X_n \parallel \bigcap_n Y_n$
 In-text matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} a & b & c \\ 1 & m & n \end{pmatrix}$.

$$a_0+\frac{1}{a_1+\frac{1}{a_2+\frac{1}{a_3+\frac{1}{a_4}}}}$$

$$\binom{p}{2}x^2y^{p-2}-\frac{1}{1-x}\frac{1}{1-x^2}=\frac{a+1}{b}\bigg/\frac{c+1}{d}.$$

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|\phi(x+iy)|^2=0$$

$$\pi(n)=\sum_{m=2}^n\left[\left(\sum_{k=1}^{m-1}\lfloor(m/k)/\lceil m/k\rceil\right)^{-1}\right].$$

$$\int_0^\infty \frac{t-ib}{t^2+b^2}e^{iat}\,dt=e^{ab}E_1(ab),\quad a,b>0.$$

$$\boldsymbol{A}:=\begin{pmatrix}x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda\end{pmatrix}.$$

$$\begin{pmatrix}a & b & c \\ d & e & f\end{pmatrix}\begin{pmatrix}u & x \\ v & y \\ w & z\end{pmatrix}$$

$$\boldsymbol{A}=\begin{pmatrix}a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn}\end{pmatrix}$$

$$\boldsymbol{M}=\begin{matrix} & C & I & C' \\ \begin{matrix} C \\ I \\ C' \end{matrix} & \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{matrix}$$

$$\sum_{n=0}^\infty a_n z^n \text{ converges if } |z|<\Big(\limsup_{n\rightarrow\infty}\sqrt[n]{|a_n|}\Big)^{-1}.$$

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \rightarrow f'(x) \quad \text{as } \Delta x \rightarrow 0.$$

$$\|u_i\| = 1, \quad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of } \begin{cases} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{cases} \text{ is } \begin{cases} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{cases}.$$

$$\begin{aligned} T(n) &\leq T(2^{\lceil \lg n \rceil}) \leq c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\ &< 3c \cdot 3^{\lg n} \\ &= 3c n^{\lg 3}. \end{aligned}$$

$$\begin{aligned} (x+y)(x-y) &= x^2 - xy + yx - y^2 \\ &= x^2 - y^2 \\ (x+y)^2 &= x^2 + 2xy + y^2. \end{aligned}$$

$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} dr d\theta \\ &= \int_0^{2\pi} \left(e^{-\frac{r^2}{2}} \Big|_{r=0}^{r=\infty} \right) d\theta \\ &= \pi. \end{aligned}$$

$$\prod_{k \geq 0} \frac{1}{(1 - q^k z)} = \sum_{n \geq 0} z^n / \prod_{1 \leq k \leq n} (1 - q^k).$$

$$\sum_{\substack{0 < i \leq m \\ 0 < j \leq n}} p(i,j) \neq \sum_{i=1}^p \sum_{j=1}^q \sum_{k=1}^r a_{ij} b_{jk} c_{ki} \neq \sum_{\substack{1 \leq i \leq p \\ 1 \leq j \leq q \\ 1 \leq k \leq r}} a_{ij} b_{jk} c_{ki}$$

$$\max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\text{Inline math: } \max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$p_1(n) = \lim_{m \rightarrow \infty} \sum_{v=0}^{\infty} (1 - \cos^{2m}(v!^n \pi/n))$$

$$\text{Inline math: } p_1(n) = \lim_{m \rightarrow \infty} \sum_{v=0}^{\infty} (1 - \cos^{2m}(v!^n \pi/n))$$

3.4 Math Test Sans Serif Bold

3.4.1 Overview **Sans Serif Bold**

Default: $a\alpha b\beta G\Gamma \varepsilon\theta\theta P\eta\Sigma\sigma; \sigma_\varepsilon, c^\alpha$

mathnormal: $\alpha\alpha b\beta G\Gamma \varepsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

 $\mathrm{a}\alpha\mathfrak{a}\mathfrak{b}\beta\mathrm{G}\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$

mathup: αααββΓΓεεθθΡΠΣσ

mathit: $\alpha\alpha\alpha b\beta G\Gamma\varepsilon\theta\theta P\Pi\Sigma\sigma$

mathbf: $\alpha\beta\Gamma\epsilon\theta\theta\rho\pi\sigma$

 $\mathbf{fit:} \alpha\beta\Gamma\epsilon\theta\theta\pi\sigma$ $\mathbf{up: \alpha\beta\Gamma\Gamma\varepsilon\theta\theta\rho\pi\sigma}$

Default: $a\alpha\alpha b\beta G\Gamma \varepsilon\theta\theta P\Pi\Sigma\sigma; \sigma_\varepsilon, c^\alpha$

mathnormal: $\alpha\alpha b\beta G\Gamma \varepsilon\epsilon\theta\theta P\Pi\Sigma\sigma$

 $\mathrm{a}\alpha\mathfrak{a}\mathfrak{b}\beta\mathsf{G}\Gamma\epsilon\varepsilon\theta\vartheta\mathsf{P}\Pi\Sigma\sigma$
$$\text{mathup: } \alpha\alpha\alpha\beta\beta\Gamma\Gamma\varepsilon\theta\theta\text{P}\Pi\Sigma\sigma$$
$$\text{mathit: } \alpha\alpha\alpha b\beta G\Gamma \varepsilon\theta\theta P\Pi\Sigma\sigma$$
$$\mathbf{mathbf{bf}}: \alpha\alpha b\beta G\Gamma \varepsilon\theta\theta P\Pi\Sigma\sigma$$
$$\mathbf{mathbf{fit:}}\alpha\alpha b\beta G\Gamma\varepsilon\theta\theta P\Pi\Sigma\sigma$$
 $\mathbf{up: \alpha\alpha\beta\beta\Gamma\Gamma\varepsilon\theta\theta\Pi\Sigma\sigma}$

Default: $a\alpha ab\beta G\Gamma \varepsilon\theta\theta P\Gamma\Sigma\sigma; \sigma_\varepsilon, c^\alpha$

mathnormal: $\alpha\alpha b\beta G\Gamma\varepsilon\theta\theta P\Pi\Sigma\sigma$

 $\mathrm{a}\alpha\mathfrak{a}\mathfrak{b}\beta\mathrm{G}\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
$$\text{mathup: } \alpha\alpha\beta\beta\Gamma\Gamma\varepsilon\theta\theta\text{P}\Pi\Sigma\sigma$$
$$\text{mathit: } a\alpha a b\beta G\Gamma \varepsilon \varepsilon \theta \theta P \Pi \Sigma \sigma$$

mathbf: *αααββγγεεθθρπςσ*

mathbf{fit}: ***αααββγγεεθθρπςσ***

 $\mathbf{\textbf{up}}$: ααββΓΓεεθθΡΠΣσ

Default: $\alpha\alpha\alpha b\beta G\Gamma\varepsilon\theta\theta P\Pi\Sigma\sigma; \sigma_\varepsilon, c^\alpha$

mathnormal: $\alpha\alpha b\beta G\Gamma\varepsilon\theta\theta P\Pi\Sigma\sigma$

 $\mathrm{a}\alpha\mathfrak{a}\mathfrak{b}\beta\mathrm{G}\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
$$\text{mathup: } \alpha\alpha\beta\beta\Gamma\Gamma\varepsilon\theta\theta\text{P}\Sigma\sigma$$
$$\text{mathit: } \alpha\alpha\alpha b\beta G\Gamma\epsilon\epsilon\theta\theta P\Pi\Sigma\sigma$$

mathbf: $\alpha\alpha b\beta G\Gamma \varepsilon\theta\theta P\Pi\Sigma\sigma$

 $\mathbf{fit:} \alpha\alpha b\beta G\Gamma \varepsilon\theta\theta P\eta\Sigma\sigma$

mathbfup: ααββΓΓεεθθΡΠΣσ

3.4.2 Formulas Sans Serif Bold

$$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \chi, \psi, \omega, f, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, \chi, \Psi, \Omega, F,$$

$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \text{f}, \Delta, \text{E}, \text{Z}, \text{H}, \Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F},$

$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \text{f}, \Delta, \text{E}, \text{Z}, \text{H}, \Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F},$

$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \text{f}, \Delta, \text{E}, \text{Z}, \text{H}, \Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F},$

$\alpha\alpha > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\alpha\alpha > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 0 O, 1 l l | l l /, i j, r n m, $\theta \theta$, $\phi \psi$, --

Latin vs. Greek: $a \alpha, d \delta, e \varepsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$\alpha\alpha > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 0 O, 1 l l | l l /, i j, r n m, $\theta \theta$, $\phi \psi$, --

Latin vs. Greek: $a \alpha, d \delta, e \varepsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$\alpha\alpha > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2} \right)$$

Disambiguation: 0 0 O, 1 l l | l l /, i j, r n m, $\theta \theta$, $\phi \psi$, --

Latin vs. Greek: $a \alpha, d \delta, e \varepsilon, i \iota, k \kappa, n \eta, o \sigma, p \rho, \beta \beta, u \upsilon, v \nu, w \omega, x \chi, y \gamma, A \Delta \Lambda, O \Theta \Omega, T \Gamma, Y \Upsilon$.

$$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$$

$$\lim_{v \rightarrow \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$\lim_{N \rightarrow \infty} \sum_{i=0}^N x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) dx = \left(\frac{27}{2}\right)$$

Disambiguation: 0 O O, 1 l l | l l /, i j, r n m, θ Θ, ϕ ψ, – –

Latin vs. Greek: a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, β β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.

3.4.3 Math Alphabets Sans Serif Bold

Default

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, Χ, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, θ, π, ρ, ζ, φ,

Math Normal (`\mathnormal`)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, Χ, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, θ, π, ρ, ζ, φ,

Math Italic (`\mathit`)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, Χ, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, θ, π, ρ, ζ, φ,

Math Roman (\mathrm)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, ϑ, ϖ, ϱ, ζ, φ,

Math Bold (\mathbf)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω,
α, β, γ, δ, ε, ζ, η, θ, ι, κ, λ, μ, ν, ξ, ο, π, ρ, σ, τ, υ, φ, χ, ψ, ω, ε, θ, π, ρ, ζ, φ,

Caligraphic (\mathcal)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

Script (\mathscr)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

Fraktur (\mathfrak)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,

Blackboard Bold (\mathbb)

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,

3.4.4 Character Sidebearings **Sans Serif Bold**

Default

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |Y| + |\Phi| + |X| + |\Psi| + |\Omega| +$
 $|\alpha| + |\beta| + |\gamma| + |\delta| + |\epsilon| + |\zeta| + |\eta| + |\theta| + |\iota| + |\kappa| + |\lambda| + |\mu| +$
 $|\nu| + |\xi| + |\omicron| + |\pi| + |\rho| + |\sigma| + |\tau| + |\upsilon| + |\phi| + |\chi| + |\psi| + |\omega| +$
 $|\varepsilon| + |\vartheta| + |\varpi| + |\varrho| + |\varsigma| + |\phi| +$

Math Roman (`\mathrm`)

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |\Upsilon| + |\Phi| + |X| + |\Psi| + |\Omega| +$

Math Bold (`\mathbf`)

$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$
 $|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$
 $|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$
 $|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$
 $|A| + |B| + |\Gamma| + |\Delta| + |E| + |Z| + |H| + |\Theta| + |I| + |K| + |\Lambda| + |M| +$
 $|N| + |\Xi| + |O| + |\Pi| + |P| + |\Sigma| + |T| + |Y| + |\Phi| + |X| + |\Psi| + |\Omega| +$

Math Calligraphic (`\mathcal`)

$|\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{D}| + |\mathcal{E}| + |\mathcal{F}| + |\mathcal{G}| + |\mathcal{H}| + |\mathcal{I}| + |\mathcal{J}| + |\mathcal{K}| + |\mathcal{L}| + |\mathcal{M}| +$
 $|\mathcal{N}| + |\mathcal{O}| + |\mathcal{P}| + |\mathcal{Q}| + |\mathcal{R}| + |\mathcal{S}| + |\mathcal{T}| + |\mathcal{U}| + |\mathcal{V}| + |\mathcal{W}| + |\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}| +$

3.4.5 Superscript Positioning Sans Serif Bold**Default**

$A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 +$
 $N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 +$
 $a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 +$
 $n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 +$
 $A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 +$
 $N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + Y^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 +$
 $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \varepsilon^2 + \zeta^2 + \eta^2 + \theta^2 + \iota^2 + \kappa^2 + \lambda^2 + \mu^2 +$
 $\nu^2 + \xi^2 + \omicron^2 + \pi^2 + \rho^2 + \sigma^2 + \tau^2 + \upsilon^2 + \phi^2 + \chi^2 + \psi^2 + \omega^2 +$
 $\varepsilon^2 + \theta^2 + \pi^2 + \rho^2 + \varsigma^2 + \phi^2 +$

Math Roman (\mathrm)

$A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 +$
 $N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 +$
 $a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 +$
 $n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 +$
 $A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 +$
 $N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 +$

Math Bold (\mathbf)

$A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 +$
 $N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 +$
 $a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 +$
 $n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 +$
 $A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 +$
 $N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + Y^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 +$

Math Calligraphic (\mathcal)

$\mathcal{A}^2 + \mathcal{B}^2 + \mathcal{C}^2 + \mathcal{D}^2 + \mathcal{E}^2 + \mathcal{F}^2 + \mathcal{G}^2 + \mathcal{H}^2 + \mathcal{I}^2 + \mathcal{J}^2 + \mathcal{K}^2 + \mathcal{L}^2 + \mathcal{M}^2 +$
 $\mathcal{N}^2 + \mathcal{O}^2 + \mathcal{P}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{S}^2 + \mathcal{T}^2 + \mathcal{U}^2 + \mathcal{V}^2 + \mathcal{W}^2 + \mathcal{X}^2 + \mathcal{Y}^2 + \mathcal{Z}^2 +$

3.4.6 Subscript Positioning **Sans Serif Bold**

Default

$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$
 $N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$
 $a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$
 $n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$
 $A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i +$
 $N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + Y_i + \Phi_i + X_i + \Psi_i + \Omega_i +$
 $\alpha_i + \beta_i + \gamma_i + \delta_i + \varepsilon_i + \zeta_i + \eta_i + \theta_i + \iota_i + \kappa_i + \lambda_i + \mu_i +$
 $\nu_i + \xi_i + o_i + \pi_i + \rho_i + \sigma_i + \tau_i + u_i + \phi_i + \chi_i + \psi_i + \omega_i +$
 $\varepsilon_i + \theta_i + \pi_i + \rho_i + \varsigma_i + \phi_i +$

Math Roman (`\mathrm`)

$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$
 $N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$
 $a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$
 $n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$
 $A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i +$
 $N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i +$

Math Bold (`\mathbf`)

$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$
 $N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$
 $a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$
 $n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$
 $A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i +$
 $N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + Y_i + \Phi_i + X_i + \Psi_i + \Omega_i +$

Math Calligraphic (`\mathcal`)

$\mathcal{A}_i + \mathcal{B}_i + \mathcal{C}_i + \mathcal{D}_i + \mathcal{E}_i + \mathcal{F}_i + \mathcal{G}_i + \mathcal{H}_i + \mathcal{I}_i + \mathcal{J}_i + \mathcal{K}_i + \mathcal{L}_i + \mathcal{M}_i +$
 $\mathcal{N}_i + \mathcal{O}_i + \mathcal{P}_i + \mathcal{Q}_i + \mathcal{R}_i + \mathcal{S}_i + \mathcal{T}_i + \mathcal{U}_i + \mathcal{V}_i + \mathcal{W}_i + \mathcal{X}_i + \mathcal{Y}_i + \mathcal{Z}_i +$

3.4.7 Accent Positioning Sans Serif Bold**Default**

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{r} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{Y} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$
 $\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{i} + \hat{k} + \hat{\lambda} + \hat{\mu} +$
 $\hat{\nu} + \hat{\xi} + \hat{o} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{u} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$
 $\hat{\epsilon} + \hat{\theta} + \hat{\pi} + \hat{\rho} + \hat{\zeta} + \hat{\phi} +$

Math Italic (\mathit)

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} + \hat{\ell} + \hat{\wp} + \hat{i} + \hat{j} + \hat{i}$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{r} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{Y} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$
 $\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{i} + \hat{k} + \hat{\lambda} + \hat{\mu} +$
 $\hat{\nu} + \hat{\xi} + \hat{o} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{u} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$
 $\hat{\epsilon} + \hat{\theta} + \hat{\pi} + \hat{\rho} + \hat{\zeta} + \hat{\phi} +$

Math Roman (\mathrm)

$\hat{O} + \hat{I} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{r} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{Y} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$

Math Bold (`\mathbf`)

$\hat{0} + \hat{1} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$
 $\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$
 $\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$
 $\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$
 $\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$
 $\hat{A} + \hat{B} + \hat{I} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$
 $\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{Y} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$

Math Calligraphic (`\mathcal`)

$\mathcal{A} + \mathcal{B} + \mathcal{C} + \mathcal{D} + \mathcal{E} + \mathcal{F} + \mathcal{G} + \mathcal{H} + \mathcal{I} + \mathcal{J} + \mathcal{K} + \mathcal{L} + \mathcal{M} +$
 $\mathcal{N} + \mathcal{O} + \mathcal{P} + \mathcal{Q} + \mathcal{R} + \mathcal{S} + \mathcal{T} + \mathcal{U} + \mathcal{V} + \mathcal{W} + \mathcal{X} + \mathcal{Y} + \mathcal{Z} +$

3.4.8 Differentials Sans Serif Bold

$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$
 $dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$
 $da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$
 $dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + dY + d\Phi + dX + d\Psi + d\Omega +$
 $d\alpha + d\beta + d\gamma + d\delta + d\varepsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$
 $dv + d\xi + do + d\pi + d\rho + d\sigma + d\tau + du + d\phi + d\chi + d\psi + d\omega +$
 $d\varepsilon + d\theta + d\pi + d\rho + d\varsigma + d\phi +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$

$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$
 $dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$
 $da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$
 $dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + dY + d\Phi + dX + d\Psi + d\Omega +$
 $d\alpha + d\beta + d\gamma + d\delta + d\varepsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$
 $dv + d\xi + do + d\pi + d\rho + d\sigma + d\tau + d\nu + d\phi + d\chi + d\psi + d\omega +$
 $d\varepsilon + d\theta + d\pi + d\rho + d\zeta + d\phi +$
 $dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$
 $dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + dY + d\Phi + dX + d\Psi + d\Omega +$

$\partial A + \partial B + \partial C + \partial D + \partial E + \partial F + \partial G + \partial H + \partial I + \partial J + \partial K + \partial L + \partial M +$
 $\partial N + \partial O + \partial P + \partial Q + \partial R + \partial S + \partial T + \partial U + \partial V + \partial W + \partial X + \partial Y + \partial Z +$
 $\partial a + \partial b + \partial c + \partial d + \partial e + \partial f + \partial g + \partial h + \partial i + \partial j + \partial k + \partial l + \partial m +$
 $\partial n + \partial o + \partial p + \partial q + \partial r + \partial s + \partial t + \partial u + \partial v + \partial w + \partial x + \partial y + \partial z +$
 $\partial A + \partial B + \partial \Gamma + \partial \Delta + \partial E + \partial Z + \partial H + \partial \Theta + \partial I + \partial K + \partial \Lambda + \partial M +$
 $\partial N + \partial \Xi + \partial O + \partial \Pi + \partial P + \partial \Sigma + \partial T + \partial Y + \partial \Phi + \partial X + \partial \Psi + \partial \Omega +$
 $\partial \alpha + \partial \beta + \partial \gamma + \partial \delta + \partial \varepsilon + \partial \zeta + \partial \eta + \partial \theta + \partial \iota + \partial \kappa + \partial \lambda + \partial \mu +$
 $\partial v + \partial \xi + \partial o + \partial \pi + \partial \rho + \partial \sigma + \partial \tau + \partial \nu + \partial \phi + \partial \chi + \partial \psi + \partial \omega +$
 $\partial \varepsilon + \partial \theta + \partial \pi + \partial \rho + \partial \zeta + \partial \phi +$
 $\partial A + \partial B + \partial \Gamma + \partial \Delta + \partial E + \partial Z + \partial H + \partial \Theta + \partial I + \partial K + \partial \Lambda + \partial M +$
 $\partial N + \partial \Xi + \partial O + \partial \Pi + \partial P + \partial \Sigma + \partial T + \partial Y + \partial \Phi + \partial X + \partial \Psi + \partial \Omega +$

3.4.9 Slash Kerning **Sans Serif Bold**

$1/A + 1/B + 1/C + 1/D + 1/E + 1/F + 1/G + 1/H + 1/I + 1/J + 1/K + 1/L + 1/M +$
 $1/N + 1/O + 1/P + 1/Q + 1/R + 1/S + 1/T + 1/U + 1/V + 1/W + 1/X + 1/Y + 1/Z +$
 $1/a + 1/b + 1/c + 1/d + 1/e + 1/f + 1/g + 1/h + 1/i + 1/j + 1/k + 1/l + 1/m +$
 $1/n + 1/o + 1/p + 1/q + 1/r + 1/s + 1/t + 1/u + 1/v + 1/w + 1/x + 1/y + 1/z +$
 $1/A + 1/B + 1/\Gamma + 1/\Delta + 1/E + 1/Z + 1/H + 1/\Theta + 1/I + 1/K + 1/\Lambda + 1/M +$
 $1/N + 1/\Xi + 1/O + 1/\Pi + 1/P + 1/\Sigma + 1/T + 1/Y + 1/\Phi + 1/X + 1/\Psi + 1/\Omega +$
 $1/\alpha + 1/\beta + 1/\gamma + 1/\delta + 1/\varepsilon + 1/\zeta + 1/\eta + 1/\theta + 1/\iota + 1/\kappa + 1/\lambda + 1/\mu +$
 $1/v + 1/\xi + 1/o + 1/\pi + 1/\rho + 1/\sigma + 1/\tau + 1/\upsilon + 1/\phi + 1/\chi + 1/\psi + 1/\omega +$
 $1/\varepsilon + 1/\theta + 1/\pi + 1/\rho + 1/\zeta + 1/\phi +$

$A/2 + B/2 + C/2 + D/2 + E/2 + F/2 + G/2 + H/2 + I/2 + J/2 + K/2 + L/2 + M/2 +$
 $N/2 + O/2 + P/2 + Q/2 + R/2 + S/2 + T/2 + U/2 + V/2 + W/2 + X/2 + Y/2 + Z/2 +$
 $a/2 + b/2 + c/2 + d/2 + e/2 + f/2 + g/2 + h/2 + i/2 + j/2 + k/2 + l/2 + m/2 +$
 $n/2 + o/2 + p/2 + q/2 + r/2 + s/2 + t/2 + u/2 + v/2 + w/2 + x/2 + y/2 + z/2 +$
 $A/2 + B/2 + \Gamma/2 + \Delta/2 + E/2 + Z/2 + H/2 + \Theta/2 + I/2 + K/2 + \Lambda/2 + M/2 +$
 $N/2 + \Xi/2 + O/2 + \Pi/2 + P/2 + \Sigma/2 + T/2 + Y/2 + \Phi/2 + X/2 + \Psi/2 + \Omega/2 +$
 $\alpha/2 + \beta/2 + \gamma/2 + \delta/2 + \varepsilon/2 + \zeta/2 + \eta/2 + \theta/2 + \iota/2 + \kappa/2 + \lambda/2 + \mu/2 +$
 $\nu/2 + \xi/2 + o/2 + \pi/2 + \rho/2 + \sigma/2 + \tau/2 + u/2 + \phi/2 + \chi/2 + \psi/2 + \omega/2 +$
 $\varepsilon/2 + \theta/2 + \pi/2 + \rho/2 + \varsigma/2 + \phi/2 +$

3.4.10 (Big) Operators **Sans Serif Bold**

$$\begin{array}{cccccccc}
 \sum_{i=1}^n x^n & \prod_{i=1}^n x^n & \coprod_{i=1}^n x^n & \int_{i=1}^n x^n & \oint_{i=1}^n x^n & \bigcup_{i=1}^n x^n & \bigcup_{i=1}^n x^n & \bigcap_{i=1}^n x^n \\
 \bigotimes_{i=1}^n x^n & \bigoplus_{i=1}^n x^n & \odot_{i=1}^n x^n & \bigwedge_{i=1}^n x^n & \bigvee_{i=1}^n x^n & \bigcup_{i=1}^n x^n & \bigcup_{i=1}^n x^n & \bigcap_{i=1}^n x^n \\
 \bigsqcup_{i=1}^n x^n & \sum_{i=1}^n x^n & \prod_{i=1}^n x^n & \coprod_{i=1}^n x^n & \int_{i=1}^n x^n & \oint_{i=1}^n x^n & & \\
 \bigotimes_{i=1}^n x^n & \bigoplus_{i=1}^n x^n & \odot_{i=1}^n x^n & \bigwedge_{i=1}^n x^n & \bigvee_{i=1}^n x^n & \bigcup_{i=1}^n x^n & \bigcup_{i=1}^n x^n & \bigcap_{i=1}^n x^n \\
 \bigsqcup_{i=1}^n x^n & & & & & & &
 \end{array}$$

3.4.11 Radicals **Sans Serif Bold**

$$\sqrt{x+y} \quad \sqrt{x^2+y^2} \quad \sqrt{x_i^2+y_j^2} \quad \sqrt{\left(\frac{\cos x}{2}\right)} \quad \sqrt{\left(\frac{\sin x}{2}\right)}$$

$$\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{x+y}}}}}}}$$

3.4.12 Over- and Underbraces **Sans Serif Bold**

$$\overbrace{x} \quad \overbrace{x+y} \quad \overbrace{x^2+y^2} \quad \overbrace{x_i^2+y_j^2} \quad \underbrace{x} \quad \underbrace{x+y} \quad \underbrace{x_i+y_j} \quad \underbrace{x_i^2+y_j^2}$$

3.4.13 Normal and Wide Accents **Sans Serif Bold**

$$\dot{x} \quad \ddot{x} \quad \vec{x} \quad \bar{x} \quad \overline{x} \quad \overline{xx} \quad \tilde{x} \quad \tilde{\tilde{x}} \quad \tilde{\tilde{\tilde{x}}} \quad \widetilde{xxx} \quad \hat{x} \quad \hat{\hat{x}} \quad \hat{\hat{\hat{x}}} \quad \widehat{xxx}$$
$$\hat{X} \quad \check{X} \quad \tilde{X} \quad \acute{X} \quad \grave{X} \quad \dot{X} \quad \ddot{X} \quad \breve{X} \quad \bar{X} \quad \vec{X}$$

3.4.14 Long Arrows Sans Serif Bold

3.4.15 Left and Right Delimiters **Sans Serif Bold**

$$-(f) - -[f] - -|f| - -[f] - -\langle f \rangle - -\{f\} -$$

Using \left and \right.

$$-(f) \quad -[f] \quad -\lfloor f \rfloor \quad -\lceil f \rceil \quad -\langle f \rangle \quad -\{f\} -$$
$$-) f (- -] f [- - / f / - - \backslash f \backslash - - / f \backslash - - \backslash f / -$$

3.4.16 Big-g-g Delimiters Sans Serif Bold

$$- \left[\left[\left[\left[\left[\left[[-] \right] \right] \right] \right] \right] \right] - \left(\left(\left(\left(\left((-) \right) \right) \right) \right) \right) \right)$$
$$-\langle\langle\langle\langle\langle\langle\langle\langle\langle\rangle\rangle\rangle\rangle\rangle\rangle\rangle\rangle- \quad -\left(\left(\left(\left(\left(\left(\left(\left(\right)\right)\right)\right)\right)\right)\right)\right)-$$

3.4.17 Binary Operators Sans Serif Bold

$x \pm y$	<code>\pm</code>	$x \cap y$	<code>\cap</code>	$x \diamond y$	<code>\diamond</code>	$x \oplus y$	<code>\oplus</code>
$x \mp y$	<code>\mp</code>	$x \cup y$	<code>\cup</code>	$x \triangle y$	<code>\bigtriangleup</code>	$x \ominus y$	<code>\ominus</code>
$x \times y$	<code>\times</code>	$x \uplus y$	<code>\uplus</code>	$x \nabla y$	<code>\bigtriangledown</code>	$x \otimes y$	<code>\otimes</code>
$x \div y$	<code>\div</code>	$x \sqcap y$	<code>\sqcap</code>	$x \triangleleft y$	<code>\triangleleft</code>	$x \oslash y$	<code>\oslash</code>
$x * y$	<code>\ast</code>	$x \sqcup y$	<code>\sqcup</code>	$x \triangleright y$	<code>\triangleright</code>	$x \odot y$	<code>\odot</code>
$x \star y$	<code>\star</code>	$x \vee y$	<code>\vee</code>	$x \lhd y$	<code>\lhd</code>	$x \bigcirc y$	<code>\bigcirc</code>
$x \circ y$	<code>\circ</code>	$x \wedge y$	<code>\wedge</code>	$x \rhd y$	<code>\rhd</code>	$x \dagger y$	<code>\dagger</code>
$x \bullet y$	<code>\bullet</code>	$x \setminus y$	<code>\setminus</code>	$x \unlhd y$	<code>\unlhd</code>	$x \ddagger y$	<code>\ddagger</code>
$x \cdot y$	<code>\cdot</code>	$x \wr y$	<code>\wr</code>	$x \unrhd y$	<code>\unrhd</code>	$x \S y$	<code>\S</code>
$x + y$	<code>+</code>	$x - y$	<code>-</code>	$x \amalg y$	<code>\amalg</code>	$x \P y$	<code>\P</code>

3.4.18 Relations Sans Serif Bold

$x \leq y$	<code>\leq</code>	$x \geq y$	<code>\geq</code>	$x \equiv y$	<code>\equiv</code>	$x \models y$	<code>\models</code>
$x < y$	<code><</code>	$x > y$	<code>></code>	$x \sim y$	<code>\sim</code>	$x \perp y$	<code>\perp</code>
$x \leq y$	<code>\preceq</code>	$x \geq y$	<code>\succeq</code>	$x \simeq y$	<code>\simeq</code>	$x \mid y$	<code>\mid</code>
$x \ll y$	<code>\ll</code>	$x \gg y$	<code>\gg</code>	$x \asymp y$	<code>\asymp</code>	$x \parallel y$	<code>\parallel</code>
$x \subset y$	<code>\subset</code>	$x \supset y$	<code>\supset</code>	$x \approx y$	<code>\approx</code>	$x \bowtie y$	<code>\bowtie</code>
$x \subseteq y$	<code>\subseteq</code>	$x \supseteq y$	<code>\supseteq</code>	$x \cong y$	<code>\cong</code>	$x \Join y$	<code>\Join</code>
$x \sqsubset y$	<code>\sqsubset</code>	$x \sqsupset y$	<code>\sqsupset</code>	$x \neq y$	<code>\neq</code>	$x \frown y$	<code>\frown</code>
$x \sqsubseteq y$	<code>\sqsubseteq</code>	$x \sqsupseteq y$	<code>\sqsupseteq</code>	$x \doteq y$	<code>\doteq</code>	$x \smile y$	<code>\smile</code>
$x \in y$	<code>\in</code>	$x \ni y$	<code>\ni</code>	$x \propto y$	<code>\propto</code>	$x = y$	<code>=</code>
$x \vdash y$	<code>\vdash</code>	$x \dashv y$	<code>\dashv</code>	$x < y$	<code><</code>	$x > y$	<code>></code>
$x : y$	<code>:</code>						

3.4.19 Punctuation Sans Serif Bold

x, y	<code>,</code>	$x; y$	<code>;</code>	$x : y$	<code>\colon</code>	$x \cdot y$	<code>\ldotp</code>	$x \bullet y$	<code>\cdotp</code>
--------	----------------	--------	----------------	---------	---------------------	-------------	---------------------	---------------	---------------------

3.4.20 Arrows Sans Serif Bold

$x \leftarrow y$	<code>\leftarrow</code>	$x \longleftarrow y$	<code>\longleftarrow</code>	$x \uparrow y$	<code>\uparrow</code>
$x \Leftarrow y$	<code>\Leftarrow</code>	$x \Longleftarrow y$	<code>\Longleftarrow</code>	$x \Uparrow y$	<code>\Uparrow</code>
$x \rightarrow y$	<code>\rightarrow</code>	$x \longrightarrow y$	<code>\longrightarrow</code>	$x \downarrow y$	<code>\downarrow</code>
$x \Rightarrow y$	<code>\Rightarrow</code>	$x \Longrightarrow y$	<code>\Longrightarrow</code>	$x \Downarrow y$	<code>\Downarrow</code>
$x \leftrightarrow y$	<code>\leftrightarrow</code>	$x \longleftrightarrow y$	<code>\longleftrightarrow</code>	$x \Updownarrow y$	<code>\Updownarrow</code>
$x \Leftrightarrow y$	<code>\Leftrightarrow</code>	$x \Longleftrightarrow y$	<code>\Longleftrightarrow</code>	$x \Updownarrow y$	<code>\Updownarrow</code>
$x \mapsto y$	<code>\mapsto</code>	$x \longmapsto y$	<code>\longmapsto</code>	$x \nearrow y$	<code>\nearrow</code>
$x \hookrightarrow y$	<code>\hookrightarrow</code>	$x \hookrightarrow y$	<code>\hookrightarrow</code>	$x \searrow y$	<code>\searrow</code>
$x \leftharpoonup y$	<code>\leftharpoonup</code>	$x \rightharpoonup y$	<code>\rightharpoonup</code>	$x \swarrow y$	<code>\swarrow</code>
$x \leftharpoondown y$	<code>\leftharpoondown</code>	$x \rightharpoondown y$	<code>\rightharpoondown</code>	$x \nwarrow y$	<code>\nwarrow</code>
$x \rightleftharpoons y$	<code>\rightleftharpoons</code>	$x \rightsquigarrow y$	<code>\rightsquigarrow</code>		

3.4.21 Miscellaneous Symbols Sans Serif Bold

$x \dots y$	<code>\ldots</code>	$x \cdots y$	<code>\cdots</code>	$x \vdots y$	<code>\vdots</code>	$x \ddots y$	<code>\ddots</code>
$x \aleph y$	<code>\aleph</code>	$x \prime y$	<code>\prime</code>	$x \forall y$	<code>\forall</code>	$x \infty y$	<code>\infty</code>
$x \hbar y$	<code>\hbar</code>	$x \emptyset y$	<code>\emptyset</code>	$x \exists y$	<code>\exists</code>	$x \Box y$	<code>\Box</code>
$x \imath y$	<code>\imath</code>	$x \nabla y$	<code>\nabla</code>	$x - y$	<code>\neg</code>	$x \Diamond y$	<code>\Diamond</code>
$x \jmath y$	<code>\jmath</code>	$x \sqrt{y}$	<code>\sqrt</code>	$x \flat y$	<code>\flat</code>	$x \Delta y$	<code>\triangle</code>
$x \ell y$	<code>\ell</code>	$x \top y$	<code>\top</code>	$x \natural y$	<code>\natural</code>	$x \clubsuit y$	<code>\clubsuit</code>
$x \wp y$	<code>\wp</code>	$x \bot y$	<code>\bot</code>	$x \sharp y$	<code>\sharp</code>	$x \diamondsuit y$	<code>\diamondsuit</code>
$x \Re y$	<code>\Re</code>	$x \ y$	<code>\ </code>	$x \backslash y$	<code>\backslash</code>	$x \heartsuit y$	<code>\heartsuit</code>
$x \Im y$	<code>\Im</code>	$x \angle y$	<code>\angle</code>	$x \partial y$	<code>\partial</code>	$x \spadesuit y$	<code>\spadesuit</code>
$x \mho y$	<code>\mho</code>	$x \cdot y$	<code>\cdot</code>	$x y$	<code> </code>	$x ! y$	<code>!</code>

3.4.22 Variable-Sized Operators Sans Serif Bold

$x \sum y$	<code>\sum</code>	$x \bigcap y$	<code>\bigcap</code>	$x \bigodot y$	<code>\bigodot</code>
$x \prod y$	<code>\prod</code>	$x \bigcup y$	<code>\bigcup</code>	$x \bigotimes y$	<code>\bigotimes</code>
$x \coprod y$	<code>\coprod</code>	$x \bigsqcup y$	<code>\bigsqcup</code>	$x \bigoplus y$	<code>\bigoplus</code>
$x \int y$	<code>\int</code>	$x \bigvee y$	<code>\bigvee</code>	$x \biguplus y$	<code>\biguplus</code>
$x \oint y$	<code>\oint</code>	$x \bigwedge y$	<code>\bigwedge</code>		

3.4.23 Log-Like Operators Sans Serif Bold

$x \arccos y$	$x \cos y$	$x \csc y$	$x \exp y$	$x \ker y$	$x \limsup y$	$x \min y$	$x \sinh y$
$x \arcsin y$	$x \cosh y$	$x \deg y$	$x \gcd y$	$x \lg y$	$x \ln y$	$x \Pr y$	$x \sup y$
$x \arctan y$	$x \cot y$	$x \det y$	$x \hom y$	$x \lim y$	$x \log y$	$x \sec y$	$x \tan y$
$x \arg y$	$x \coth y$	$x \dim y$	$x \inf y$	$x \liminf y$	$x \max y$	$x \sin y$	$x \tanh y$

3.4.24 Delimiters **Sans Serif Bold**

$x(y$	$($	$x)y$	$)$	$x \uparrow y$	<code>\uparrow</code>	$x \Uparrow y$	<code>\Uparrow</code>
$x[y$	$[$	$x]y$	$]$	$x \downarrow y$	<code>\downarrow</code>	$x \Downarrow y$	<code>\Downarrow</code>
$x\{y$	<code>\{</code>	$x\}y$	<code>\}</code>	$x \updownarrow y$	<code>\updownarrow</code>	$x \Updownarrow y$	<code>\Updownarrow</code>
$x\lfloor y$	<code>\lfloor</code>	$x\rfloor y$	<code>\rfloor</code>	$x\lceil y$	<code>\lceil</code>	$x\rceil y$	<code>\rceil</code>
$x\langle y$	<code>\langle</code>	$x\rangle y$	<code>\rangle</code>	x/y	<code>/</code>	$x\backslash y$	<code>\backslash</code>
$x y$	<code> </code>	$x y$	<code> </code>				

3.4.25 Large Delimiters **Sans Serif Bold**

$\left($	<code>\rmoustache</code>	$\int$	<code>\lmoustache</code>	$\right)$	<code>\rgroup</code>	$\left($	<code>\lgroup</code>
$\Big $	<code>\arrowvert</code>	$\Big\ $	<code>\Arrowvert</code>	$\Big $	<code>\bracevert</code>		

3.4.26 Math Mode Accents **Sans Serif Bold**

$\hat{a}$	<code>\hat{a}</code>	$\acute{a}$	<code>\acute{a}</code>	$\bar{a}$	<code>\bar{a}</code>	$\dot{a}$	<code>\dot{a}</code>	$\breve{a}$	<code>\breve{a}</code>
$\check{a}$	<code>\check{a}</code>	$\grave{a}$	<code>\grave{a}</code>	$\vec{a}$	<code>\vec{a}</code>	$\ddot{a}$	<code>\ddot{a}</code>	$\tilde{a}$	<code>\tilde{a}</code>

3.4.27 Miscellaneous Constructions **Sans Serif Bold**

$\widetilde{abc}$	<code>\widetilde{abc}</code>	$\widehat{abc}$	<code>\widehat{abc}</code>
$\overleftarrow{abc}$	<code>\overleftarrow{abc}</code>	$\overrightarrow{abc}$	<code>\overrightarrow{abc}</code>
$\overline{abc}$	<code>\overline{abc}</code>	$\underline{abc}$	<code>\underline{abc}</code>
$\overbrace{abc}$	<code>\overbrace{abc}</code>	$\underbrace{abc}$	<code>\underbrace{abc}</code>
$\sqrt{abc}$	<code>\sqrt{abc}</code>	$\sqrt[n]{abc}$	<code>\sqrt[n]{abc}</code>
f'	<code>f'</code>	$\frac{abc}{xyz}$	<code>\frac{abc}{xyz}</code>

3.4.28 AMS Delimiters **Sans Serif Bold**

$\ulcorner$	<code>\ulcorner</code>	$\urcorner$	<code>\urcorner</code>	$\llcorner$	<code>\llcorner</code>	$\lrcorner$	<code>\lrcorner</code>
-------------	------------------------	-------------	------------------------	-------------	------------------------	-------------	------------------------

3.4.29 AMS Arrows Sans Serif Bold

$x \dashrightarrow y$	<code>\dashrightarrow</code>	$x \dashleftarrow y$	<code>\dashleftarrow</code>
$x \leftrightsquigarrow y$	<code>\leftrightsquigarrow</code>	$x \rightleftarrows y$	<code>\rightleftarrows</code>
$x \Lleftarrow y$	<code>\Lleftarrow</code>	$x \twoheadleftarrow y$	<code>\twoheadleftarrow</code>
$x \leftarrowtail y$	<code>\leftarrowtail</code>	$x \looparrowleft y$	<code>\looparrowleft</code>
$x \leftrightharpoons y$	<code>\leftrightharpoons</code>	$x \curvearrowleft y$	<code>\curvearrowleft</code>
$x \circlearrowleft y$	<code>\circlearrowleft</code>	$x \lsh y$	<code>\lsh</code>
$x \upuparrows y$	<code>\upuparrows</code>	$x \upharpoonleft y$	<code>\upharpoonleft</code>
$x \downharpoonleft y$	<code>\downharpoonleft</code>	$x \multimap y$	<code>\multimap</code>
$x \leftrightsquigarrow y$	<code>\leftrightsquigarrow</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \rightleftarrows y$	<code>\rightleftarrows</code>	$x \rightrightarrows y$	<code>\rightrightarrows</code>
$x \rightleftarrows y$	<code>\rightleftarrows</code>	$x \twoheadrightarrow y$	<code>\twoheadrightarrow</code>
$x \rightarrowtail y$	<code>\rightarrowtail</code>	$x \looparrowright y$	<code>\looparrowright</code>
$x \rightleftharpoons y$	<code>\rightleftharpoons</code>	$x \curvearrowright y$	<code>\curvearrowright</code>
$x \circlearrowright y$	<code>\circlearrowright</code>	$x \rsh y$	<code>\rsh</code>
$x \downdownarrows y$	<code>\downdownarrows</code>	$x \upharpoonright y$	<code>\upharpoonright</code>
$x \downharpoonright y$	<code>\downharpoonright</code>	$x \rightsquigarrow y$	<code>\rightsquigarrow</code>

3.4.30 AMS Negated Arrows Sans Serif Bold

$x \nleftarrow y$	<code>\nleftarrow</code>	$x \nrightarrow y$	<code>\nrightarrow</code>
$x \nLleftarrow y$	<code>\nLleftarrow</code>	$x \nRrightarrow y$	<code>\nRrightarrow</code>
$x \nleftrightarrow y$	<code>\nleftrightarrow</code>	$x \nLleftrightarrow y$	<code>\nLleftrightarrow</code>

3.4.31 AMS Greek Sans Serif Bold

$x \digamma y$ `\digamma` $x \varkappa y$ `\varkappa`

3.4.32 AMS Hebrew Sans Serif Bold

$x \beth y$ `\beth` $x \daleth y$ `\daleth` $x \gimel y$ `\gimel`

3.4.33 AMS Miscellaneous **Sans Serif Bold**

$x\hbar y$	<code>\hbar</code>	$x\hslash y$	<code>\hslash</code>
$x\triangle y$	<code>\vartriangle</code>	$x\nabla y$	<code>\triangledown</code>
$x\square y$	<code>\square</code>	$x\lozenge y$	<code>\lozenge</code>
$x\circledcirc y$	<code>\circledcirc</code>	$x\angle y$	<code>\angle</code>
$x\measuredangle y$	<code>\measuredangle</code>	$x\nexists y$	<code>\nexists</code>
$x\mho y$	<code>\mho</code>	$x\Finv y$	<code>\Finv</code>
$x\Game y$	<code>\Game</code>	$x\Bbbk y$	<code>\Bbbk</code>
$x\backprime y$	<code>\backprime</code>	$x\varnothing y$	<code>\varnothing</code>
$x\blacktriangle y$	<code>\blacktriangle</code>	$x\blacktriangledown y$	<code>\blacktriangledown</code>
$x\blacksquare y$	<code>\blacksquare</code>	$x\blacklozenge y$	<code>\blacklozenge</code>
$x\bigstar y$	<code>\bigstar</code>	$x\angle y$	<code>\sphericalangle</code>
$x\complement y$	<code>\complement</code>	$x\eth y$	<code>\eth</code>
$x\diagup y$	<code>\diagup</code>	$x\diagdown y$	<code>\diagdown</code>

" Not defined in `amssymb.sty`, define using the `\newsymbol` command.

3.4.34 AMS Binary Operators **Sans Serif Bold**

$x\dotplus y$	<code>\dotplus</code>	$x\smallsetminus y$	<code>\smallsetminus</code>
$x\cap y$	<code>\Cap</code>	$x\cup y$	<code>\Cup</code>
$x\barwedge y$	<code>\barwedge</code>	$x\veebar y$	<code>\veebar</code>
$x\doublebarwedge y$	<code>\doublebarwedge</code>	$x\boxminus y$	<code>\boxminus</code>
$x\boxtimes y$	<code>\boxtimes</code>	$x\boxdot y$	<code>\boxdot</code>
$x\boxplus y$	<code>\boxplus</code>	$x\divideontimes y$	<code>\divideontimes</code>
$x\ltimes y$	<code>\ltimes</code>	$x\rtimes y$	<code>\rtimes</code>
$x\leftthreetimes y$	<code>\leftthreetimes</code>	$x\rightthreetimes y$	<code>\rightthreetimes</code>
$x\curlywedge y$	<code>\curlywedge</code>	$x\curlyvee y$	<code>\curlyvee</code>
$x\circleddash y$	<code>\circleddash</code>	$x\circledast y$	<code>\circledast</code>
$x\circledcirc y$	<code>\circledcirc</code>	$x\centerdot y$	<code>\centerdot</code>
$x\intercal y$	<code>\intercal</code>		

3.4.35 AMS Relations Sans Serif Bold

$x \leqslant y$	<code>\leqslant</code>
$x \lesssim y$	<code>\lesssim</code>
$x \approx y$	<code>\approx</code>
$x \lll y$	<code>\lll</code>
$x \lesseqgtr y$	<code>\lesseqgtr</code>
$x \doteqdot y$	<code>\doteqdot</code>
$x \fallingdotseq y$	<code>\fallingdotseq</code>
$x \backsimeq y$	<code>\backsimeq</code>
$x \Subset y$	<code>\Subset</code>
$x \preccurlyeq y$	<code>\preccurlyeq</code>
$x \precapprox y$	<code>\precapprox</code>
$x \vartriangleleft y$	<code>\vartriangleleft</code>
$x \vDash y$	<code>\vDash</code>
$x \smallsmile y$	<code>\smallsmile</code>
$x \bumpeq y$	<code>\bumpeq</code>
$x \geqq y$	<code>\geqq</code>
$x \gtrless y$	<code>\gtrless</code>
$x \gtrapprox y$	<code>\gtrapprox</code>
$x \ggg y$	<code>\ggg</code>
$x \gtreqless y$	<code>\gtreqless</code>
$x \eqcirc y$	<code>\eqcirc</code>
$x \trianglelefteq y$	<code>\trianglelefteq</code>
$x \thickapprox y$	<code>\thickapprox</code>
$x \supseteq y$	<code>\supseteq</code>
$x \succcurlyeq y$	<code>\succcurlyeq</code>
$x \succapprox y$	<code>\succapprox</code>
$x \vartriangleright y$	<code>\vartriangleright</code>
$x \Vdash y$	<code>\Vdash</code>
$x \parallel y$	<code>\parallel</code>
$x \pitchfork y$	<code>\pitchfork</code>
$x \blacktriangleleft y$	<code>\blacktriangleleft</code>
$x \backepsilon y$	<code>\backepsilon</code>
$x \because y$	<code>\because</code>

3.4.36 AMS Negated Relations **Sans Serif Bold**

$x \not< y$	<code>\nless</code>	$x \not\leq y$	<code>\nleq</code>
$x \not<slant y$	<code>\nleqslant</code>	$x \not\leqq y$	<code>\nleqq</code>
$x \lesssim y$	<code>\lneq</code>	$x \lesseqgtr y$	<code>\lneqq</code>
$x \not\equiv y$	<code>\lvertneqq</code>	$x \not\approx y$	<code>\lnsim</code>
$x \not\approx y$	<code>\lnapprox</code>	$x \not\prec y$	<code>\nprec</code>
$x \not\preceq y$	<code>\npreceq</code>	$x \not\precnsim y$	<code>\precnsim</code>
$x \not\sim y$	<code>\precnapprox</code>	$x \sim y$	<code>\nsim</code>
$x \nmid y$	<code>\nshortmid</code>	$x \nmid y$	<code>\nmid</code>
$x \nvdash y$	<code>\nvdash</code>	$x \nVdash y$	<code>\nVDash</code>
$x \ntriangleleft y$	<code>\ntriangleleft</code>	$x \ntrianglelefteq y$	<code>\ntrianglelefteq</code>
$x \not\subseteq y$	<code>\nsubseteq</code>	$x \subsetneq y$	<code>\subsetneq</code>
$x \subsetneqq y$	<code>\varsubsetneq</code>	$x \subsetneqq y$	<code>\subsetneqq</code>
$x \not\supseteq y$	<code>\varsubsetneqq</code>	$x \not> y$	<code>\ngtr</code>
$x \ngeq y$	<code>\ngeq</code>	$x \not>slant y$	<code>\ngeqslant</code>
$x \ngeqq y$	<code>\ngeqq</code>	$x \gtrsim y$	<code>\gtrsim</code>
$x \gtrless y$	<code>\gtrless</code>	$x \gtrsim y$	<code>\gtrsim</code>
$x \gtrapprox y$	<code>\gnsim</code>	$x \gtrapprox y$	<code>\gnapprox</code>
$x \nprec y$	<code>\nsucc</code>	$x \nprecneq y$	<code>\nsucceq</code>
$x \nprecneqq y$	<code>\nsucceq</code>	$x \precnsim y$	<code>\succnsim</code>
$x \not\sim y$	<code>\succnapprox</code>	$x \not\cong y$	<code>\ncong</code>
$x \nparallel y$	<code>\nshortparallel</code>	$x \nparallel y$	<code>\nparallel</code>
$x \nVdash y$	<code>\nVDash</code>	$x \nVdash y$	<code>\nVDash</code>
$x \ntriangleright y$	<code>\ntriangleright</code>	$x \ntrianglerighteq y$	<code>\ntrianglerighteq</code>
$x \not\supseteq y$	<code>\nsupseteq</code>	$x \not\supseteq y$	<code>\nsupseteq</code>
$x \supsetneq y$	<code>\supsetneq</code>	$x \supsetneqq y$	<code>\varsupsetneq</code>
$x \supsetneqq y$	<code>\supsetneqq</code>	$x \supsetneqq y$	<code>\varsupsetneqq</code>

3.4.37 Math “Torture” Test **Sans Serif Bold**

Most of the following examples are taken from *The T_EXbook* (Knuth 1984, see <https://ctan.org/pkg/texbook>) and were adapted for L^AT_EX from Karl Berry’s torture test for plain T_EX math fonts.

$x + y - z, \quad x + y * z, \quad z * y / z, \quad (x + y)(x - y) = x^2 - y^2,$
 $x \times y \cdot z = [xyz], \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,$
 $x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y / z, \quad x := y, \quad x \leq y \neq z, \quad x \sim y \simeq z \equiv y \not\equiv z, \quad x \subset y \subseteq z$
 $\sin 2\theta = 2 \sin \theta \cos \theta, \quad O(n \log n \log n), \quad \Pr(X > x) = \exp(-x/\mu),$
 $(x \in A(n) \mid x \in B(n)), \quad \bigcup_n X_n \parallel \bigcap_n Y_n$
 In-text matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} a & b & c \\ 1 & m & n \end{pmatrix}$.

$$a_0+\frac{1}{a_1+\frac{1}{a_2+\frac{1}{a_3+\frac{1}{a_4}}}}$$

$$\binom{p}{2}x^2y^{p-2}-\frac{1}{1-x}\frac{1}{1-x^2}=\frac{a+1}{b}\bigg/\frac{c+1}{d}.$$

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|\phi(x+iy)|^2=0$$

$$\pi(n)=\sum_{m=2}^n\left[\left(\sum_{k=1}^{m-1}\lfloor(m/k)/\lceil m/k\rceil\rfloor\right)^{-1}\right].$$

$$\int_0^\infty \frac{t-\mathrm{i}b}{t^2+b^2}e^{\mathrm{i}at}\,\mathrm{d}t=e^{ab}E_1(ab),\quad a,b>0.$$

$$A:=\begin{pmatrix}x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda\end{pmatrix}.$$

$$\begin{pmatrix}a & b & c \\ d & e & f\end{pmatrix}\begin{pmatrix}u & x \\ v & y \\ w & z\end{pmatrix}$$

$$A=\begin{pmatrix}a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn}\end{pmatrix}$$

$$M=\begin{matrix} & C & I & C' \\ \begin{matrix} C \\ I \\ C' \end{matrix} & \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{matrix}$$

$$\sum_{n=0}^\infty a_nz^n \text{ converges if } |z|<\Big(\limsup_{n\rightarrow\infty}\sqrt[n]{|a_n|}\Big)^{-1}.$$

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \rightarrow f'(x) \quad \text{as } \Delta x \rightarrow 0.$$

$$\|u_i\| = 1, \quad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of } \begin{Bmatrix} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{Bmatrix} \text{ is } \begin{Bmatrix} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{Bmatrix}.$$

$$\begin{aligned} T(n) &\leq T(2^{\lceil \lg n \rceil}) \leq c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\ &< 3c \cdot 3^{\lg n} \\ &= 3cn^{\lg 3}. \end{aligned}$$

$$\begin{aligned} (x+y)(x-y) &= x^2 - xy + yx - y^2 \\ &= x^2 - y^2 \\ (x+y)^2 &= x^2 + 2xy + y^2. \end{aligned}$$

$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} dr d\theta \\ &= \int_0^{2\pi} \left(e^{-\frac{r^2}{2}} \Big|_{r=0}^{r=\infty} \right) d\theta \\ &= \pi. \end{aligned}$$

$$\prod_{k \geq 0} \frac{1}{(1 - q^k z)} = \sum_{n \geq 0} z^n / \prod_{1 \leq k \leq n} (1 - q^k).$$

$$\sum_{\substack{0 < i \leq m \\ 0 < j \leq n}} p(i,j) \neq \sum_{i=1}^p \sum_{j=1}^q \sum_{k=1}^r a_{ij} b_{jk} c_{ki} \neq \sum_{\substack{1 \leq i \leq p \\ 1 \leq j \leq q \\ 1 \leq k \leq r}} a_{ij} b_{jk} c_{ki}$$

$$\max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\text{Inline math: } \max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$p_1(n) = \lim_{m \rightarrow \infty} \sum_{v=0}^{\infty} (1 - \cos^{2m}(v!^n \pi/n))$$

$$\text{Inline math: } p_1(n) = \lim_{m \rightarrow \infty} \sum_{v=0}^{\infty} (1 - \cos^{2m}(v!^n \pi/n))$$

Lebenslauf

Geboren am 24. Januar 1995 in Summacumlaudeville, wuchs ich in Neustadt (Nordrhein-Westfalen) sowie in Newcastle (Nova Landia, Neufundland) auf. Im Jahr 2013 erlangte ich am Gymnasium Neustadt die allgemeine Hochschulreife. Im Wintersemester 2013/2014 habe ich zunächst das Studium der Kunstgeschichte an der Rheinischen Friedrich-Wilhelms-Universität Bonn begonnen. Im Sommersemester 2014 nahm ich dann das Studium der Volkswirtschaftslehre auf, das ich im August 2018 mit dem Abschluss Master of Science (M. Sc.) beendete (Gesamtnote: 1,3). Meine Masterarbeit „The Influence of Stress on the Performance of BGSE Graduate Students“ wurde von Prof. Dr. Lorem Ipsum betreut. Während des Masterstudiums besuchte ich im Herbst 2016 die Universität Tel Aviv in Israel als Austauschstudent. Im Oktober 2018 habe ich das Promotionsstudium an der Bonn Graduate School of Economics aufgenommen.